Immunology

Career Avenues

Table of Contents

INTRODUCTION TO THE IMMUNE SYSTEM	4
PRIMARY AND SECONDARY LYMPHOID ORGAN	13
ANTIGEN	22
CELLS OF IMMUNE SYSTEM	34
MAJOR-HISTOCOMPATIBILITY COMPLEX (MHC)	49
ANTIGEN PROCESSING AND PRESENTATION	53
IMMUNOGLOBULINS	55
ORIGIN OF ANTIBODY DIVERSITY	68
POLYCLONAL AND MONOCLONAL ANTIBODIES	75
COMPLEMENT	79
ANTIGEN-ANTIBODY REACTION	88
REGULATION OF IMMUNE RESPONSES	91
IMMUNE TOLERANCE	96
HYPERSENSITIVITY	99
AUTO-IMMUNITY	104
HIV	107
PRINCIPLE AND TECHNIQUES OF IMMUNOLOGY	111
KEY NOTES FOR IMMUNOLOGY	121

Career Avenues

INTRODUCTION TO THE IMMUNE SYSTEM

The immune system is remarkably versatile defence system that has evolved to protect animals from invading pathogenic microorganisms and cancer. It is able to generate an enormous variety of cells and molecules capable of specifically recognizing and eliminating an apparently limitless variety of foreign invaders. These cells and molecules act together in a dynamic network whose complexity rivals that of the nervous system.

Functionally, an immune response can be divided into two related activities — recognition and response. Immune recognition is remarkable for its specificity. The immune system is able to recognize subtle chemical differences that distinguish one foreign pathogen from another. Furthermore, the system is able to discriminate between foreign molecules and the body's own cells and proteins. Once a foreign organism has been recognized, the immune system recruits a variety of cells and molecules to mount an appropriate response, called an **effector response,** to eliminate or neutralize the organism. In this way the system is able to convert the initial recognition event into a variety of effector responses, each uniquely suited for eliminating a particular type of pathogen. Later exposure to the same foreign organism induces a **memory response,** characterized by a more rapid and heightened immune reaction that serves to eliminate the pathogen and prevent disease.

TYPES OF IMMUNITY

Innate Immunity: Innate immunity can be seen to comprise four types of defensive barriers: anatomic, physiologic, phagocytic, and inflammatory.

Career Avenues

TYPE	MECHANISM
Anatomic barriers	
Skin	Mechanical barriers retards entry of microbes. Acidic environment (pH 3-5) retards growth of microbes.
Mucous membranes	Normal flora compete with microbes for attachment sites and nutrients. Mucus entraps foreign microorganisms. Cilia propel microorganisms out of body.
Physiologic barriers	
Temperature	Normal body temperature inhibits growth of some pathogens. Fever response inhibits growth of some pathogens.
Low pH	Acidity of stomach contents kills most ingested microorganisms.
Chemical mediators	Lysozyme cleaves bacterial cell wall. Interferon induces antiviral state in uninfected cells. Complement lyses microorganisms or facilitates phagocytosis. Toll-like receptors recognize microbial molecules, signal cell to secrete immunostimulatory cytokines. Collectins disrupt cell wall of pathogen.
Phagocytic/endocytic barriers	Various cells internalize(endocytose) and break down foreign macromolecules. Specialized cells(blood monocytes, neutrophils, tissue macrophages) internalize(phagocytose), kill, and digest whole microorganisms.
Inflammatory barriers	Tissue damage and infection induce leakage of vascular fluid, containing serum proteins with antibacterial activity, and influx of phagocytic cells into the affected area.

Career Avenues

ADAPTIVE IMMUNITY

Adaptive immunity or acquired immunity, obtained in some manner other than heredity. It provides specific, nonhereditary defense and protection after exposure to a specific pathogen.

(A) **Naturally acquired adaptive immunity** is obtained by having a specific disease. The immune system respond to antigens, activates T cells, produces antibodies to protect against the same disease. **Colostrum** is the first fluid secreted by the mammary glands after childbirth. It contains a lot of antibodies. Only protect for a short time then disappear.

Naturally acquire passive immunity refers to the antibodies across the placenta and in breast milk.

(B) **Artificially acquired adaptive immunity** is obtained by the injections of **vaccine** that produces immunity.

All animals have an innate immune system. Only vertebrates have an adaptive immune system.

ACTIVE AND PASSIVE IMMUNITY

- In **active immunity,** an individual's own immune system makes antibodies.
- In **passive immunity**, ready-made antibodies are introduced into the body.
  - **Naturally acquired passive immunity** is by receiving antibodies from planceta or in colostrums.
  - **Artificially acquired passive immunity is** by receiving injection of gamma globulin or immune serum.

OTHER CLASSIFICATIONS

TYPES OF IMMUNITY

A. **Natural (innate) immunity/natural resistance** – the inherited immunity individual may possess due to race, species, or ethnic background (some groups naturally resistant to certain diseases, just as some are more susceptible to certain diseases).

B. **Acquired Immunity –** the body develops specific immunity (antibodies and cells) against invading agents such as lethal bacteria, viruses, toxins, and even foreign tissues from other organisms. This is further classified in two types:

- Active Humoral immunity:
  - Active natural immunity: comes from having the diseases (ex. Mumps) acquired during bacterial and viral infections.
  - Active artificial immunity: comes from receiving a vaccination containing a modified pathogen or toxin, which stimulates production of antibodies.

Career Avenues

- Passive Humoral immunity:
    - Passive natural immunity – antibodies received through maternal-fetus transmission (antibodies are temporary – antibodies will protect for up to 6 months or longer if the mother continues nursing)
    - Passive artificial immunity – injection of antibodies (antitoxins, gamma globulin)

C. Cell Mediated Immune Response

- Lymphocytes
- MHC (major histocompatibility complex: Cell surface proteins that mark everybody cell as self.

IMMUNE RESPONSE

TWO TYPES OF IMMUNE RESPONSES

- **Cell-mediated immunity**: Direct response of T cells to antigens. T cells are responsible for cell medicated immunity
    - Killer cells bind to surface of invading cell, disrupt the membrane, and destroy it by altering its internal environment
    - Helper cells stimulate B cells to mature into plasma cells, which begin to synthesize and secrete immunoglobulin (proteins with known antibody activity)
        - Suppressor cells reduce the humoral immunity
- **Humoral immunity**: Also called immunoglobulin-mediated immunity. Associated with circulating antibodies. B cells are responsible for humoral immunity
    - B lymphocytes enlarge and divide to become mature plasma cells.
    - The plasma cells secrete antibodies into blood and lymph

IMMUNOGENICITY VERSUS ANTIGENICITY

Immunogenicity and antigenicity are related but distinct immunologic properties.

Immunogenesity is the ability to induce immune response. For example: humeral or cell mediated immunity.

Although a substance that induces a specific immune response is usually called an antigen, it is more appropriately called an **immunogen.**

Antigenicity is the ability to combine specifically with the final products of the above responses (i.e., antibodies and/or cell-surface receptors).

All the molecules that are immunogen are antigen but all antigens cannot be immunogen. For example, heptanes are antigenic but not immunogenic.

Career Avenues

FACTORS THAT INFLUENCE IMMUNOGENICITY

Proteins are the most potent immunogens, with polysaccharides ranking second. In contrast, lipids and nucleic acids of an infectious agent generally do not serve as immunogens unless they are complexed with proteins or polysaccharides. For cell-mediated immunity, only proteins and some lipids and glycolipids serve as immunogens. These molecules are not recognized directly. Proteins must first be processed into small peptides and then presented together with MHC molecules on the membrane of a cell before they can be recognized as immunogens.

IMMUNOGENESITY OF IMMUNOGEN

Immunogenicity of an immunogen is determined by four properties

- Foreignness
- Molecular size
- Chemical composition and heterogeneity
- Ability to be processed and presented with an MHC molecule on the surface of an antigen-presenting cell or altered self-cell.

FOREIGNNESS

Immunogenicity is directly proportional to foreignness of antigen. Generally the greater the phylogenetic distance between two species, the greater the structural (and therefore the antigenic) difference between them. For example: Bovine serum albumin (BSA) is not immunogenic for cow but strongly immunogenic for rabbits.

Some exceptions are also there. Some macromolecules like collagen and cytochrome-C are highly conserved throughout evolution that is why express little immunogenicity. Some self-components (e.g. corneal tissue and sperm) are effectively sequestered from the immune system, so that if these tissues are injected even into the animal from which they originated, they will function as immunogens.

MOLECULAR SIZE

There is a correlation between the size of a macromolecule and its immunogenicity. The most active immunogens tend to have a molecular mass of 100,000 daltons (Da). Thus, if molecular weight decreases, immunogenicity also decreases.

CHEMICAL COMPOSITION AND HETEROGENEITY

Chemical composition plays an important role to make a molecule immunogenic. For example, synthetic homopolymers (polymers composed of a single amino acid or sugar) tend to lack immunogenicity regardless of their size. Copolymers composed of different amino acids or sugars are usually more immunogenic than homopolymers of their constituents. Immunogenicity is directly proportional to complexity of immunogen.

Career Avenues

Thus the immunogenicity in proteins (if we consider all four levels of protein organization i.e. primary, secondary, tertiary, and quaternary):

Quaternary > Tertiary > Secondary > Primary

SUSCEPTIBILITY TO ANTIGEN PROCESSING AND PRESENTATION

The development of both humoral and cell-mediated immune responses requires interaction of T cells with antigen that has been processed and presented together with MHC molecules. Large, insoluble macromolecules generally are more immunogenic than small, soluble ones because the larger molecules are more readily phagocytosed and processed. Macromolecules that cannot be degraded and presented with MHC molecules are poor immunogens.

Ex. Polymers of D-amino acids, which are stereoisomers of the naturally occurring L-amino acids. Because the degradative enzymes within antigen-presenting cells can degrade only proteins containing L-amino acids, polymers of D-amino acids cannot be processed and thus are poor immunogens.

HAPTANES

Those substances who fail to induce immune responses in their native form because of their low molecular weight and their less chemical complexity called haptens. Haptens are antigenic but not immunogenic. These molecules become immunogenic and can induce immune response when they are conjugated to high molecular weight, physiochemically complex carriers.

ANTIGEN- ANTIBODY INTERACTIONS

Ag:Ab interaction is highly specific and occurs in similar way as enzyme substrate interaction. The binding between antigens and immune components (intigen:antibody binding) involves weak non-covalent interactions. The binding forces are relatively weak and reversible and consist mainly of van der walls forces, electrostatic forces, and hydrophobic interactions.

The smallest unit of antigen that is capable of binding with antibodies is called antigenic determinant or epitope. The corresponding area on antibody molecule combining with the epitope is called paratope. The number of epitops on the surface of antigen is its valence. The valence determines the number of antibody molecules that can combine with the antigen at one time. If one epitope is present, the antigen is monovalent. Most antigen, however, have more than one copy of same epitope and are termed polyvalent.

AFFINITY AND AVIDITY

The intrinsic association constant that characterizes the non-covalent interaction between single antigen binding sites of an antibody with an epitope is termed affinity.

When the antigen consists of many repeating identical epitope or when antigens are multivalent, the association between entire antigen molecules and antibodies depend not only on affinity between each epitope and its corre0.sponding antibody but also depend on the sum of affinities of all the epitops involved.

Career Avenues

The term avidity is used to represent the overall binding between antibodies and a multivalent antigen. So, when complex antigen having multiple repeating epitops, are mixed with antibody having multiple binding sites, the interaction of such type between multivalent Ab and Ag is called, the avidity.

Thus, in general, IgM antibodies with 10 antigen binding sites are of higher avidity the IgG antibodies, although the binding of each Fab in the IgM antibody with ligand may be of the same affinity as that of the Fab from IgM.

ADJUVANTS

Adjuvants are a substance that when mixed with an immunogen and injected with it, enhances the immune response against the immunogen. Adjuvants are often used to boost the immune response in case of weak immunogen or when antigens are available in small amount.

A adjuvant and a carrier molecule for a hapten is different. Heptanes will become immunogenic when conjugated covalently to a carrier; it will not immunogenic if mixed with an adjuvant. Thus an adjuvant enhances the immune response to immunogens but does not confer the immunogenicity to haptens.

While many adjuvants have been developed in animal models and tested experimentally in humans, only one has been used for routine vaccination. Currently aluminium potassium sulfate (alum) is a common adjuvant used for human vaccines. Alum prolongs the persistence of antigen. When an antigen mixed with alum, the salt precipitates the antigen.

Another commonly used adjuvants is Freund's complete and incomplete adjuvant. Freund's incomplete adjuvant contains antigen in aqueous solution, oil, and an emulsifying agent such as mannide monooleate. Freund's complete adjuvant contains heat-killed mycobacteria as an additional ingredient.

CROSS REACTIVE ANTIGENS

Cross reactive antigen is one which is capable of binding to antibody produced in response to a different antigen. This is due to sharing of antigenic determinants by the two antigen. One such cross reactive antigen is hetrophyle antigen.

HETROPHYLE ANTIGEN OR HETROGENETIC ANTIGEN

These are the antigen which present on the surface of tissue cells of unrelated plants, animals and bacteria but are identical so that the antibody to one will cross reacts with other. Very often these hetrogenetic antigens are polysaccharides. For example: human blood group 'a' antigen is cross reactive with antibody of pneumococcal capsular polysaccharide 'type XIV'. Human blood group is cross reactive with certain strain of E. coli, common colon bacteria.

Forssman Antigen

This antigen is truly hetrogenetic as it is found on the RBCs of many species such as horse, sheep, dog, cat, wheal, toad and many other organisms. Humans are generally forssman negative.

Career Avenues

COMPARISON OF ANTIGEN RECOGNITION BY T CELLS AND B CELLS

The recognition of antigen by T cells and B cells is fundamentally different. B cell recognize soluble antigens whereas most T cells recognize only peptide combined with MHC molecules.

Table: Antigen Recognition By B And T Cells

CHARACTERISTICS	B CELLS	T CELLS
Antigen interaction	B cell receptor (membrane Ig) binds antigen	T cell receptor binds antigen presented with MHC
Nature of antigens	Protein, polysaccharide, glycolipid	Peptide
Binding to soluble antigens	Yes	No
Epitops recognized	Accessible areas of protein structure containing sequential amino acids and nonsequential amino acids	Antigen processed internally and presented as linear peptides bound to MHC molecules
MHC molecules		Required to display processed antigen

T- INDEPENDENT AND T-DEPENDENT ANTIGENS

T-dependent (TD) antigens are those that requires direct contact with T helper (Th) cells, not simply exposure to Th – derived cytokines to stimulate B-cells for the production of antibody. Proteins are TD antigens.

T- independent (TI) antigens are antigens that can activate B cells in the absence of any direct contact with Th cells. Examples of TI antigens are lipopolycasscahride (LPS) of bacterial outer membrane, bacterial flagellin and cell-wall and capsular polysaccharides.
Unlike the TD antigens, the TI antigens:

- Do not produce isotype switching (IgM is exclusively produced)
- Do not demonstrate affinity maturation (in which antibodies of progressively higher affinity are produced).
- Do not show immunological memory and secondary immune responses.

SUPERANTIGENS

Superantigens are viral or bacterial proteins that bind simultaneously to the variable domain of β of a T cell receptor and to the α chain of class II MHC molecule. Because of their unique binding ability, superantigens can activate large number of T cells irrespective of their antigenic specificity. Superantigen can be exogenous or endogenous.

Career Avenues



GATE QUESTIONS

GATE 2007

Q.1 In addition to adjuvants, generation of anti-hapten antibodies will require injection of
(A) the hapten to a mice
(B) a mixture of hapten and protein to a mice
(C) the hapten covalently linked to a protein to a mice
(D) a mixture of hapten and lipid to a mice

GATE 2009

Q.2 Which of these can be an antigen but cannot induce immune responses ?
(A) Hemacyanin (B) Influenza virus
(C) Azobenzene arsonate (D) Corynebacteria

Answers :

1-C. 2-C.

Career Avenues

PRIMARY AND SECONDARY LYMPHOID ORGAN

ORGANS OF THE IMMUNE SYSTEM

A number of morphologically and functionally diverse organs and tissues have various functions in the development of immune responses. These can be distinguished by function as the primary and secondary lymphoid organs. The thymus and bone marrow are the primary (or central) lymphoid organs, where maturation of lymphocytes takes place. The lymph nodes, spleen, and various mucosal associated lymphoid tissues (MALT) such as gut-associated lymphoid tissue (GALT) are the secondary (or peripheral) lymphoid organs, which trap antigen and provide sites for mature lymphocytes to interact with that antigen. In addition, tertiary lymphoid tissues, which normally contain fewer lymphoid cells than secondary lymphoid organs, can import lymphoid cells during an inflammatory response. Most prominent of these are cutaneous-associated lymphoid tissues. Once mature lymphocytes have been generated in the primary lymphoid organs, they circulate in the blood and lymphatic system, a network of vessels that collect fluid that has escaped into the tissues from capillaries of the circulatory system and ultimately return it to the blood.

The lymphoid cell line traces its origin to a pluripotential stem cell that in the fetus is found within the yolk sac, bone marrow, and liver and in the adult in the bone marrow. The differentiation of the stem cell to lymphoid or myeloid pathways is dependent upon the microenvironments in which these stem cells develop. These environments are provided in what is called the lymphoid system. The lymphoid system involves the organs and tissues in which lymphocytic cells originate as lymphocyte precursors and then mature and differentiate; the cells finally lodge in the lymphoid organs or move throughout the body.

THE LYMPHOID ORGANS

The lymphoid tissues are divided functionally into primary and secondary organs.

- The central (primary) lymphoid organs are the thymus and the bursa or bone marrow.
- The peripheral (secondary) lymphoid tissues are the lymph nodes, spleen, diffuse lymphoid tissues, and lymphoid follicles.

Stem cells from bone marrow or embryonic tissues are deposited in the primary lymphoid organs, where the cells mature into lymphocytes. It is here where the precursors acquire the ability to recognize antigens through the development of specific surface receptors. Lymphocytes are precursor cells of immunologic function as well as regulators and effectors of immunity. T cells have a long life span of months or years, where most B cells have a shorter life span of 5 to 7 days.

Career Avenues

The immunocompetent cells produced in the thymus are called T lymphocytes (T cells), and those produced and maturate in the bone marrow are the B lymphocyte (B cells). The site in the lymph node where the plasma cell produces the antibodies is called germinal centers or follicles. The primary follicles contain small B lymphocytes, following antigenic stimulation; secondary follicles develop which contain many large dividing lymphocytes and plasma cells.

In order to carry out the functions of the immunity, a ubiquitous immunologic cell system has appeared with in the vertebrates; the lymphoreticular system. This collection of cellular elements is distributed strategically throughout the tissues as well as lining lymphatic and vascular channels. Its cells are housed within the blood, tissues, thymus, lymph nodes, and spleen (internal secretory system), and in those body tracts exposed to the external environment – the respiratory, gastrointestinal, and genitourinary systems (external secretory system).

PRIMARY LYMPHOID ORGANS

Immature lymphocytes generated in hematopoiesis mature and become committed to a particular antigenic specificity within the primary lymphoid organs. Only after a lymphocyte has matured within a primary lymphoid organ is the cell immunocompetent (capable of mounting an immune response). T cells arise in the thymus, and in many mammals—humans and mice for example—B cells originate in bone marrow.

THYMUS

The thymus is the site of T-cell development and maturation. It is a flat, bilobed organ situated above the heart. Each lobe is surrounded by a capsule and is divided into lobules, which are separated from each other by strands of connective tissue called trabeculae.

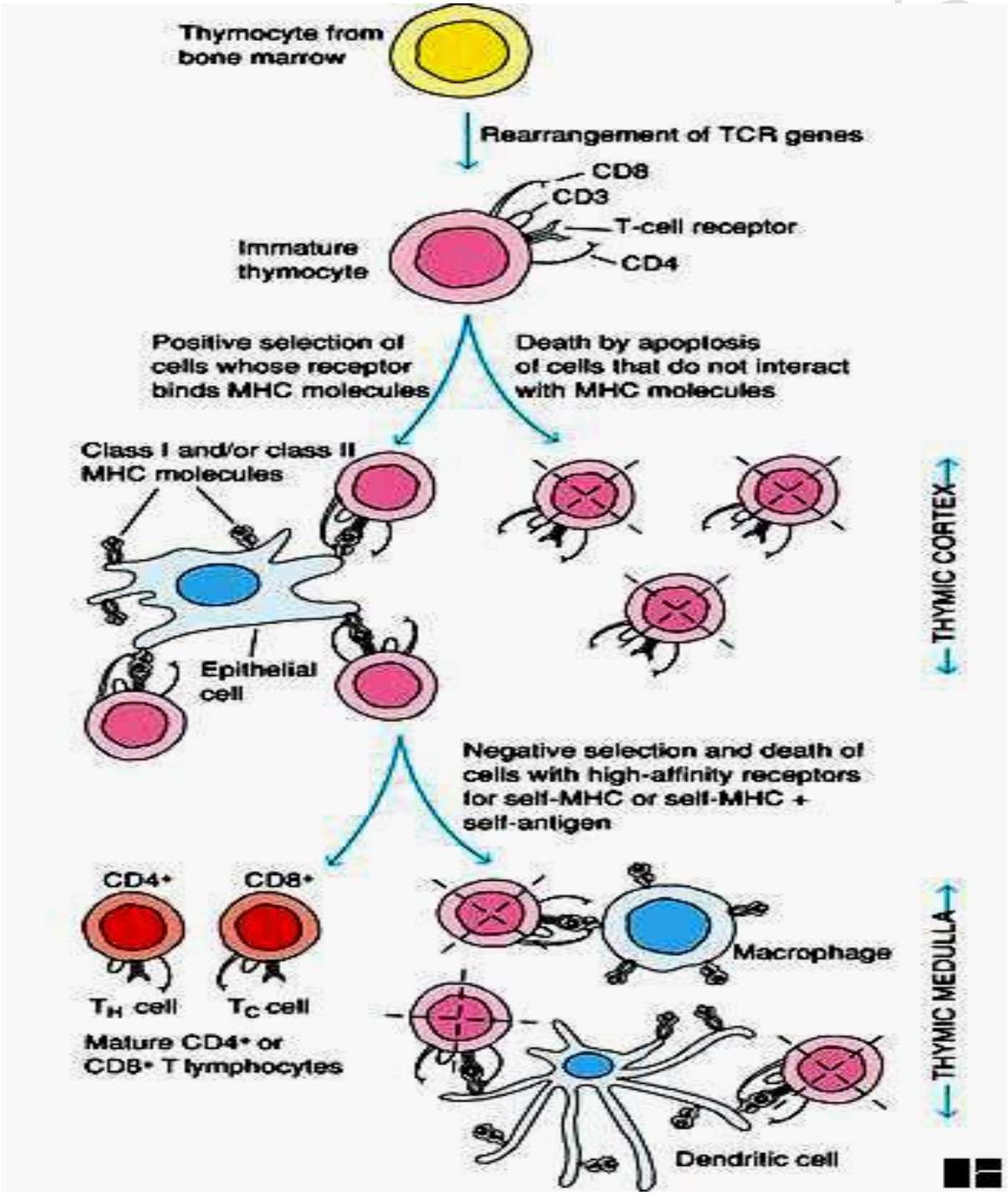
Each lobule is organized into two compartments:

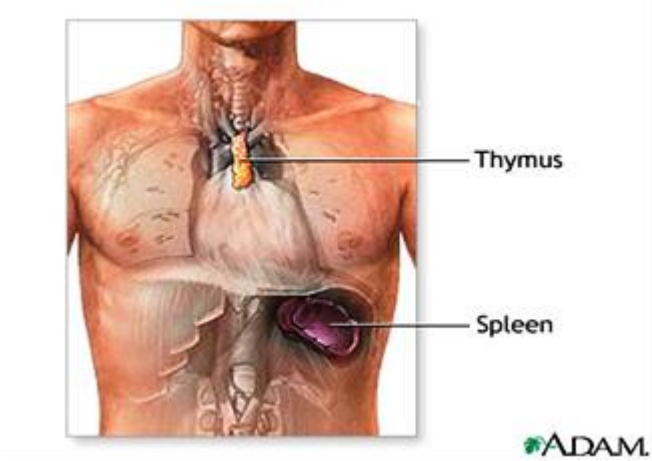
- The outer compartment, or cortex, is densely packed with immature T cells, called thymocytes.
- The inner compartment, or medulla, is sparsely populated with thymocytes.

Both the cortex and medulla of the thymus are crisscrossed by a three-dimensional stromal-cell network composed of epithelial cells, dendritic cells, and macrophages, which make up the framework of the organ and contribute to the growth and maturation of thymocytes. Many of these stromal cells interact physically with the developing thymocytes. Some thymic epithelial cells in the outer cortex, called **nurse cells,** have long membrane extensions that surround as many as 50thymocytes, forming large multicellular complexes. Other cortical epithelial cells have long interconnecting cytoplasmic extensions that form a network and have been shown to interact with numerous thymocytes as they traverse the cortex. The function of the thymus is to generate and select a repertoire of T cells that will protect the body from infection.

Career Avenues

As thymocytes develop, an enormous diversity of T-cell receptors is generated by a random process that produces some T cells with receptors capable of recognizing antigen-MHC complexes. However, most of the T-cell receptors produced by this random process are incapable of recognizing antigen-MHC complexes and a small portion react with combinations of self antigen-MHC complexes. The thymus induces the death of those T cells that cannot recognize antigen-MHC complexes and those that react with self-antigen– MHC and pose a danger of causing autoimmune disease. More than 95% of all thymocytes die by apoptosis in the thymus without ever reaching maturity.



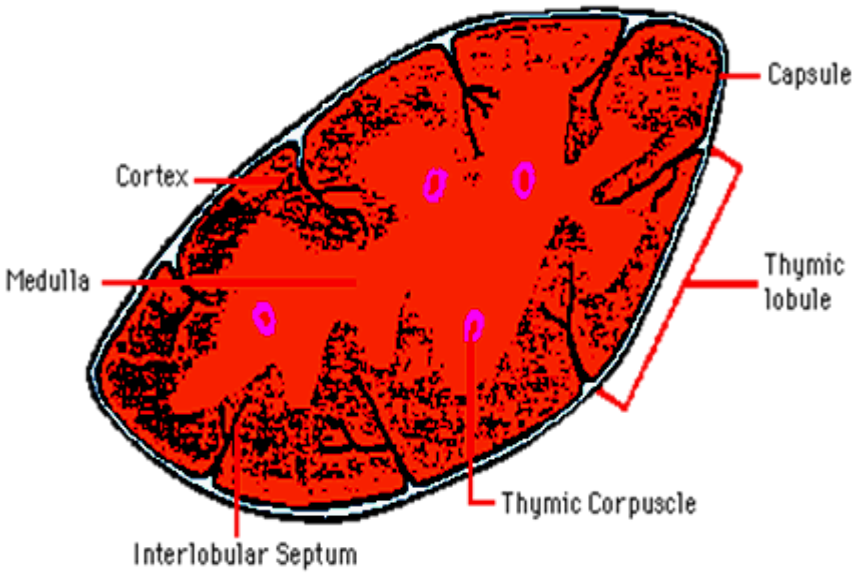


THE THYMUS AND IMMUNE FUNCTION

The role of the thymus in immune function can be studied in mice by examining the effects of neonatal thymectomy, a procedure in which the thymus is surgically removed from newborn mice. These thymectomized mice show a dramatic decrease in circulating lymphocytes of the T-cell lineage and an absence of cell-mediated immunity. Other evidence of the importance of the thymus comes from studies of a congenital birth defect in humans (DiGeorge's syndrome) and in certain mice (nude mice) in which the thymus fails to develop. In both cases, there is an absence of circulating T cells and of cell-mediated immunity and an increase in infectious disease. Aging is accompanied by a decline in thymic function. This decline may play some role in the decline in immune function during aging in humans and mice.

The thymus reaches its maximal size at puberty and then atrophies, with a significant decrease in both cortical and medullary cells and an increase in the total fat content of the organ. Whereas the average weight of the thymus is 70 g in infants, its age-dependent involution leaves an organ with an average weight of only 3g in the elderly. A number of experiments have been designed to look at the effect of age on the immune function of the thymus. In one experiment, the thymus from a 1-day-old or 33-month old mouse was grafted into thymectomized adults. Mice receiving the newborn thymus graft showed a significantly larger improvement in immune function than mice receiving the 33- month-old thymus.

Career Avenues



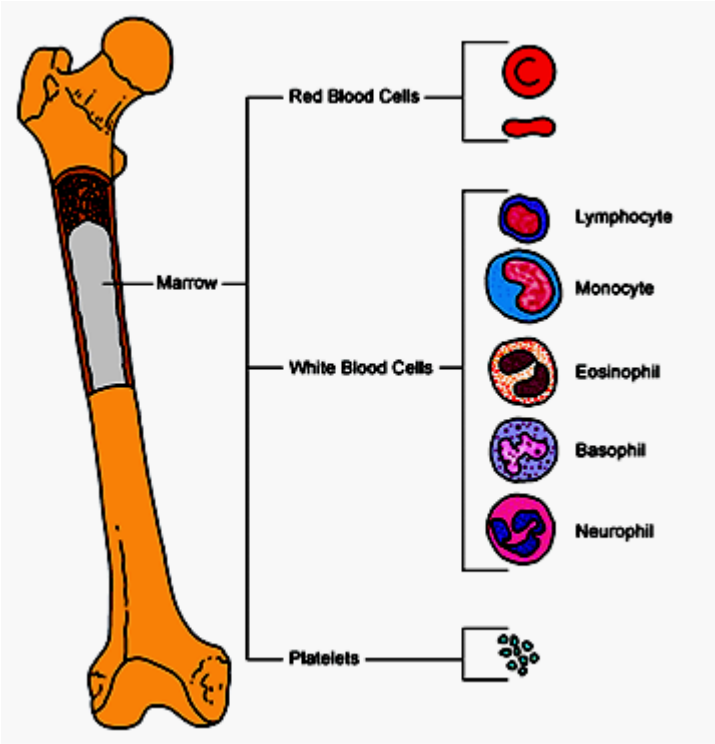
BONE MARROW

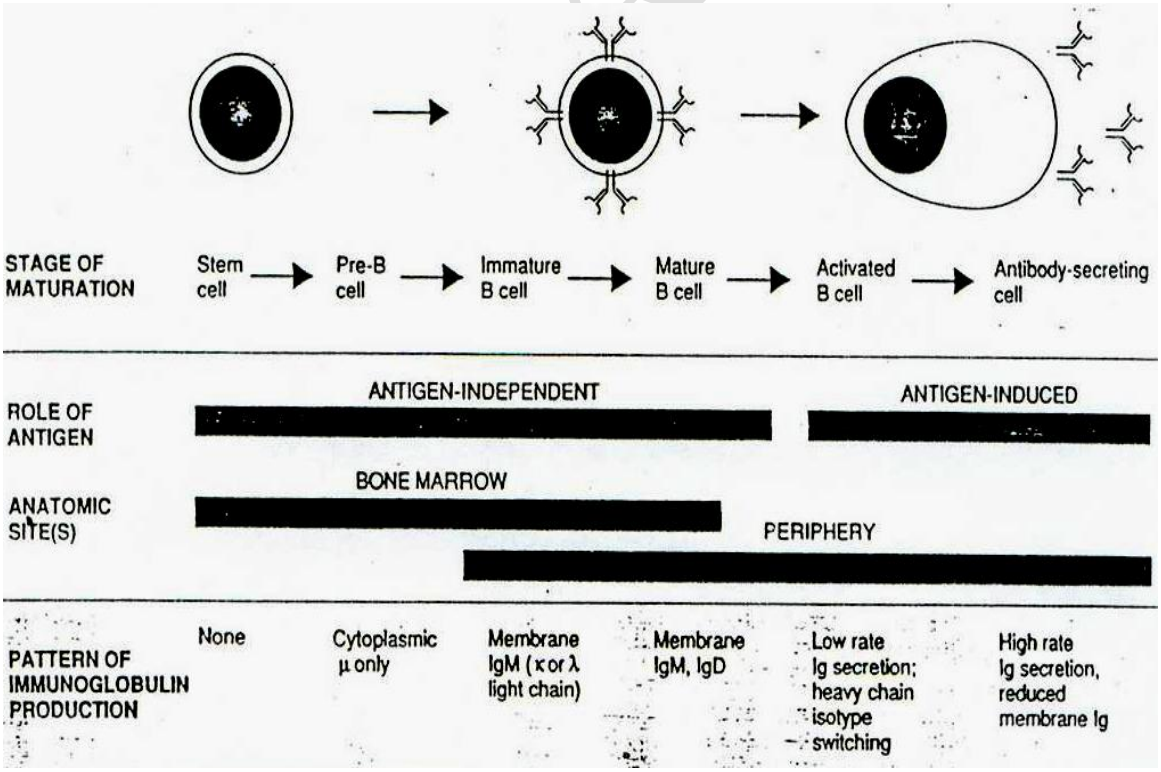
In humans and mice, bone marrow is the site of B-cell origin and development. Arising from lymphoid progenitors, immature B cells proliferate and differentiate within the bone marrow, and stromal cells within the bone marrow interact directly with the B cells and secrete various cytokines that are required for development. Like thymic selection during T cell maturation, a selection process within the bone marrow eliminates B cells with self-reactive antibody receptors. Bone marrow is not the site of B-cell development in all species. In birds, a lymphoid organ called the bursa of Fabricius, a lymphoid tissue associated with the gut, is the primary site of B-cell maturation.

In mammals such as primates and rodents, there is no bursa and no single counterpart to it as a primary lymphoid organ. In cattle and sheep, the primary lymphoid tissue hosting the maturation, proliferation, and diversification of B cells early ingestation is the fetal spleen. Later in gestation, this function is assumed by a patch of tissue embedded in the wall of the intestine called the ileal Peyer's patch, which contains a large number ($>10^{10}$) B cells. The rabbit, too, uses gut-associated tissues such as the appendix as primary lymphoid tissue for important steps in the proliferation and diversification of B cells.

Bone marrow is the site of hematopoiesis as we age it occurs mainly in the flat bones-sternum,vertebra, iliac (hip) bones, and ribs. All the blood cells originate from a common pluripotent stem cell that becomes committed to differentiate along a particular lineage.T cells within the bone marrow produce IL3 which acts on all stem cells and GM-CSF which stimulates the formation of granulocytes and monocytes. MO and marrow stromal cells produce GM-CSF and G-CSF and MCSF and IL1 and IL6 which enhance cell differentiation. Interleukin-7 also produced by marrow stromal cells stimulates the maturation of B cells from precursor cells. However, little is known about the nature of the uncommitted stem cell or the mechanisms that regulate its commitment to specific lineages.

Career Avenues





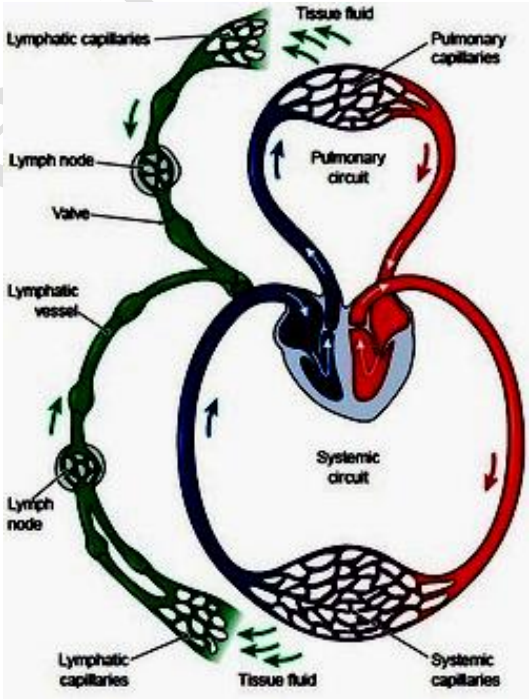
Career Avenues

LYMPHATIC SYSTEM

As blood circulates under pressure, its fluid component(plasma) seeps through the thin wall of the capillaries into the surrounding tissue. Much of this fluid, called interstitial fluid, returns to the blood through the capillary membranes. The remainder of the interstitial fluid, now called lymph, flows from the spaces in connective tissue into a network of tiny open lymphatic capillaries and then into a series of progressively larger collecting vessels called lymphatic vessels. The largest lymphatic vessel, the thoracic duct, empties into the left subclavian vein near the heart. In this way, the lymphatic system captures fluid lost from the blood and returns it to the blood, thus ensuring steady-state levels of fluid within the circulatory system.

The heart does not pump the lymph through the lymphatic system; instead the flow of lymph is achieved as the lymph vessels are squeezed by movements of the body's muscles. A series of one-way valves along the lymphatic vessels ensures that lymph flows only in one direction. When a foreign antigen gains entrance to the tissues, it is picked up by the lymphatic system (which drains all the tissues of the body) and is carried to various organized lymphoid tissues such as lymph nodes, which trap the foreign antigen. As lymph passes from the tissues to lymphatic vessels, it becomes progressively enriched in lymphocytes.

Thus, the lymphatic system also serves as a means of transporting lymphocytes and antigen from the connective tissues to organized lymphoid tissues where the lymphocytes may interact with the trapped antigen and undergo activation.



Career Avenues

SECONDARY LYMPHOID ORGANS

Various types of organized lymphoid tissues are located along the vessels of the lymphatic system. Some lymphoid tissue in the lung and lamina propria of the intestinal wall consists of diffuse collections of lymphocytes and macrophages. Other lymphoid tissue is organized into structures called lymphoid follicles, which consist of aggregates of lymphoid and non-lymphoid cells surrounded by a network of draining lymphatic capillaries.

Until it is activated by antigen, a lymphoid follicle—called a primary follicle—comprises a network of follicular dendritic cells and small resting B cells. After an antigenic challenge, a primary follicle becomes a larger secondary follicle—a ring of concentrically packed B lymphocytes surrounding a center (the germinal center) in which one finds a focus of proliferating B lymphocytes and an area that contains non dividing B cells, and some helper T cells interspersed with macrophages and follicular dendritic cells.

Most antigen-activated B cells divide and differentiate into antibody-producing plasma cells in lymphoid follicles, but only a few B cells in the antigen-activated population find their way into germinal centers. Those that do undergo one or more rounds of cell division, during which the genes that encode their antibodies mutate at an unusually high rate. Following the period of division and mutation, there is a rigorous selection process in which more than 90% of these B cells die by apoptosis. In general, those B cells producing antibodies that bind antigen more strongly have a much better chance of surviving than do their weaker companions. The small number of B cells that survive the germinal center's rigorous selection differentiate into plasma cells or memory cells and emerge.

Lymph nodes and the spleen are the most highly organized of the secondary lymphoid organs; they comprise not only lymphoid follicles, but additional distinct regions of T cell and B-cell activity, and they are surrounded by a fibrous capsule. Less-organized lymphoid tissue, collectively called mucosal-associated lymphoid tissue (MALT), is found in various body sites. MALT includes Peyer's patches (in the small intestine), the tonsils, and the appendix, as well as numerous lymphoid follicles within the lamina propria of the intestines and in the mucous membranes lining the upper airways, bronchi, and genital tract.

LYMPH NODES

Lymph nodes are the sites where immune responses are mounted to antigens in lymph. They are encapsulated bean shaped structures containing a reticular network packed with lymphocytes, macrophages, and dendritic cells. Clustered at junctions of the lymphatic vessels, lymph nodes are the first organized lymphoid structure to encounter antigens that enter the tissue spaces. As lymph percolates through a node, any particulate antigen that is brought in with the lymph will be trapped by the cellular network of phagocytic cells and dendritic cells (follicular and interdigitating).

Career Avenues

The overall architecture of a lymph node supports an ideal microenvironment for lymphocytes to effectively encounter and respond to trapped antigens. Morphologically, a lymph node can be divided into three roughly concentric regions: the cortex, the paracortex, and the medulla, each of which supports a distinct microenvironment. The outermost layer, the cortex, contains lymphocytes (mostly B cells), macro-phages, and follicular dendritic cells arranged in primary follicles. After antigenic challenge, the primary follicles enlarge into secondary follicles, each containing a germinal center. In children with B-cell deficiencies, the cortex lacks primary follicles and germinal centers.

Beneath the cortex is the paracortex, which is populated largely by T lymphocytes and also contains interdigitating dendritic cells thought to have migrated from tissues to the node. These interdigitating dendritic cells express high levels of class II MHC molecules, which are necessary for presenting antigen to TH cells. Lymph nodes taken from neonatally thymectomized mice have unusually few cells in the paracortical region; the paracortex is therefore sometimes referred to as a thymus-dependent area in contrast to the cortex, which is a thymus-independent area.

The innermost layer of a lymph node, the medulla, is more sparsely populated with lymphoid-lineage cells; of those present, many are plasma cells actively secreting antibody molecules. As antigen is carried into a regional node by the lymph, it is trapped, processed, and presented together with class II MHC molecules by interdigitating dendritic cells in the paracortex, resulting in the activation of TH cells. The initial activation of B cells is also thought to take place within the T-cell-rich paracortex.

Once activated, TH and B cells form small foci consisting largely of proliferating B cells at the edges of the paracortex. Some B cells within the foci differentiate into plasma cells secreting IgM and IgG. These foci reach maximum size within 4–6 days of antigen challenge. Within 4–7 days of antigen challenge, a few B cells and TH cells migrate to the primary follicles of the cortex. It is not known what causes this migration. Within a primary follicle, cellular interactions between follicular dendritic cells, B cells, and TH cells take place, leading to development of a secondary follicle with a central germinal center.

Some of the plasma cells generated in the germinal center move to the medullary areas of the lymph node, and many migrate to bone marrow. Afferent lymphatic vessels pierce the capsule of a lymph node at numerous sites and empty lymph into the sub capsular sinus
Lymph coming from the tissues percolates slowly inward through the cortex, paracortex, and medulla, allowing phagocytic cells and dendritic cells to trap any bacteria or particulate material (e.g.antigen-antibody complexes) carried by the lymph. After infection or the introduction of other antigens into the body, th elymph leaving a node through its single efferent lymphatic vessel is enriched with antibodies newly secreted by medullary plasma cells and also has a fifty-fold higher concentration of lymphocytes than the afferent lymph.

The increase in lymphocytes in lymph leaving a node is due in part to lymphocyte proliferation within the node in response to antigen. Most of the increase, however, represents

Career Avenues

blood-borne lymphocytes that migrate into the node by passing between specialized endothelial cells that line the post capillary venules of the node. Estimates are that 25% of the lymphocytes leaving a lymph node have migrated across this endothelial layer and entered the node from the blood. Because antigenic stimulation within a node can increase this migration tenfold, the concentration of lymphocytes in a node that is actively responding can increase greatly, and the node swells visibly. Factors released in lymph nodes during antigen stimulation are thought to facilitate this increased migration.

SPLEEN

The spleen plays a major role in mounting immune responses to antigens in the blood stream. It is a large, ovoid secondary lymphoid organ situated high in the left abdominal cavity. While lymph nodes are specialized for trapping antigen from local tissues, the spleen specializes in filtering blood and trapping blood-borne antigens; thus, it can respond to systemic infections. Unlike the lymph nodes, the spleen is not supplied by lymphatic vessels. Instead, blood-borne antigens and lymphocytes are carried into the spleen through the splenic artery. Experiments with radioactively labeled lymphocytes show that more recirculating lymphocytes pass daily through the spleen than through all the lymph nodes combined.

The spleen is surrounded by a capsule that extends a number of projections (trabeculae) into the interior to form a compartmentalized structure. The compartments are of two types, the red pulp and white pulp, which are separated by a diffuse marginal zone. The splenic red pulp consists of a network of sinusoids populated by macrophages and numerous red blood cells (erythrocytes) and few lymphocytes; it is the site where old and defective red blood cells are destroyed and removed. Many of the macrophages within the red pulp contain engulfed red blood cells or iron pigments from degraded hemoglobin.

The splenic white pulp surrounds the branches of the splenic artery, forming a periarteriolar lymphoid sheath (PALS) populated mainly by T lymphocytes. Primary lymphoid follicles are attached to the PALS. These follicles are rich in B cells and some of them contain germinal centers. The marginal zone, located peripheral to the PALS, is populated by lymphocytes and macrophages. Blood-borne antigens and lymphocytes enter the spleen through the splenic artery, which empties into the marginal zone. In the marginal zone, antigen is trapped by interdigitating dendritic cells, which carry it to the PALS. Lymphocytes in the blood also enter sinuses in the marginal zone and migrate to the PALS. The initial activation of B and T cells takes place in the T cell- rich PALS.

Here interdigitating dendritic cells capture antigen and present it combined with class II MHC molecules to TH cells. Once activated, these TH cells can then activate B cells. The activated B cells, together with some TH cells, then migrate to primary follicles in the marginal zone. Upon antigenic challenge, these primary follicles develop into characteristic secondary follicles containing germinal centers(like those in the lymph nodes), where rapidly dividing B cells(centroblasts) and plasma cells are surrounded by dense clusters

Career Avenues

of concentrically arranged lymphocytes. The effects of splenectomy on the immune response depend on the age at which the spleen is removed. In children, splenectomy often leads to an increased incidence of bacterial sepsis caused primarily by Streptococcus pneumoniae, Neisseria meningitidis, and Haemophilus influenzae. Splenectomy in adults has less adverse effects, although it leads to some increase in blood-borne bacterial infections (bacteremia).

MUCOSAL-ASSOCIATED LYMPHOID TISSUE (MALT)

The mucous membranes lining the digestive, respiratory, and urogenital systems have a combined surface area of about 400 m^2 (nearly the size of a basketball court) and are the major sites of entry for most pathogens. These vulnerable membrane surfaces are defended by a group of organized lymphoid tissues mentioned earlier and known collectively as mucosal-associated lymphoid tissue (MALT). Structurally, these tissues range from loose, barely organized clusters of lymphoid cells in the lamina propria of intestinal villi to well-organized structures such as the familiar tonsils and appendix, as well as Peyer's patches, which are found within the submucosal layer of the intestinal lining. The functional importance of MALT in the body's defense is attested to by its large population of antibody-producing plasma cells, whose number far exceeds that of plasma cells in the spleen, lymph nodes, and bone marrow combined.

The tonsils are found in three locations: lingual at the base of the tongue; palatine at the sides of the back of the mouth; and pharyngeal (adenoids) in the roof of the nasopharynx. All three tonsil groups are nodular structures consisting of a meshwork of reticular cells and fibers interspersed with lymphocytes, macrophages, granulocytes, and mast cells. The B cells are organized into follicles and germinal centers; the latter are surrounded by regions showing T-cell activity. The tonsils defend against antigens entering through the nasal and oral epithelial routes. The best studied of the mucous membranes is the one that lines the gastrointestinal tract. This tissue, like that of the respiratory and urogenital tracts, has the capacity to endocytose antigen from the lumen.

Immune reactions are initiated against pathogens and antibody can be generated and exported to the lumen to combat the invading organisms. Lymphoid cells are found in various regions within this tissue. The outer mucosal epithelial layer contains so-called intraepithelial lymphocytes (IELs). Many of these lymphocytes are T cells that express unusual receptors (_T-cell receptors, or _ TCRs), which exhibit limited diversity for antigen. Although this population of T cells is well situated to encounter antigens that enter through the intestinal mucous epithelium, their actual function remains largely unknown. The lamina propria, which lies under the epithelial layer, contains large numbers of B cells, plasma cells, activated TH cells, and macrophages in loose clusters. Histologic sections have revealed more than 15,000 lymphoid follicles within the intestinal lamina propria of a healthy child. The submucosal layer beneath the lamina propria contains Peyer'spatches, nodules of 30–40 lymphoid follicles.

Career Avenues

Like lymphoid follicles in other sites, those that compose Peyer'spatches can develop into secondary follicles with germinal centers. The epithelial cells of mucous membranes play an important role in promoting the immune response by delivering small samples of foreign antigen from the lumina of the respiratory, digestive, and urogenital tracts to the underlying mucosal-associated lymphoid tissue. This antigen transport is carried out by specialized M cells. The structure of the M cell is striking: these are flattened epithelial cells lacking the microvilli that characterize the rest of the mucous epithelium. In addition, M cells have a deep invagination, or pocket, in the basolateral plasma membrane; this pocket is filled with a cluster of B cells, T cells, and macrophages. Luminal antigens are endocytosed into vesicles that are transported from the luminal membrane to the underlying pocket membrane. The vesicles then fuse with the pocket membrane, delivering the potentially response-activating antigens to the clusters of lymphocytes contained within the pocket.

M cells are located in so-called inductive sites—small regions of a mucous membrane that lie over organized lymphoid follicles. Antigens transported across the mucous membrane by M cells can activate B cells within these lymphoid follicles. The activated B cells differentiate into plasma cells, which leave the follicles and secrete the IgA class of antibodies. These antibodies then are transported across the epithelial cells and released as secretory IgA into the lumen, where they can interact with antigens. The mucous membranes are an effective barrier to the entrance of most pathogens, which thereby contributes to nonspecific immunity. One reason for this is that the mucosal epithelial cells are cemented to one another by tight junctions that make it difficult for pathogens to penetrate. Interestingly, some enteric pathogens, including both bacteria and viruses, have exploited the M cell as an entry route through the mucous-membrane barrier. In some cases, the pathogen is internalized by the M cell and transported into the pocket. In other cases, the pathogen binds to the M cell and disrupts the cell, thus allowing entry of the pathogen. Among the pathogens that use M cells in these ways are several invasive Salmonella species, Vibrio cholerae, and the polio virus.

CUTANEOUS-ASSOCIATED LYMPHOID TISSUE (CALT)

The skin is an important anatomic barrier to the external environment, and its large surface area makes this tissue important in nonspecific (innate) defenses. The epidermal (outer) layer of the skin is composed largely of specialized epithelial cells called keratinocytes. These cells secrete a number of cytokines that may function to induce a local inflammatory reaction. In addition, keratinocytes can be induced to express class II MHC molecules and may function as antigen-presenting cells. Scattered among the epithelial-cell matrix of the epidermis are Langerhans cells, a type of dendritic cell, which internalize antigen by phagocytosis or endocytosis. The Langerhans cells then migrate from the epidermis to regional lymph nodes, where they differentiate into interdigitating dendritic cells.

These cells express high levels of class II MHC molecules and function as potent activators of naive TH cells. The epidermis also contains so-called intra epidermal lymphocytes. These are similar to the intraepithelial lymphocytes of MALT in that most of them are CD8+ T

Career Avenues

cells, many of which express T-cell receptors, which have limited diversity for antigen. These intra epidermal T cells are well situated to encounter antigens that enter through the skin and some immunologists believe that they may play a role in combating antigens that enter through the skin. The underlying dermal layer of the skin contains scattered CD4_ and CD8+ T cells and macrophages. Most of these dermal T cells were either previously activated cells or are memory cells.

SYSTEMIC FUNCTION OF THE IMMUNE SYSTEM

The many different cells, organs, and tissues of the immune system are dispersed throughout the body, yet the various components communicate and collaborate to produce an effective response to an infection. An infection that begins in one area of the body initiates processes that eventually involve cells, organs, and tissues distant from the site of pathogen invasion. Consider what happens when the skin is broken, allowing bacteria to enter the body and initiate infection. The tissue damage associated with the injury and infection results in an inflammatory response that causes increased blood flow, vasodilation, and an increase in capillary permeability. Chemotactic signals are generated that can cause phagocytes and lymphocytes to leave the blood stream and enter the affected area.

Factors generated during these early stages of the infection stimulate the capacity of the adaptive immune system to respond. Langerhans cells (dendritic cells found throughout the epithelial layers of the skin and the respiratory, gastrointestinal, urinary, and genital tracts) can capture antigens from invading pathogens and migrate into a nearby lymphatic vessel, where the flow of lymph carries them to nearby lymph nodes. In the lymph nodes these class II MHC–bearing cells can become members of the interdigitating dendritic-cell population and initiate adaptive immune responses by presenting antigen to TH cells. The recognition of antigen by TH cells can have important consequences, including the activation and proliferation of TH cells within the node as the TH cells recognize the antigen, and the secretion by the activated T cells of factors that support T-cell– dependent antibody production by B cells that may already have been activated by antigen delivered to the lymph node by lymph.

The antigen-stimulated TH cells also release chemotactic factors that cause lymphocytes to leave the blood circulation and enter the lymph node through the endothelium of the post capillary venules. Lymphocytes that respond to the antigen are retained in the lymph node for 48 hours or so as they undergo activation and proliferation before their release via the node's efferent lymphatic vessel. Once in the lymph, the newly released activated lymphocytes can enter the bloodstream via the subclavian vein. Eventually, the circulation carries them to blood vessels near the site of the infection, where the inflammatory process makes the vascular endothelium of the nearby blood vessels more adherent for activated T cells and other leukocytes. Chemotactic factors that attract lymphocytes, macrophages, and neutrophils are also generated during the inflammatory process, promoting leukocyte adherence to nearby vascular epithelium and leading leukocytes to the site of the infection.

Career Avenues

Later in the course of the response, pathogen-specific antibodies produced in the node are also carried to the bloodstream. Inflammation aids the delivery of the anti-pathogen antibody by promoting increased vascular permeability, which increases the flow of antibody-containing plasma from the blood circulation to inflamed tissue. The result of this network of interactions among diffusible molecules, cells, organs, the lymphatic system, and the circulatory system is an effective and focused immune response to an infection.

Career Avenues

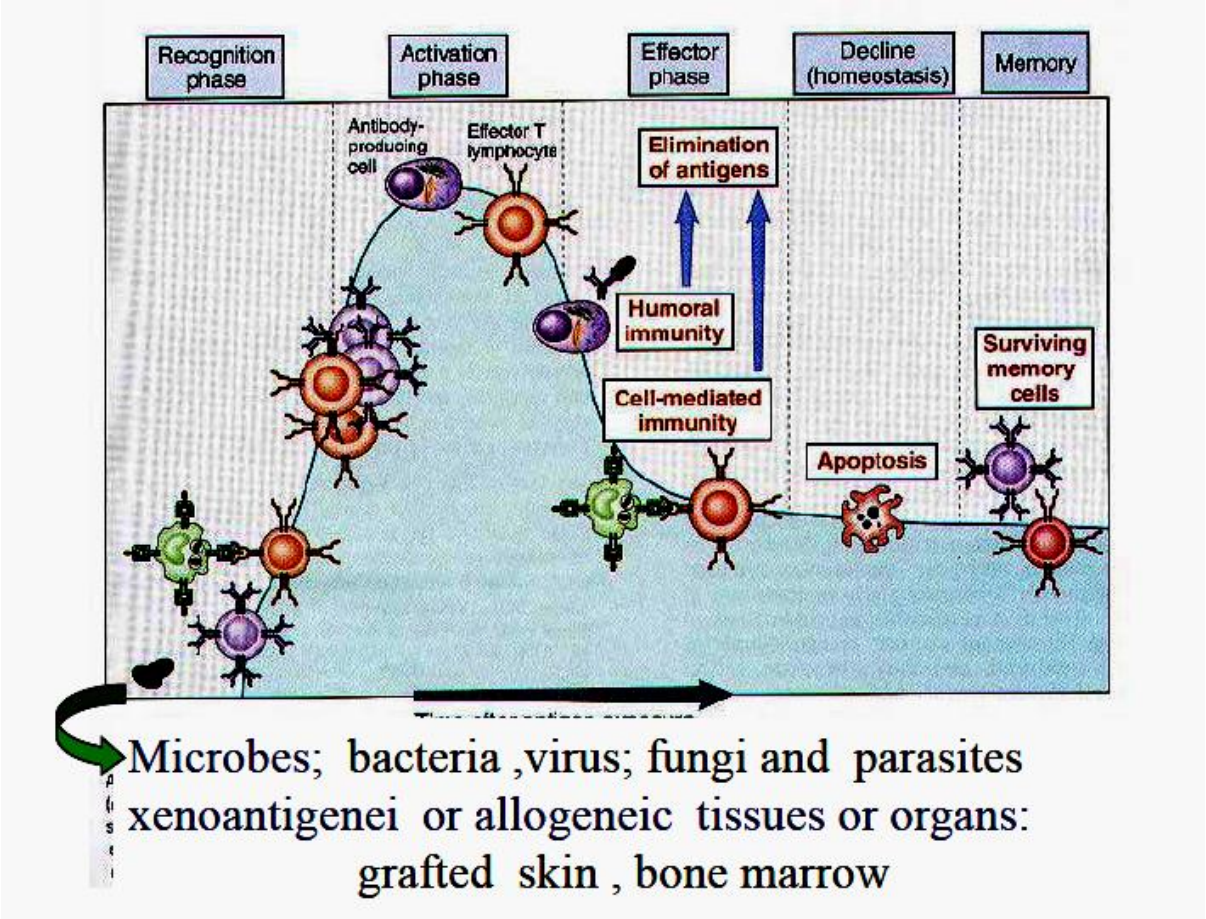
ANTIGEN

ANTIGEN

Antigen (Ag) can be described as a substance that reacts with the products of a specific immune response. The word originated from the notion that they can stimulate antibody generation. This phenomenon is known as antigenicity of a molecule.

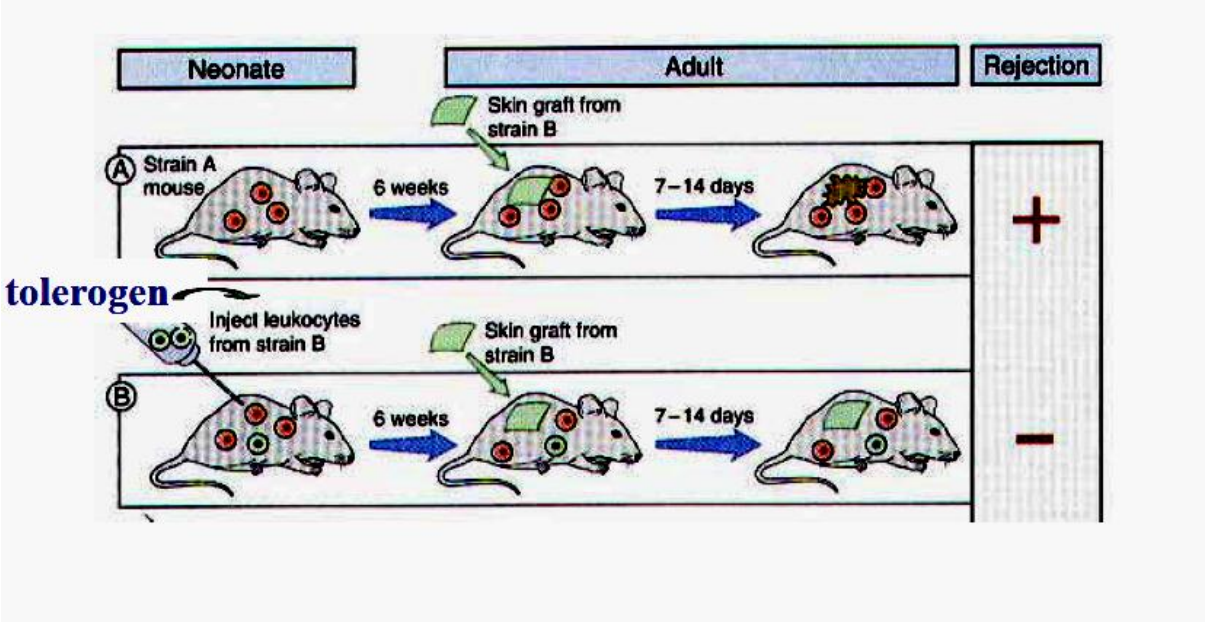
IMMUNOGEN

Immunogen on the other hand is a substance that induces a specific immune response and the phenomenon is known as immunogenicity. Immunogens are usually proteins or polysaccharides. This includes parts (coats, capsules, cell walls, flagella, fimbrae, and toxins) of bacteria, viruses, and other microorganisms. Lipids and nucleic acids are antigenic only when combined with proteins and polysaccharides. Non-microbial exogenous (non-self) antigens can include pollen, egg white, and proteins from transplanted tissues and organs or on the surface of transfused blood cells.



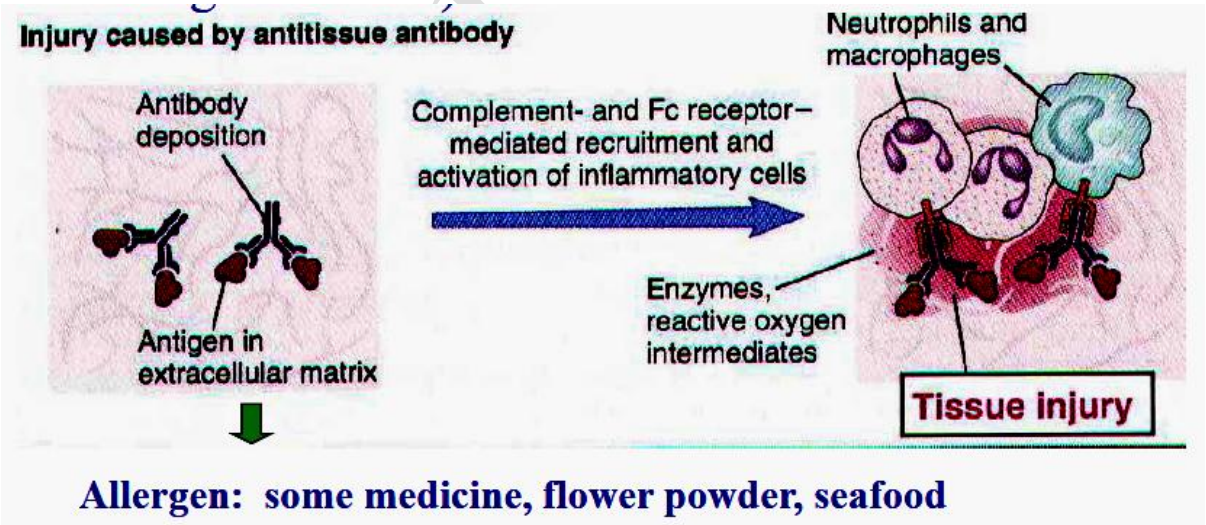
TOLEROGEN

Tolerogen is an antigen that invokes a specific immune non-responsiveness due to its molecular form. If its molecular form is changed a tolerogen can become an immunogen.



ALLERGEN

An allergen is a substance that causes the allergic reaction. The (detrimental) reaction may result after exposure via ingestion, inhalation, injection or contact with skin.



Cells present their antigens to the immune system via a histocompatibility molecule. Depending on the antigen presented and the type of the histocompatibility molecule, several types of immune cells can become activated.

ANTIGEN CLASSIFICATION

CLASSIFICATION BY ORIGIN

EXOGENOUS ANTIGENS

Exogenous antigens are antigens that have entered the body from the outside, for example by inhalation, ingestion, or injection. By endocytosis or phagocytosis, these antigens are taken into the antigen-presenting cells (APCs) and processed into fragments. APCs then present the fragments to T helper cells (CD4+) by the use of class II histocompatibility molecules on their surface. Some T cells are specific for the peptide:MHC complex. They become activated and start to secrete cytokines. Cytokines are substances that can activate cytotoxic T lymphocytes (CTL), antibody-secreting B cells, macrophages and other particles.

ENDOGENOUS ANTIGENS

Endogenous antigens are antigens that have been generated within the cell, as a result of normal cell metabolism, or because of viral or intracellular bacterial infection. The fragments are then presented on the cell surface in the complex with MHC class I molecules. If activated cytotoxic CD8+T cells recognize them, the T cells begin to secrete different toxins that cause the lysis or apoptosis of the infected cell. In order to keep the cytotoxic cells from killing cells just for presenting self-proteins, self-reactive T cells are deleted from the repertoire as a result of tolerance (also known as negative selection which occurs in the thymus).

AUTOANTIGENS

An autoantigen is usually a normal protein or complex of proteins (and sometimes DNA or RNA) that is recognized by the immune system of patients suffering from a specific autoimmune disease. These antigens should under normal conditions not be the target of the immune system, but due to mainly genetic and environmental factors the normal immunological tolerance for such an antigen has been lost in these patients.

CLASSIFICATION ACCORDING TO THE CELLULAR RESPONSE GENERATED

T-INDEPENDENT ANTIGENS

T-independent antigens are antigens which can directly stimulate the B cells to produce antibody without the requirement for T cell help In general, polysaccharides are T-independent antigens. The responses to these antigens differ from the responses to other antigens.

Properties of T-independent antigens

1. **Polymeric structure:** These antigens are characterized by the same antigenic determinant repeated many times.

2. **Polyclonal activation of B cells:** Many of these antigens can activate B cell clones specific for other antigens (polyclonal activation). T-independent antigens can be subdivided

Career Avenues

into Type 1 and Type 2 based on their ability to polyclonally activate B cells. Type 1 T-independent antigens are polyclonal activators while Type 2 are not.

3. **Resistance to degradation:** T-independent antigens are generally more resistant to degradation and thus they persist for longer periods of time and continue to stimulate the immune system.

T-DEPENDENT ANTIGENS

T-dependent antigens are those that do not directly stimulate the production of antibody without the help of T cells. Proteins are T-dependent antigens. Structurally these antigens are characterized by a few copies of many different antigenic determinants.

FACTORS INFLUENCING IMMUNOGENICITY OF AN ANTIGEN

1. Foreignness
The immune system normally discriminates between self and non-self such that only foreign molecules are immunogenic.

2. Size
There is not absolute size above which a substance will be immunogenic. However, in general, the larger the molecule the more immunogenic it is likely to be.

3. Chemical Composition
In general, the more complex the substance is chemically the more immunogenic it will be. The antigenic determinants are created by the primary sequence of residues in the polymer and/or by the secondary, tertiary or quaternary structure of the molecule.

4. Physical form
In general particulate antigens are more immunogenic than soluble ones and denatured antigens more immunogenic than the native form.

5. Degradability
Antigens that are easily phagocytosed are generally more immunogenic. This is because for most antigens (T-dependant antigens, see below) the development of an immune response requires that the antigen be phagocytosed, processed and presented to helper T cells by an antigen presenting cell (APC).

ANTIGENIC DETERMINANTS

DETERMINANTS RECOGNIZED BY B CELLS

1. **Composition:** Antigenic determinants recognized by B cells and the antibodies secreted by B cells are created by the primary sequence of residues in the polymer (linear or sequence determinants) and/or by the secondary, tertiary or quaternary structure of the molecule (conformational determinants).

2. **Size:** In general antigenic determinants are small and are limited to approximately 4-8 residues, (amino acids and or sugars). The combining site of an antibody will accommodate an antigenic determinant of approximately 4-8 residues.

Career Avenues

3. **Number:** Although, in theory, each 4-8 residues can constitute a separate antigenic determinant, in practice, the number of antigenic determinants per antigen is much lower than what would theoretically be possible. Usually the antigenic determinants are limited to those portions of the antigen that are accessible to antibodies.

DETERMINANTS RECOGNIZED BY T CELLS

1. **Composition:** Antigenic determinants recognized by T cells are created by the primary sequence of amino acids in proteins. T cells do not recognize polysaccharide or nucleic acid antigens. This is why polysaccharides are generally T-independent antigens and proteins are generally T-dependent antigens. The determinants need not be located on the exposed surface of the antigen since recognition of the determinant by T cells requires that the antigen be proteolytically degraded into smaller peptides. Free peptides are not recognized by T cells, rather the peptides associate with molecules coded for by the major histocompatibility complex (MHC) and it is the complex of MHC molecules + peptide that is recognized by T cells.

2. **Size:** In general antigenic determinants are small and are limited to approximately 8-15 amino acids.

3. **Number:** Although, in theory, each 8-15 residues can constitute a separate antigenic determinant, in practice, the number of antigenic determinants per antigen is much less than what would theoretically be possible. The antigenic determinants are limited to those portions of the antigen that can bind to MHC molecules. This is why there can by differences in the responses of different individuals.

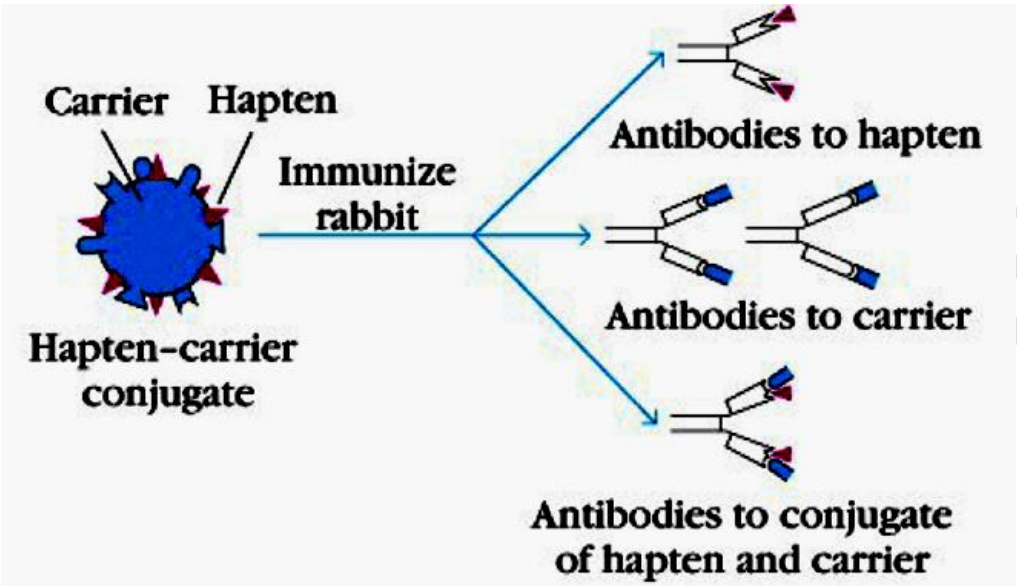
HAPTENS

A Hapten is a substance that is non-immunogenic but which can react with the products of a specific immune response. It is a small molecule which can elicit an immune response only when attached to a large carrier such as a protein, the carrier may be one which also does not elicit an immune response by itself (Generally, only large molecules, infectious agents, or insoluble foreign matter can elicit an immune response in the body.). On their ownthey can never induce an immune response. Free haptens, however, can react with products of the immune response after such products have been elicited. Haptens have the property of antigenicity but not immunogenicity.

Once the body has generated antibodies to a hapten-carrier adduct, the small-molecule hapten may also be able to bind to the antibody, but it will usually not initiate an immune response; usually only the hapten-carrier adduct can do this. Sometimes the small-molecule hapten can even block immune response to the hapten-carrier adduct by preventing the adduct from binding to the antibody. A well known example of a hapten is urushiol, which is the toxin found in poison ivy. When absorbed through the skin from a poison ivy plant, urushiol undergoes oxidation in the skin cells to generate the actual hapten, a reactive molecule called a quinone which then reacts with skin proteins to form hapten adducts. Typically the first exposure only causes sensitization, in which there is a proliferation of B cells able to make antibody to the hapten adduct. After a second exposure later, the proliferated B cells can become activated, generating an immune reaction producing the typical blisters of poison ivy exposure.

Career Avenues

Some haptens can induce autoimmune disease. An example is hydralazine, a blood pressure lowering drug which occasionally can produce drug-induced lupus erythematosus in certain individuals. This also appears to be the mechanism by which the anaesthetic gas halothane can cause a life-threatening hepatitis, as well as the mechanism by which penicillin-class drugs causes autoimmune hemolytic anemia.



Injection with:	Antibodies formed:
Hapten (DNP)	None
Protein carrier (BSA)	Anti-BSA
Hapten–carrier conjugate (DNP-BSA)	Anti-DNP (major) Anti-BSA (minor) Anti-DNP/BSA (minor)

MITOGENS

Unlike immunogens which activate only lymphocytes bearing specific receptors, mitogens activate many clones of T or B cells to undergo cell division irrespective of their antigen specificity. Because of this ability, mitogens are known as polyclonal activators.
A number of common mitogens are proteins called lectins that are derived from plants and bind sugars. Some mitogens selectively activate T cells or B cells whereas some activate both populations. ConA and PHA are T cell mitogens, PWM is mitogenic for both populations and LPS (component of gm- bacterial cell wall) is mitogenic for B cells.

SUPER ANTIGENS

The most potent type of T cell mitogens. The superantigens such as staphylococcal enterotoxins bind on the outside of the antigen binding cleft. In this way a superantigen crosslinks a T cell to a class II MHC molecule in an antigen independent manner resulting in

T cell activation. Estimates are that one out of every five T cells can be activated by SAs resulting in the release of abnormally high levels of cytokines. The high levels of cytokines released can lead to shock and death such as seen with toxic shock syndrome.



GATE QUESTIONS

GATE 2013

Q.3 Endogenous antigens are presented on to the cell surface along with

(A) MHC-II (B) MHC-I

(C) FCγ receptor (D) complement receptor

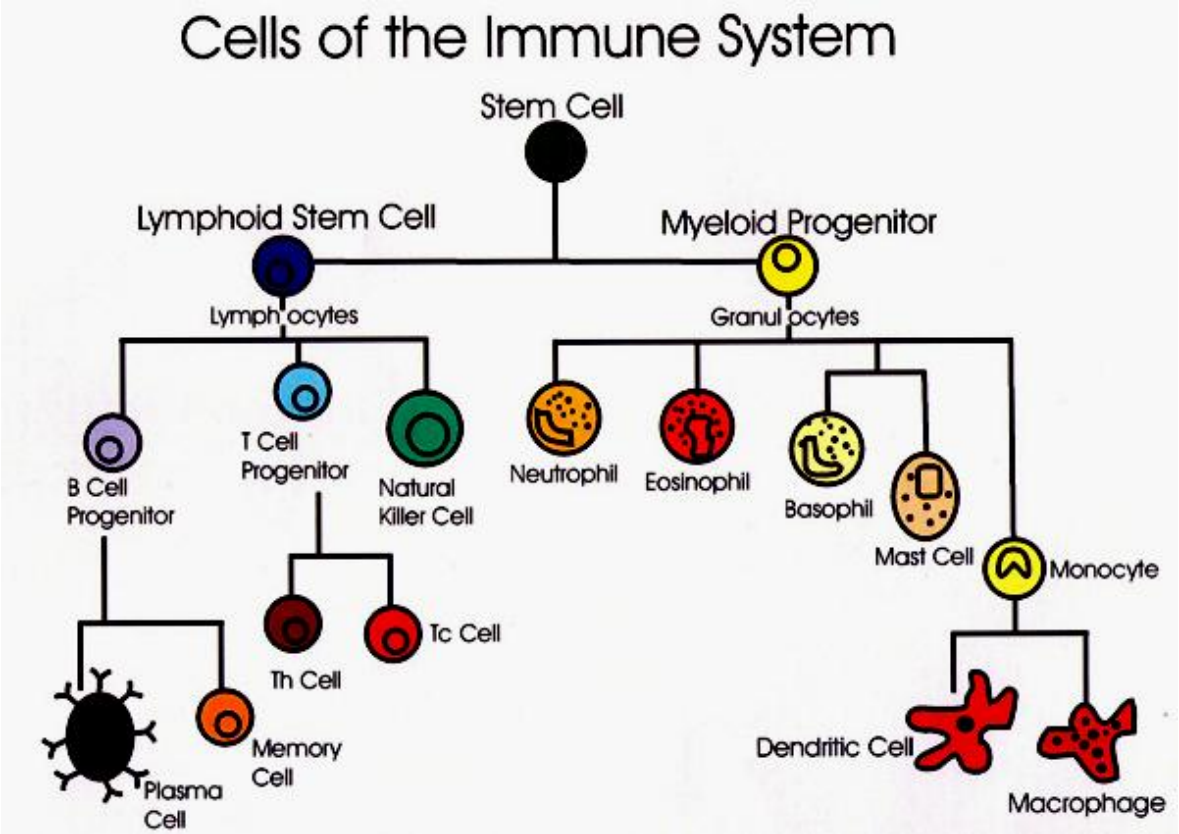
Answers:

3-B.

Career Avenues

CELLS OF IMMUNE SYSTEM

The immune system has developed to protect the host from pathogens and other foreign substances. Self/non-self-discrimination is one of the hallmarks of the immune system. There are two main sites where pathogens may reside: extra-cellularly in tissue spaces or intra-cellularly within a host cell, and the immune system has different ways of dealing with pathogens at these sites. Although immune responses are tailored to the pathogen and to where the pathogen resides, most pathogens can elicit both an antibody and a cell-mediated response, both of which may contribute to ridding the host of the pathogen. However, for any particular pathogen an antibody or a cell-mediated response may be more important for defense against the pathogen.



T-LYMPHOCYTES

T lymphocytes also arise in the bone marrow. Unlike B cells, which mature within the bone marrow, T cells migrate to the thymus gland to mature. During its maturation within the thymus, the T cell comes to express a unique antigen-binding molecule, called the **T-cell receptor,** on its membrane. Unlike membrane-bound antibodies on B cells, which can recognize antigen alone, T-cell receptors can recognize only antigen that is bound to cell-membrane proteins called **major histocompatibility complex (MHC) molecules.** MHC molecules that function in this recognition event, which is termed "antigen presentation," are polymorphic (genetically diverse) glycoproteins found on cell membranes. There are two major types of MHC molecules: Class I MHC molecules, which are expressed by nearly all nucleated cells of vertebrate species, consist of a heavy chain linked to a small invariant protein called β2-microglobulin. Class II MHC molecules, which consist of an alpha and a beta glycoprotein chain, are expressed only by antigen-presenting cells. When a naive T cell encounters antigen combined with a MHC molecule on a cell, the T cell proliferates and differentiates into memory T cells and various effector T cells. There are two well-defined subpopulations of T cells: **T helper (T_H)** and **T cytotoxic (T_C) cells.** Although a third type of T cell, called a T suppressor (T_S) cell, has been postulated, recent evidence suggests that it may not be distinct from T_H and T_C subpopulations. T helper and T cytotoxic cells can be distinguished from one another by the presence of either **CD4** or **CD8** membrane glycoproteins on their surfaces. T cells displaying CD4 generally function as TH cells, whereas those displaying CD8 generally function as TC cells. After a TH cell recognizes and interacts with an antigen– MHC class II molecule complex, the cell is activated—it becomes an effector cell that secretes various growth factors known collectively as **cytokines.** The secreted cytokines play an important role in activating B cells, TC cells, macrophages, and various other cells that participate in the immune response. Differences in the pattern of cytokines produced by activated TH cells result in different types of immune response. Under the influence of TH-derived cytokines, a TC cell that recognizes an antigen–MHC class I molecule complex proliferates and differentiates into an effector cell called a **cytotoxic T lymphocyte (CTL).** In contrast to the TC cell, the CTL generally does not secrete many cytokines and instead exhibits cell-killing or cytotoxic activity. The CTL has a vital function in monitoring the cells of the body and eliminating any that display antigen, such as virus-infected cells, tumor cells, and cells of a foreign tissue graft. Cells that display foreign antigen complexed with a class I MHC molecule are called *altered self-cells*; these are targets of CTLs.

MARKERS OF T-CELLS

All T-cell subpopulations express the T-cell receptor, a complex of polypeptides that includes CD3; and most can be distinguished by the presence of one or the other of two membrane molecules, CD4 and CD8. In addition, most mature.

T-CELLS EXPRESS THE FOLLOWING MEMBRANE MOLECULES

CD28, a receptor for the co-stimulatory B7 family of molecules present on B cells and other antigen presenting cells.

CD45, a signal-transduction molecule T-cells that express the membrane glycoprotein molecule.

Career Avenues

CD4 are restricted to recognizing antigen bound to class II MHC molecules, whereas T cells expressing CD8, a dimeric membrane glycoprotein, are restricted to recognition of antigen bound to class I MHC molecules. Thus the expression ofCD4 versus CD8 corresponds to the MHC restriction of the T cell. In general, expression of CD4 and of CD8 also defines two major functional subpopulations of T lymphocytes.

CD4+ T cells generally function as T helper (TH) cells and are class-II restricted; CD8+ T cells generally function as T cytotoxic(TC) cells and are class-I restricted.

T-HELPER CELLS RESPONSES

The **TH1 response** produces a cytokine profile that supports inflammation and activates mainly certain T cells and macrophages, whereas the **TH2 response** activates mainly B cells and immune responses that depend upon antibodies.

T suppressor (TS) cells — has been postulated. It is clear that some T cells help to suppress the humoral and the cell-mediated branches of the immune system.

T-CELL SELECTION AND CENTRAL TOLERANCE

Tolerance is state of the immune system characterized by unresponsiveness to a particular antigen - ability to distinguish self from non self.

- Freemartin Condition in Twin Cattle: Burnet Hypothesis
    - Dizygotic twins sharing a placenta will therefore share RBC's (hematopoitic chimera [share stem cells])
    - Developing/immature lymphocytes are subject to CLONAL DELETION in presence if self-antigens are recognized.
- Clonal Deletion Model
    - Self-tolerance learned by developing immune system –occurs during fetal and neonatal development.
- Tolerance decreases rapidly with time
- If exposed to antigen in embryo → tolerance in adult → rejection
- If receptor of immature lymphocyte engages a self-antigen → deletion of that clone
- T-Cells are more important than B-cells for tolerance

Central Tolerance – generated during lymphocyte development, involves clonal deletion

Peripheral Tolerance – generated after lymphocytes have matured involves anergy (inactivation).

Career Avenues

T Cell Development (occurs in thymus in 3 stages)

- TCR gene rearrangements resulting in productive α/β dimers on cell surface
    - α/β have many more V/J segments than γ/δ
    - δ locus resides within the α locus (between J and V) → this prevents α/β and γ/δ co-expression since rearrangement of Vα to Jα to form a productive α chain completely deletes the δ locus.

The complete rearrangement of γ/δ actually occurs first on a small proportion of thymocytes prior to expression of the αβ TCR. Expression of a complete surface γδ TCR appears to inhibit further recomingation of the TCR αβ genes. Consequently, T cells which express the γδ receptor are committed to a distinct T cell lineage and are exported to the peripheral lymphoid organs where they constitute 1-5% of total T cells (used for innate immunity since they are double negative).

The majority of thymocytes continue to rearrange α and β genes. This can occur if there is an out-of-frame rearrangement of the γ and δ genes precluding the syn of viable γ or δ protein products or if a V to J rearrangement of the αlocus occurs prior to rearrangement of the δ locus, thereby deleting it.

- V/J rearrangements in α/β and γ/δ→ usually α/β wins
- βgene rearrangement completes first (double negative state) and is required to progress to double positive! (*can induce SCID mouse with TCRβ transgene)*
- β pre TCR α complex associated with CD3 signals successful completion of TCRβ gene rearrangement
    - Without good TCRB cannot place BpreTCRa or progress to the next phase
    - *Similar to μ-λ5-vpreB complex in developing B cells (λ5 is surrogate like pre TCRα)*
- CD4 and CD8 co-expression (but not in γ/δ)
- Ifα gene rearrangement is successful →α/β TCR expressed on surface if unsuccessful → cell dies.

CELLULAR SELECTION

- Negative Selection – for self-reactive "thymocytes"
- Positive Selection – recognize foreign antigens in presence of SELF MHC

ACQUISITION OF MATURE EFFECTOR FUNCTION

Either CD4 or CD8 is inactivated.

Selection

- SUBCAPSULAR VESSELS (double negative)
- CORTEX (double positive)
- Medulla (single positive)

Career Avenues

Negative Selection

- Major mechanism of central tolerance
- Occurs after TCR receptor expression BEFORE mature T-cell effector function
- Proof: recall that mice express two MHC class II molecules (I-A and I-E)
- Most mice express an endogenous retrovirally-encoded super Ag - binds Vβ17 on TCR and I-E on MHCII.
- I-E mice did not have single positive T cells to express the Vβ17+ TCRs, but did have double positives that did.
- Physical deletion of thymocytes expressing self-reactive TCRs at the immature double positive state.
- Findings have shown that CD4+, CD8+ cells express self-reactive TCR's → clonal deletion

  CD4+, CD8- (or vice-versa) do not

Positive Selection

- Chooses TCR's that react to a self-MHC in presence of foreign antigen
- TCR cannot recognize MHC → no development after CD4+, CD8+ stage non selection
- TCR can recognize MHC and self-antigen →thymocytes deleted at CD4+, CD8+negative selection
- TCR recognizes MHC and FOREIGN antigen →thymocytes progress to CD8+ positive selection

B LYMPHOCYTES

B lymphocytes mature within the bone marrow; when they leave it, each expresses a unique antigen-binding receptor on its membrane. This antigen-binding or B-cell receptor is a membrane-bound **antibody molecule.** Antibodies are glycoproteins that consist of two identical heavy polypeptide chains and two identical light polypeptide chains. Each heavy chain is joined with a light chain by disulfide bonds, and additional disulfide bonds hold the two pairs together. The amino-terminal ends of the pairs of heavy and light chains form a cleft within which antigen binds. When a naive B cell (one that has not previously encountered antigen) first encounters the antigen that matches its membrane bound antibody, the binding of the antigen to the antibody causes the cell to divide rapidly; its progeny differentiate into **memory B cells** and **effector B cells** called **plasma cells.** Memory B cells have a longer life span than naive cells, and they express the same membrane-bound antibody as their parent B cell. Plasma cells produce the antibody in a form that can be secreted and have little or no membrane-bound antibody. Although plasma cells live for only a few days, they secrete enormous amounts of antibody during this time. It has been estimated that a single plasma cell can secrete more than 2000 molecules of antibody per second. Secreted antibodies are the major effector molecules of humoral immunity.

Career Avenues

B-CELL MARKERS

Among the other molecules expressed on the membrane of mature B cells are the following:

- **B220** (a form of CD45) is frequently used as a marker for B cells and their precursors. However, unlike antibody, it is not expressed uniquely by B-lineage cells.
- **Class II MHC molecules** permit the B cell to function as an antigen-presenting cell (APC).
- **CR1** (CD35) and **CR2** (CD21) are receptors for certain complement products.
- **FcγRII**(CD32) is a receptor for IgG, a type of antibody.
- **B7-1** (CD80) and **B7-2** (CD86) are molecules that interact with CD28 and CTLA-4, important regulatory molecules on the surface of different types of T cells, including TH cells.
- **CD40** is a molecule that interacts with CD40 ligand on the surface of helper T cells. In most cases this interaction is critical for the survival of antigen stimulated B cells and for their development into antibody-secreting plasma cells or memory B cells.

B-CELL ACTIVATION

- **Native B** cells are fully differentiated B cells that have not yet encountered a complementary antigen; they bear membrane–bound IgM and IgD on their outer surface.
- Clonal selection begins when native B cells bind to their **complementary antibodies**: Then they multiply; produce soluble IgM and IgD, and undergo a high rate of mutation in their DNA-encoding region, resulting in affinity maturation.
- Later, class switching occurs: Various activated B cells begin to make all classes of antibodies.
- Some members of the clone become short-lived effectors (plasma) cells, which make large quantities of antibody; others become long-lived memory cells, which fight future infections by the same pathogen.
- Lymphocytes, including interleukins-1,-2,-4,-5, and -6, produced by NK cells help B-cell activation stimulated by most antigens. Some T-independent antigens stimulate B-cell activation without help.
- B cells are selected to respond to specific antigens in accordance with the particular antibody present on the B cell membrane prior to encountering an antigen.
- When a B cell detects an antigen with which it can react, it binds to the antigen (sensitized or activates) and divides many times to produce a clone of many plasma cells and some memory cells.
- Many B cells require helper T cells to proliferate and differentiate into both plasma and memory B cells.

Career Avenues

B cell Development

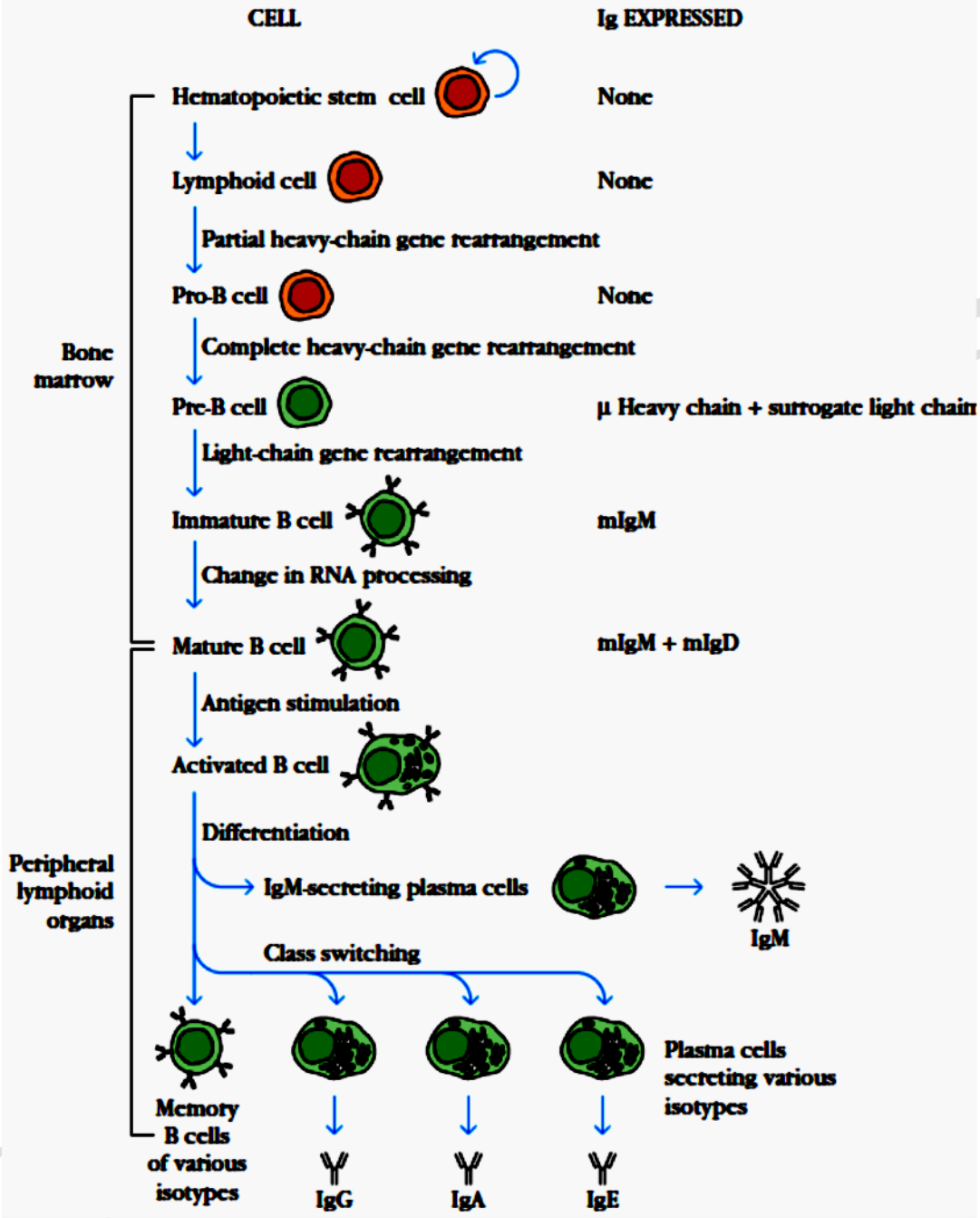


FIG: Overview of B-cell development. The events that occur during maturation in the bone marrow do not require antigen, whereas activation and differentiation of mature B cells in peripheral lymphoid organs require antigen. The labels mIgM and mIgD refer to membrane-associated Igs. IgG, IgA, and IgE are secreted immunoglobulins.

B cells differentiation in the periphery. Immature B cells that leave the bone marrow move to peripheral lymphoid organs, in particular the spleen where most of peripheral B cell development takes place. T1 cells differentiate to T2 cells in the PALS of the spleen in 2 days. T2 cells respond to BCR stimulation by activating survival and proliferative signals. Survival signals are mediated by a btk/PKC-dependent pathway to the transcription factor NF-κB. These cells may also be receptive to co-stimulatory signals via BAFF/BAFFr

signaling pathway. Further differentiation of T2 cells to mature B cells requires progressively higher thresholds of BCR signaling.

T2 cells may be the target of B cell positive selection. There are 3 major subsets of mature B cells: 1) follicular cells are the conventional naïve B cells. They reside in the follicles (splenic white pulp), participate in germinal center rxns, present antigen to T cells, proliferate in response to BCR signals and poorly to innate signals (via the TLRs). 2) marginal zone B cells located at the border of spleen white and red pulp. They do not participate in the GC reaction (being T-independent) and proliferate efficiently to TLR signals but not BCR. Overall, their response has been likened to an innate like response (no affinity maturation). 3) B1-cells are enriched in the peritoneal cavities (not spleen), but their development requires spleen.

- **Plasma cells** synthesize and release large numbers of antibodies.
- **Memory cells** remain in lymphoid tissues ready to respond to subsequent exposure to the same antigen.

RECOGNITION SPECIFICITY OF B-CELLS AND T-CELLS

B cells recognize soluble antigen when it binds to their membrane-bound antibody. Because B cells bind antigen that is free in solution, the epitopes they recognize tend to be highly accessible sites on the exposed surface of the immunogen.

Most T cells recognize only peptides combined with MHC molecules on the surface of antigen presenting cells and altered self-cells; T-cell epitopes, as a rule, cannot be considered apart from their associated MHC molecules.

MONONUCLEAR PHAGOCYTES

The mononuclear phagocytic system consists of **monocytes** circulating in the blood and **macrophages** in the tissues. During hematopoiesis in the bone marrow, granulocyte-monocyte progenitor cells differentiate into promonocytes, which leave the bone marrow and enter the blood, where they further differentiate into mature monocytes. Monocytes circulate in the bloodstream for about 8 h, during which they enlarge; they then migrate into the tissues and differentiate into specific tissue macrophages or, as discussed later, into dendritic cells. Differentiation of a monocyte into a tissue macrophage involves a number of changes: The cell enlarges five- to tenfold; its intracellular organelles increase in both number and complexity; and it acquires increased phagocytic ability, produces higher levels of hydrolytic enzymes, and begins to secrete a variety of soluble factors. Macrophages are dispersed throughout the body. Some take up residence in particular tissues, becoming fixed macrophages, whereas others remain motile and are called free, or wandering, macrophages. Free macrophages travel by amoeboid movement throughout the tissues. Although normally in a resting state, macrophages are activated by a variety of stimuli in the course of an immune response. Phagocytosis of particulate antigens serves as an initial activating stimulus. However, macrophage activity can be further enhanced by cytokines secreted by activated TH cells, by mediators of the inflammatory response, and by components of bacterial cell walls. One of the most potent activators of macrophages is interferon gamma (IFN-γ) secreted by activated TH cells.

Career Avenues

Activated macrophages are more effective than resting ones in eliminating potential pathogens, because they exhibit greater phagocytic activity, an increased ability to kill ingested microbes, increased secretion of inflammatory mediators, and an increased ability to activate T cells. In addition, activated macrophages, but not resting ones, secrete various cytotoxic proteins that help them eliminate a broad range of pathogens, including virus-infected cells, tumor cells, and intracellular bacteria. Activated macrophages also express higher levels of class II MHC molecules, allowing them to function more effectively as antigen-presenting cells. Thus, macrophages and TH cells facilitate each other's activation during the immune response.

MACROPHAGES ARE DISPERSED THROUGHOUT THE BODY

- **Alveolar macrophages** in the lung
- **Histiocytes**in connective tissues
- **Kupffer cells** in the liver
- **Mesangial cells** in the kidney
- **Microglial cells** in the brain
- **Osteoclasts** in bone

KILLING PROCESS OF MACROPHAGES

PHAGOCYTOSIS

Macrophages are capable of ingesting and digesting exogenous antigens, such as whole microorganisms and insoluble particles, and endogenous matter, such as injured or dead host cells, cellular debris, and activated clotting factors. In the first step in phagocytosis, macrophages are attracted by and move toward a variety of substances generated in an immune response; this process is called **chemotaxis.** The next step in phagocytosis is adherence of the antigen to the macrophage cell membrane. Complex antigens, such as whole bacterial cells or viral particles, tend to adhere well and are readily phagocytosed; isolated proteins and encapsulated bacteria tend to adhere poorly and are less readily phagocytosed. Adherence induces membrane protrusions, called **pseudopodia,** to extend around the attached material (Figure). Fusion of the pseudopodia encloses the material within a membrane-bounded structure called a **phagosome,** which then enters the endocytic processing pathway (Figure). In this pathway, a phagosome moves toward the cell interior, where it fuses with a **lysosome** to form a **phagolysosome.** Lysosomes contain lysozyme and a variety of other hydrolytic enzymes that digest the ingested material. The digested contents of the phagolysosome are then eliminated in a process called **exocytosis** (see Figure). The macrophage membrane has receptors for certain classes of antibody. If an antigen (e.g., a bacterium) is coated with the appropriate antibody, the complex of antigen and antibody binds to antibody receptors on the macrophage membrane more readily than antigen alone and phagocytosis is enhanced. In one study, for example, the rate of phagocytosis of an antigen was 4000-fold higher in the presence of specific antibody to the antigen than in its absence. Thus, antibody functions as an **opsonin,** a molecule that binds to both antigen and macrophage and enhances phagocytosis. The process by which particulate antigens are rendered more susceptible to phagocytosis is called **opsonization.**

Career Avenues

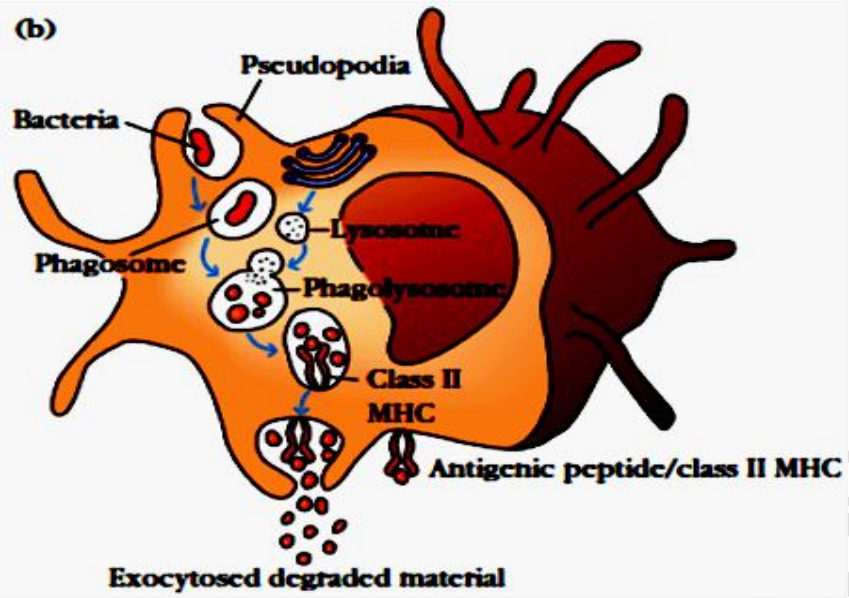


Fig: Phagocytosis and processing of exogenous antigen by macrophages.

ANTIMICROBIAL AND CYTOTOXIC ACTIVITIES

A number of antimicrobial and cytotoxic substances produced by activated macrophages can destroy phagocytosed microorganisms. Many of the mediators of cytotoxicity listed in Table 2-6 are reactive forms of oxygen.

OXYGEN-DEPENDENT KILLING MECHANISMS

Activated phagocytes produce a number of **reactive oxygen intermediates (ROIs)** and **reactive nitrogen intermediates** that have potent antimicrobial activity. During phagocytosis, a metabolic process known as the **respiratory burst** occurs in activated macrophages. This process results in the activation of a membrane-bound oxidase that catalyzes the reduction of oxygen to superoxide anion, reactive oxygen intermediate that is extremely toxic to ingested microorganisms. The superoxide anion also generates other powerful oxidizing agents, including hydroxyl radicals and hydrogen peroxide.

OXYGEN-INDEPENDENT KILLING MECHANISMS

Activated macrophages also synthesize **lysozyme** and various hydrolytic enzymes whose degradative activities do not require oxygen. In addition, activated macrophages produce a group of antimicrobial and cytotoxic peptides, commonly known as **defensins.** These molecules are cysteine-rich cationic peptides containing 29–35 amino-acid residues. Each peptide, which contains six invariant cysteines, forms a circular molecule that is stabilized by intra-molecular disulfide bonds.

These circularized defensin peptides have been shown to form ion-permeable channels in bacterial cell membranes.

Career Avenues

OTHER CELLS OF IMMUNE SYSTEM

NEUTROPHILS

Neutrophils are produced by hematopoiesis in the bone marrow. They are released into the peripheral blood and circulate for 7–10 h before migrating into the tissues, where they have a life span of only a few days. In response to many types of infections, the bone marrow releases more than the usual number of neutrophils and these cells generally are the first to arrive at a site of inflammation. The resulting transient increase in the number of circulating neutrophils, called **leukocytosis,** is used medically as an indication of infection. Movement of circulating neutrophils into tissues, called **extravasation,** takes several steps: the cell first adheres to the vascular endothelium, then penetrates the gap between adjacent endothelial cells lining the vessel wall, and finally penetrates the vascular basement membrane, moving out into the tissue spaces. A number of substances generated in an inflammatory reaction serve as **chemotactic factors** that promote accumulation of neutrophils at an inflammatory site. Among these chemotactic factors are some of the complement components, components of the blood-clotting system, and several cytokines secreted by activated TH cells and macrophages. Like macrophages, neutrophils are active phagocytic cells. Phagocytosis by neutrophils is similar to that described for macrophages, except that the lytic enzymes and bactericidal substances in neutrophils are contained within primary and secondary granules. Both primary and secondary granules fuse with phagosomes, whose contents are then digested and eliminated much as they are in macrophages.

Neutrophils also employ both oxygen-dependent and oxygen-independent pathways to generate antimicrobial substances. Neutrophils are in fact much more likely than macrophages to kill ingested microorganisms. Neutrophils exhibit a larger respiratory burst than macrophages and consequently are able to generate more reactive oxygen intermediates and reactive nitrogen intermediates. In addition, neutrophils express higher levels of defensins than macrophages do.

EOSINOPHILS

Eosinophils, like neutrophils, are motile phagocytic cells that can migrate from the blood into the tissue spaces. Their phagocytic role is significantly less important than that of neutrophils, and it is thought that they play a role in the defense against parasitic organisms. The secreted contents of eosinophilic granules may damage the parasite membrane.

BASOPHILS

Basophils are nonphagocytic granulocytes that function by releasing pharmacologically active substances from their cytoplasmic granules. These substances play a major role in certain allergic responses.

MAST CELLS

Mast-cell precursors, which are formed in the bone marrow by hematopoiesis, are released into the blood as undifferentiated cells; they do not differentiate until they leave the blood and enter the tissues. Mast cells can be found in a wide variety of tissues, including the skin,

Career Avenues

connective tissues of various organs, and mucosal epithelial tissue of the respiratory, genitourinary, and digestive tracts. Like circulating basophils, these cells have large numbers of cytoplasmic granules that contain histamine and other pharmacologically active substances. Mast cells, together with blood basophils, play an important role in the development of allergies.

DENDRITIC CELLS

The dendritic cell (DC) acquired its name because it is covered with long membrane extensions that resemble the dendrites of nerve cells. Dendritic cells can be difficult to isolate because the conventional procedures for cell isolation tend to damage their long extensions. The development of isolation techniques that employ enzymes and gentler dispersion has facilitated isolation of these cells for study in vitro. There are many types of dendritic cells, although most mature dendritic cells have the same major function, the presentation of antigen to TH cells. Four types of dendritic cells are known: Langerhans cells, interstitial dendritic cells, myeloid cells, and lymphoid dendritic cells. Each arises from hematopoietic stem cells via different pathways and in different locations.

Despite their differences, they all constitutively express high levels of both class II MHC molecules and members of the co-stimulatory B7 family. For this reason, they are more potent antigen-presenting cells than macrophages and B cells, both of which need to be activated before they can function as antigen-presenting cells (APCs). Immature or precursor forms of each of these types of dendritic cells acquire antigen by phagocytosis or endocytosis; the antigen is processed, and mature dendritic cells present it to TH cells. Following microbial invasion or during inflammation, mature and immature forms of Langerhans cells and interstitial dendritic cells migrate into draining lymph nodes, where they make the critical presentation of antigen to TH cells that is required for the initiation of responses by those key cells.

Another type of dendritic cell, the **follicular dendritic cell**, does not arise in bone marrow and has a different function from the antigen-presenting dendritic cells described above. Follicular dendritic cells do not express class II MHC molecules and therefore do not function as antigen presenting cells for TH-cell activation. These dendritic cells were named for their exclusive location in organized structures of the lymph node called lymph follicles, which are rich in B cells. Although they do not express class II molecules, follicular dendritic cells express high levels of membrane receptors for antibody, which allows the binding of antigen-antibody complexes. The interaction of B cells with this bound antigen can have important effects on B cell responses.

NATURAL KILLER CELLS

The natural killer cell was first described in 1976, when it was shown that the body contains a small population of large, granular lymphocytes that display cytotoxic activity against a wide range of tumor cells in the absence of any previous immunization with the tumor. NK cells were subsequently shown to play an important role in host defense both against tumor cells and against cells infected with some, though not all, viruses. These cells, which constitute 5%–10% of lymphocytes in human peripheral blood, do not express the membrane molecules and receptors that distinguish T- and B-cell lineages. Although NK cells do not have T-cell receptors or immunoglobulin incorporated in their plasma membranes, they can recognize potential target cells in two different ways. In some cases, an NK cell employs NK cell

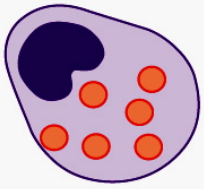
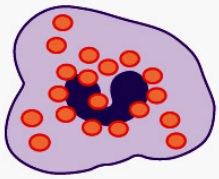
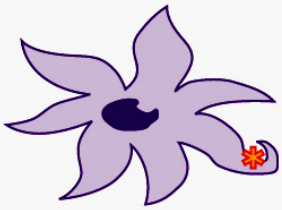
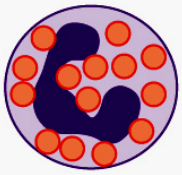
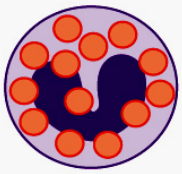
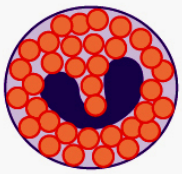
Career Avenues

receptors to distinguish abnormalities, notably a reduction in the display of class I MHC molecules and the unusual profile of surface antigens displayed by some tumor cells and cells infected by some viruses. Another way in which NK cells recognize potential target cells depends upon the fact that some tumor cells and cells infected by certain viruses display antigens against which the immune system has made an antibody response, so that antitumor or antiviral antibodies are bound to their surfaces. Because NK cells express CD16, a membrane receptor for the carboxyl-terminal end of the IgG molecule, called the Fc region, they can attach to these antibodies and subsequently destroy the targeted cells. This is an example of a process known as **antibody-dependent cellmediated cytotoxicity (ADCC).**

Several observations suggest that NK cells play an important role in host defense against tumors. For example, in humans the **Chediak-Higashi syndrome**—an autosomal recessive disorder—is associated with impairment in neutrophils, macrophages, and NK cells and an increased incidence of lymphomas. Likewise, mice with an autosomal mutation called *beige* lack NK cells; these mutants are more susceptible than normal mice to tumor growth following injection with live tumor cells.

There has been growing recognition of a cell type, the **NK1-T cell,** that has some of the characteristics of both T cells and NK cells. Like T cells, NK1-T cells have T cell receptors (TCRs).Unlike most T cells, the TCRs of NK1-T cells interact with MHC-like molecules called CD1 rather than with class I or class II MHC molecules. Like NK cells, they have variable levels of CD16 and other receptors typical of NK cells, and they can kill cells. A population of triggered NK1-T cells can rapidly secrete large amounts of the cytokines needed to support antibody production by B cells as well as inflammation and the development and expansion of cytotoxic T cells. Some immunologists view this cell type as a kind of rapid response system that has evolved to provide early help while conventional TH responses are still developing.

Career Avenues

Cell type	Characteristics	Location	Image
Mast cell	Dilates blood vessels and induces inflammation through release of histamines and heparin. Recruits macrophages and neutrophils. Involved in wound healing and defense against pathogens but can also be responsible for allergic reactions.	Connective tissues, mucous membranes	
Macrophage	Phagocytic cell that consumes foreign pathogens and cancer cells. Stimulates response of other immune cells.	Migrates from blood vessels into tissues.	
Natural killer cell	Kills tumor cells and virus-infected cells.	Circulates in blood and migrates into tissues.	
Dendritic cell	Presents antigens on its surface, thereby triggering adaptive immunity.	Present in epithelial tissue, including skin, lung and tissues of the digestive tract. Migrates to lymph nodes upon activation.	
Monocyte	Differentiates into macrophages and dendritic cells in response to inflammation.	Stored in spleen, moves through blood vessels to infected tissues.	
Neutrophil	First responders at the site of infection or trauma, this abundant phagocytic cell represents 50-60 percent of all leukocytes. Releases toxins that kill or inhibit bacteria and fungi and recruits other immune cells to the site of infection.	Migrates from blood vessels into tissues.	
Basophil	Responsible for defense against parasites. Releases histamines that cause inflammation and may be responsible for allergic reactions.	Circulates in blood and migrates to tissues.	
Eosinophil	Releases toxins that kill bacteria and parasites but also causes tissue damage.	Circulates in blood and migrates to tissues.	



GATE QUESTIONS

GATE 2008

Q.4 One common feature between B and T cells is that

(A) both cells produce antibodies

(B) both cells possess MHC class II

(C) both B cell receptor and T cell receptor undergo rearrangement

(D) both cells can produce cytokines

Q.5 Nucleated cells tends to be more resistant to complement mediated lysis than RBC because

(A) many nucleated cells can endocytose the membrane attack complex

(B) membrane attack complex cannot get inserted in the nucleated cell membrane

(C) membrane attack complex can get inactivated by the nucleated cells

(D) membrane attack complex get inactivated hence cannot get inserted in the nucleated cell membrane

GATE 2012

Q.6 Pick the correctly matched pairs.

P. Immature B cells – Terminal deoxynucleotidyl transferase

Q. Activated B cells – Class Switching

R. Pre B cells - Surrogate light chain

S. Mature B cells - Recombination activating gene I

(A) P and R (B) Q and R

(C) Q and S (D) Q and P

Answers		
4-C.	**5-D.**	**6-C.**

Career Avenues

MAJOR-HISTOCOMPATIBILITY COMPLEX (MHC)

The major histocompatibility complex is a collection of genes arrayed within a long continuous stretch of DNA on chromosome 6 in humans and on chromosome 17 in mice. The MHC is referred to as the **HLA complex** in humans and as the **H-2 complex** in mice.

MHC is tightly linked cluster of genes present in every vertebrate species, the product of which play a role in intercellular recognition and in discrimination between self and nonself. MHC is highly polymorphic i.e. many alternative forms of gene, or alleles, exist at each locus. Different individual within a species have different allele of MHC gene- that is, at a single MHC locus, different individuals have different type of prototypical genes. The phenomenon of having multiple stable forms of one gene in the population is known as genetic polymorphism.

Mouse H-2 complex

Complex	H-2						
MHC class	I	II		III		I	
Region	K	IA	IE	S		D	
Gene products	H-2K	IA αβ	IE αβ	C′ proteins	TNF-α TNF-β	H-2D	H-2L

Human HLA complex

Complex	HLA							
MHC class	II			III		I		
Region	DP	DQ	DR	C4, C2, BF		B	C	A
Gene products	DP αβ	DQ αβ	DR αβ	C′ proteins	TNF-α TNF-β	HLA-B	HLA-C	HLA-A

MHC genes are codominantly expressed- that is every cell that expresses MHC molecules expresses proteins transcribed from both the maternal and paternal chromosome.

The genes of the MHC loci lie together. A group of linked MHC genes is generally inherited as a unit from parents; these linked groups are known as haplotypes.

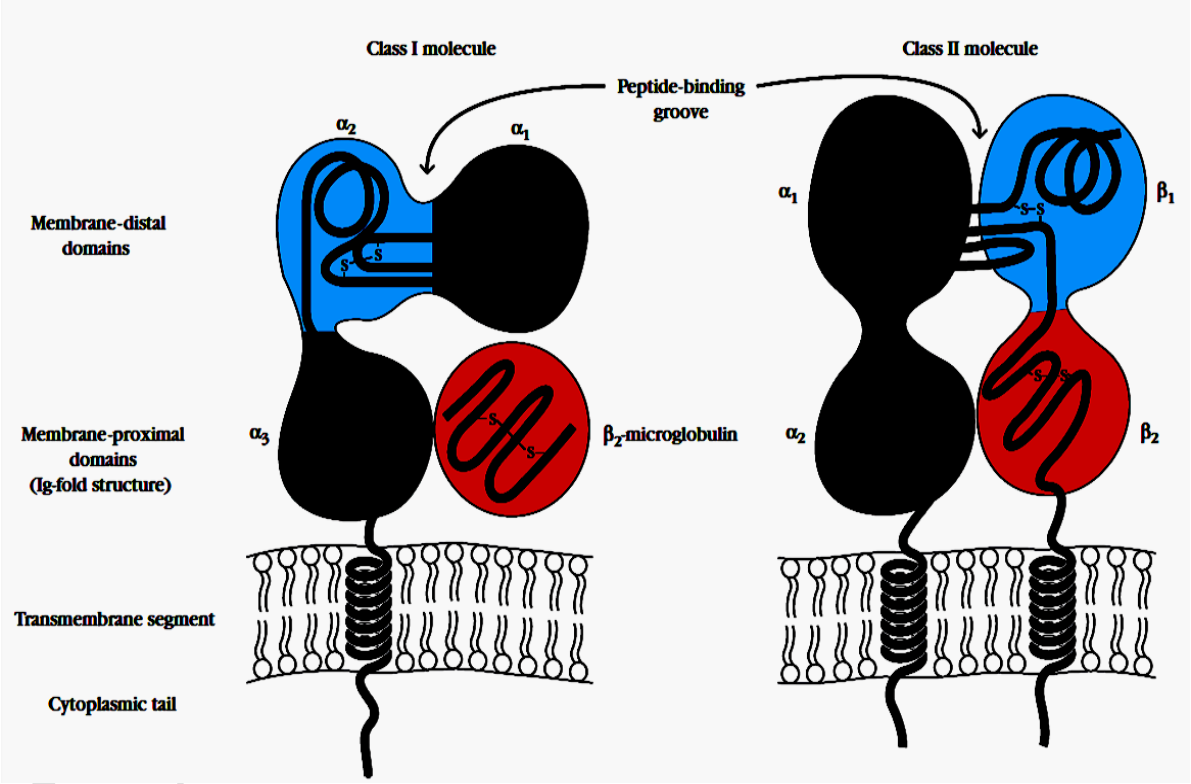
Two major set of MHC gene are present:

CLASS I MHC GENES

Class I MHC genes express glycoprotein called MHC molecules. Class I MHC molecules are expressed on the surface of all nucleated cells. Class I MHC molecule consists of two polypeptide chains:

- Heavy chain or H chain or α chain: α chain has three domain (α1, α2, α3). The α1 and α2 domains comprise a highly polymorphic peptide binding region. The β2- microglobulin associates with α chain and helps maintain the proper conformation of the molecule and help it to display on cell surface.
- Light chain of L chain of β2 microglobulin: β2-Microglobulin is a protein encoded by a highly conserved gene located on a different chromosome.

Thus, α chain shows polymorphism while β microglobulin doesn't show polymorphism. α chain responsible for variability while β microglobulin has no role on variability. Out of α chain and β microglobulin, only α chain is encoded by MHC gene.



CLASS II MHC GENES

Class I MHC gene code for two chains i.e. α and β. MHC class II molecule, like MHC class I molecules, are transmembrane glycoprotein with cytoplasmic tail and extracellular Ig-like domains; the domain are referred to as α_1,α_2, β_1, and β_2. Both α and β chains are encoded by MHC gene. No β2 microglobulin is present in class II MHC molecules. Unlike to class I MHC molecules (where only α chain contribute to peptide binding groove), in class II MHC molecules both α and β chain contribute to peptide binding groove.

Class II MHC gene express MHC molecules constituvely (i.e. under all conditions) only one antigen presenting cells (APCs). Unlike the MHC class I, class II MHC molecules are present only some specific cell type. For example: B lymphocytes, macrophages, dendritic cells, sperm, fibrblast and vascular endothelial cells.

In general, in the absence of inducing factors, most cells express MHC class I molecules without expressing MHC class II molecules. Certain cells, such as B cells, constitutively express both MHC class I and class II molecules.

MHC MOLECULES AND ANTIGEN PRESENTATIONS

Both type of membrane glycoproteins (MHC class I and class II) function as highly specialized antigen presenting molecules. Both class I and class II MHC molecule present antigen to T cells. T cells recognize only peptides combined with MHC molecules. Class I molecules present processed endogenous antigen to CD8 molecule expressing T-cells (cytotoxic T cells or Tc cells). Class II MHC molecules present processed exogenous antigen to CD4 molecule expressing T – cells (helper T – cells or Th cells).

TABLE: FEATURES OF CLASS I AND CLASS II MHC MOLECULES

Feature	Class I MHC	Class II MHC
Structure	α chain and β2 microglobulin	α and β chain
Polymorphic domain	α_1 and α_2 domain	α_1 and β_1 domain
Binding site for T cell receptor	α_3 region binds CD8	β_2 region binds CD4
Constitutive cellular expression	Nearly all nucleated cells	APCs
Peptide derived from	Endogenous antigen	Exogenous antigen
Peptide presented to	$CD8^+$ T cells	$CD4^+$ T cells
General size of presented peptide	8-10 amino acids	13 – 18 amino acids
Nomenclature		
Human	HLA-A, HLA-B, HLA-C	HLA-DR, HLA-DQ, HLA-DP
mouse	H-2K, H-2D, H-2L	I-A, I-E

IMPORTANT ASPECTS OF MHC

- Although there is a high degree of polymorphism for a species, an individual has maximum of six different class I MHC products and only slightly more class II MHC products (considering only the major loci).
- Each MHC molecule has only one binding site. The different peptides a given MHC molecule can bind all bind to the same site, but only one at a time.
- Because each MHC molecule can bind many different peptides, binding is termed degenerate.
- MHC polymorphism is determined only in the germline. There are no recombinational mechanisms for generating diversity.

Career Avenues

- MHC molecules are membrane-bound; recognition by T cells requires cell-cell contact.
- Alleles for MHC genes are co-dominant. Each MHC gene product is expressed on the cell surface of an individual nucleated cell.
- A peptide must associate with a given MHC of that individual, otherwise no immune response can occur. That is one level of control.
- Mature T cells must have a T cell receptor that recognizes the peptide associated with MHC. This is the second level of control.
- Cytokines (especially interferon-γ) increase level of expression of MHC.
- Peptides from the cytosol associate with class I MHC and are recognized by Tc cells. Peptides from within vesicles associate with class II MHC and are recognized by Th cells.
- Polymorphism in MHC is important for survival of the species.

Career Avenues

ANTIGEN PROCESSING AND PRESENTATION

Recognition of foreign proteins by a T cell requires that peptides derived from the antigen be displayed within the cleft of an MHC molecule on the membrane of a cell. The formation of these peptide-MHC complexes requires that a protein antigen be degraded into peptides by a sequence of events called **antigen processing.** The degraded peptides then associate with MHC molecules within the cell interior, and the peptide-MHC complexes are transported to the membrane, where they are displayed (**antigen presentation**).

ENDOGENOUS ANTIGENS: THE CYTOSOLIC PATHWAY

The endogenous pathway is used to present cellular peptide fragments on the cell surface on MHC class I molecules. If a virus had infected the cell, viral peptides would also be presented, allowing the immune system to recognize and kill the infected cell. Worn out proteins within the cell become ubiquitinated, marking them for proteasome degradation. Proteasomes break the protein up into peptides that include some around nine amino acids long (suitable for fitting within the peptide binding cleft of MHC class I molecules). Transporter associated with antigen presenting (TAP), a protein that spans the membrane of the rough endoplasmic reticulum, transports the peptides into the lumen of the rough endoplasmic reticulum (ER). Also within the rough ER, a series of chaperone proteins, including calnexin, calreticulin, ERp57, and Binding immunoglobulin protein (BiP) facilitates the proper folding of class I MHC and its association with β2 microglobulin. The partially folded MHC class I molecule then interacts with TAP via tapasin (the complete complex also contains calreticulin and Erp57 and, in mice, calnexin). Once the peptide is transported into the ER lumen it binds to the cleft of the awaiting MHC class I molecule, stabilizing the MHC and allowing it to be transported to the cell surface by the golgi apparatus.

EXOGENOUS ANTIGENS: THE ENDOCYTIC PATHWAY

The exogenous pathway is utilized by specialized antigen presenting cells to present peptides derived from proteins that the cell has endocytosed. The peptides are presented on MHC class II molecules. Proteins are endocytosed and degraded by acid-dependent proteases in endosomes; this process takes about an hour.

The nascent MHC class II protein in the rough ER has its peptide-binding cleft blocked by Ii (the invariant chain; a trimer) to prevent it from binding cellular peptides or peptides from the endogenous pathway. The invariant chain also facilitates MHC class II's export from the ER in a vesicle. This fuses with a late endosome containing the endocytosed, degraded proteins. It is then broken down in stages, leaving only a small fragment called CLIP which still blocks the peptide binding cleft. An MHC class II-like structure, HLA-DM, removes CLIP and replaces it with a peptide from the endosome. The stable MHC class-II is then presented on the cell surface.

In Cross-presentation, peptides derived from extracellular proteins are presented in the context of MHC class I.

Career Avenues

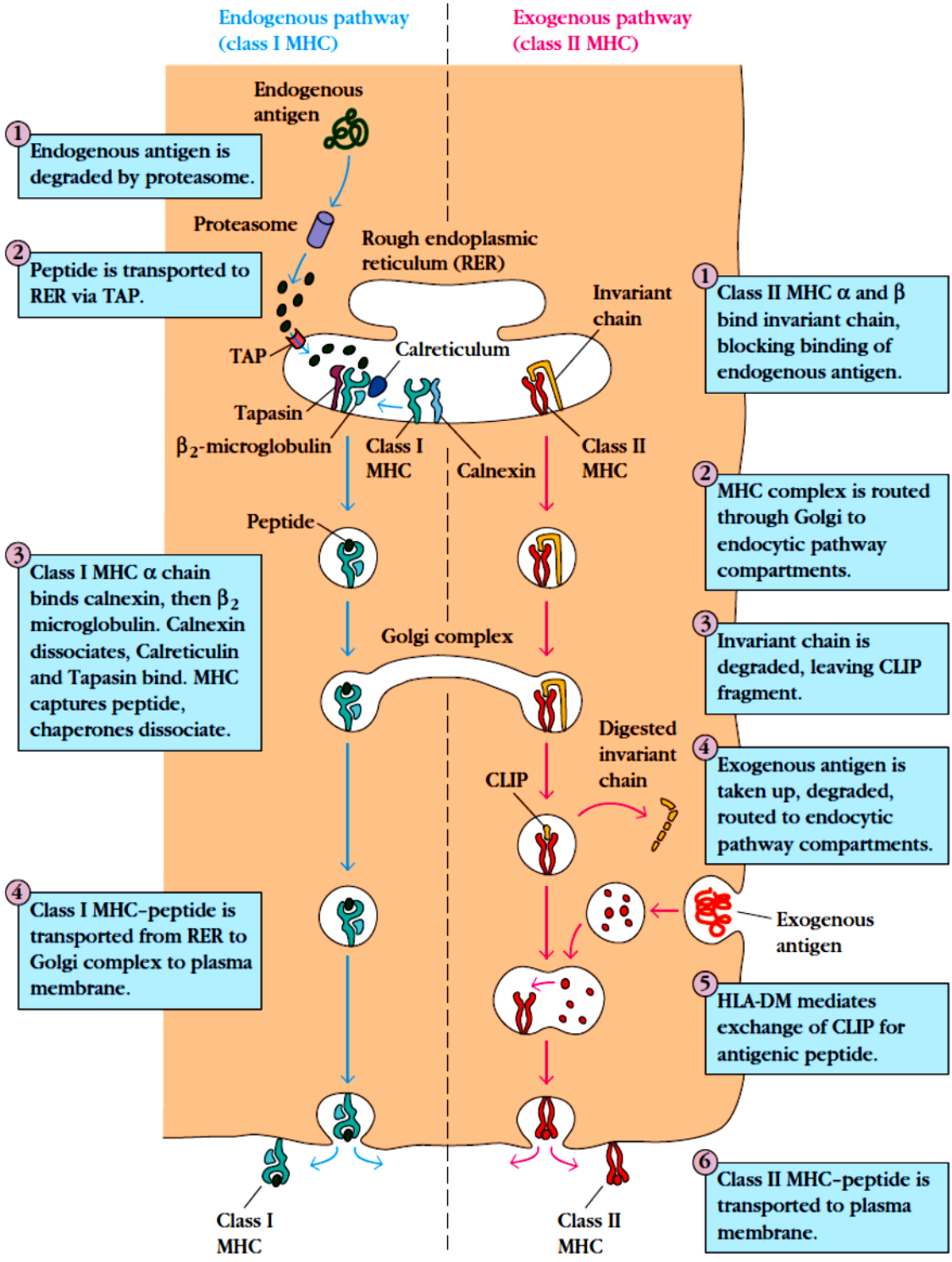


Figure: *Separate antigen-presenting pathways are utilized for endogenous (green) and exogenous (red) antigens. The mode of antigen entry into cells and the site of antigen processing determine whether antigenic peptides associate with class I MHC molecules in the rough endoplasmic reticulum or with class II molecules in endocytic compartments.*

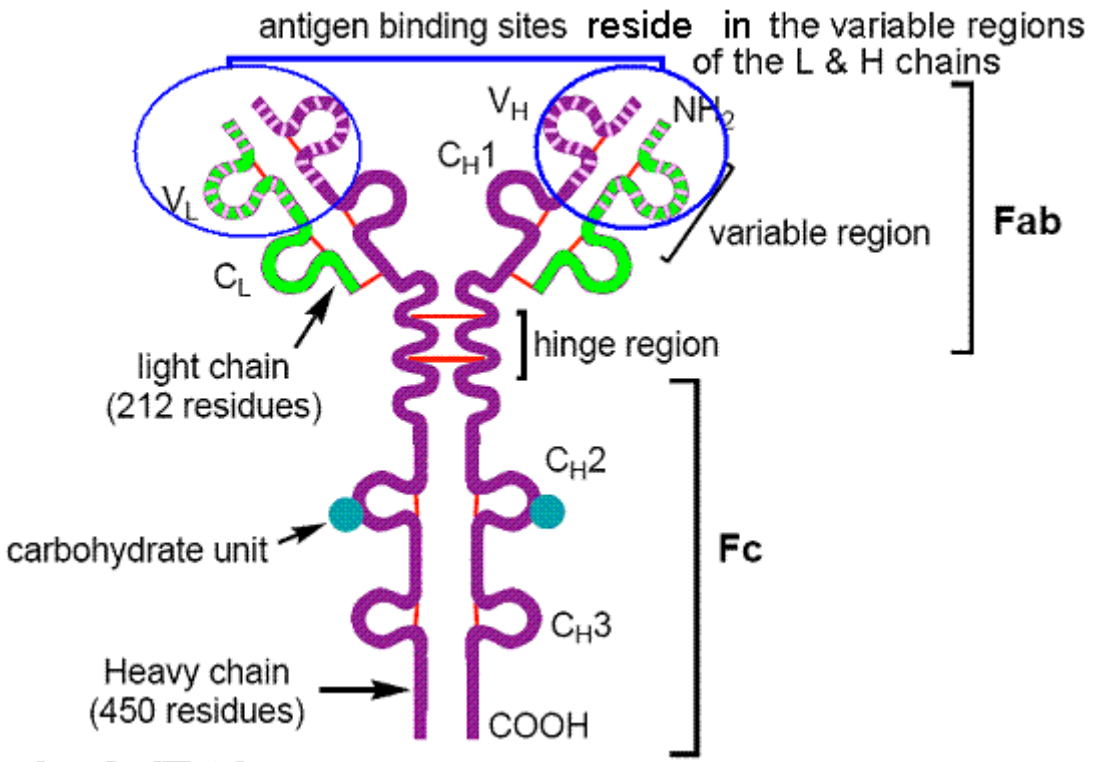
IMMUNOGLOBULINS

IMMUNOGLOBULIN (IG)

Immunoglobulins are glycoprotein molecules that are produced by plasma cells in response to an immunogen and which function as antibodies.

BASIC STRUCTURE OF IMMUNOGLOBULINS

The basic structure of the immunoglobulins is illustrated in figure. Although different immunoglobulins can differ structurally, they all are built from the same basic units.



HEAVY AND LIGHT CHAINS

All immunoglobulins have a four chain structure as their basic unit. They are composed of two identical light chains (23kD) and two identical heavy chains (50-70kD).

DISULFIDE BONDS

1. *Inter-chain disulfide bonds* - The heavy and light chains and the two heavy chains are held together by inter-chain disulfide bonds and by non-covalent interactions The number of inter-chain disulfide bonds varies among different immunoglobulin molecules.
2. *Intra-chain disulfide binds* - Within each of the polypeptide chains there are also intra-chain disulfide bonds.

VARIABLE (V) AND CONSTANT (C) REGIONS

When the amino acid sequences of many different heavy chains and light chains were compared, it became clear that both the heavy and light chain could be divided into two regions based on variability in the amino acid sequences. These are the:

1. Light Chain - V_L (110 amino acids) and C_L (110 amino acids)
2. Heavy Chain - V_H (110 amino acids) and C_H (330-440 amino acids)

HINGE REGION

This is the region at which, the arms of the antibody molecule forms a Y. It is called the hinge region because there is some flexibility in the molecule at this point.

DOMAINS

Three dimensional images of the immunoglobulin molecule show that it is not straight as depicted in figure 2A. Rather, it is folded into globular regions each of which contains an intra-chain disulfide bond (figure 2B-D). These regions are called domains.

1. Light Chain Domains - V_L and C_L
2. Heavy Chain Domains - V_H, C_{H1} - C_{H3} (or C_{H4})

OLIGOSACCHARIDES

Carbohydrates are attached to the C_{H2} domain in most immunoglobulins. However, in some cases carbohydrates may also be attached at other locations.

STRUCTURE OF THE VARIABLE REGION

HYPERVARIABLE (HVR) OR COMPLEMENTARITY DETERMINING REGIONS (CDR)

Comparisons of the amino acid sequences of the variable regions of immunoglobulins show that most of the variability resides in three regions called the hypervariable regions or the complementarity determining regions. Antibodies with different specificities (i.e. different combining sites) have different complementarity determining regions while antibodies of the exact same specificity have identical complementarity determining regions (*i.e.* CDR is the antibody combining site). Complementarity determining regions are found in both the H and the L chains.

FRAMEWORK REGIONS

The regions between the complementarity determining regions in the variable region are called the framework regions (figure 3). Based on similarities and differences in the framework regions the immunoglobulin heavy and light chain variable regions can be divided into groups and subgroups. These represent the products of different variable region genes.

Career Avenues

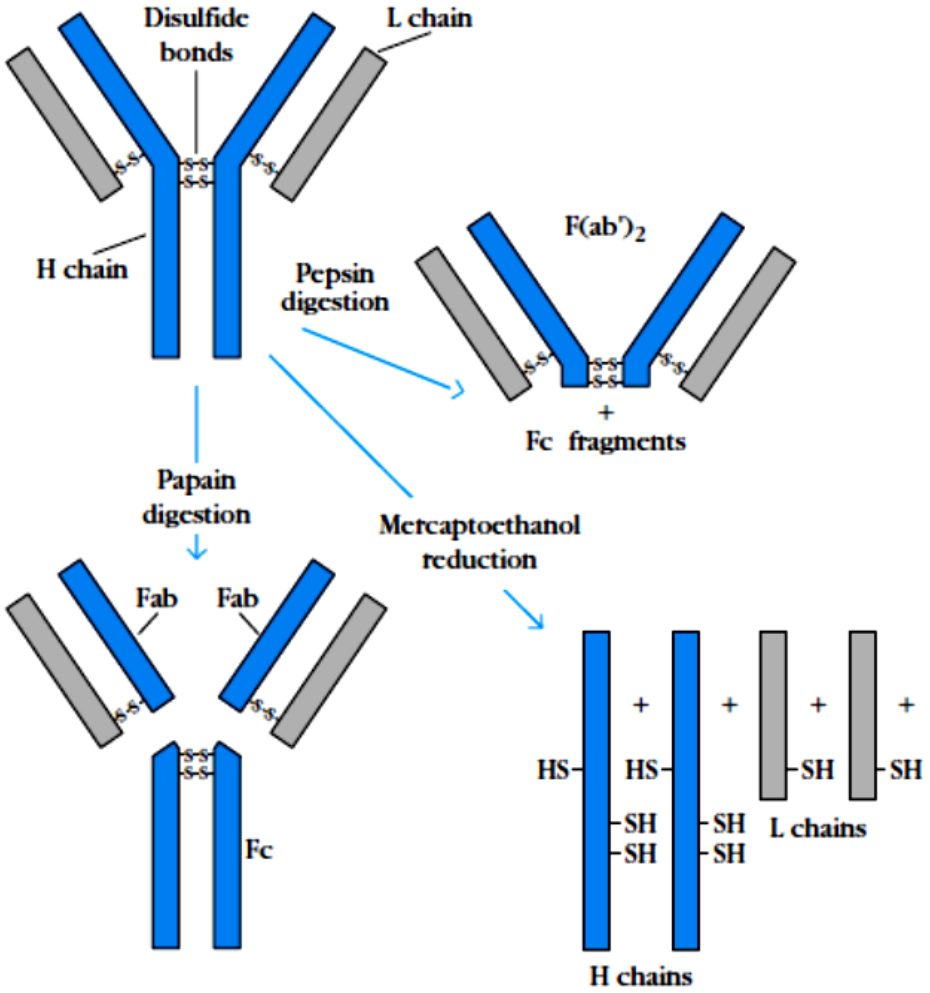
IMMUNOGLOBULIN FRAGMENTS: STRUCTURE / FUNCTION RELATIONSHIPS

Immunoglobulin fragments produced by proteolytic digestion have proven very useful in elucidating structure/function relationships in immunoglobulins.

FAB

Digestion with papain breaks the immunoglobulin molecule in the hinge region before the H-H inter-chain disulfide bond Figure 4. This results in the formation of two identical fragments that contain the light chain and the V_H and C_{H1} domains of the heavy chain.

Antigen binding - These fragments were called the Fab fragments because they contained the antigen binding sites of the antibody. Each Fab fragment is monovalent whereas the original molecule was divalent. The combining site of the antibody is created by both V_H and V_L. An antibody is able to bind a particular antigenic determinant because it has a particular combination of V_H and V_L. Different combinations of a V_H and V_L result in antibodies that can bind a different antigenic determinants.



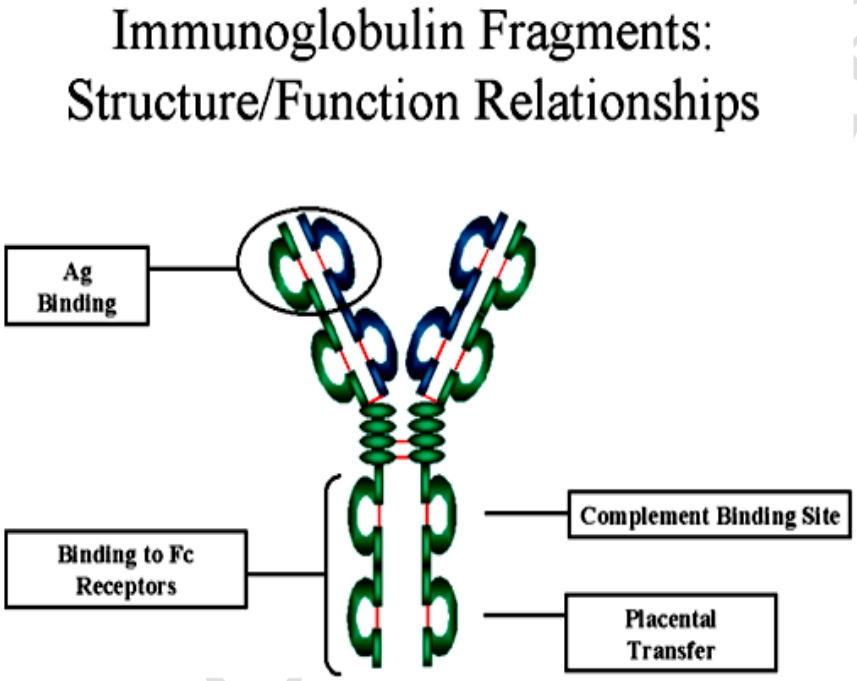
FC

Digestion with papain also produces a fragment that contains the remainder of the two heavy chains each containing a C_{H2} and C_{H3} domain. This fragment was called Fc because it was easily crystallized.

Effector functions - The effector functions of immunoglobulins are mediated by this part of the molecule. Different functions are mediated by the different domains in this fragment. Normally the ability of an antibody to carry out an effector function requires the prior binding of an antigen; however, there are exceptions to this rule.

$F(AB')_2$

Treatment of immunoglobulins with pepsin results in cleavage of the heavy chain after the H-H inter-chain disulfide bonds resulting in a fragment that contains both antigen binding sites. This fragment was called $F(ab')_2$ because it is divalent. The Fc region of the molecule is digested into small peptides by pepsin. The $F(ab')_2$ binds antigen but it does not mediate the effector functions of antibodies.



IMMUNOGLOBULIN CLASSES, SUBCLASSES, TYPES AND SUBTYPES

A. IMMUNOGLOBULIN CLASSES

The immunoglobulins can be divided into five different classes, based on differences in the amino acid sequences in the constant region of the heavy chains. All immunoglobulins within a given class will have very similar heavy chain constant regions. These differences can be detected by sequence studies or more commonly by serological means (*i.e.* by the use of antibodies directed to these differences).

- IgG - Gamma heavy chains
- IgM - Mu heavy chains
- IgA - Alpha heavy chains
- IgD - Delta heavy chains
- IgE - Epsilon heavy chains

B. IMMUNOGLOBULIN SUBCLASSES

The classes of immunoglobulins can be divided into subclasses based on small differences in the amino acid sequences in the constant region of the heavy chains. All immunoglobulins within a subclass will have very similar heavy chain constant region amino acid sequences. Again these differences are most commonly detected by serological means.

- IgG Subclasses
    - IgG1 - Gamma 1 heavy chains
    - IgG2 - Gamma 2 heavy chains
    - IgG3 - Gamma 3 heavy chains
    - IgG4 - Gamma 4 heavy chains
- IgA Subclasses
    - IgA1 - Alpha 1 heavy chains
    - IgA2 - Alpha 2 heavy chains

IMMUNOGLOBULIN TYPES

Immunoglobulins can also be classified by the type of light chain that they have. Light chain types are based on differences in the amino acid sequence in the constant region of the light chain. These differences are detected by serological means.

- Kappa light chains
- Lambda light chains

IMMUNOGLOBULIN SUBTYPES

The light chains can also be divided into subtypes based on differences in the amino acid sequences in the constant region of the light chain.

- Lambda subtypes
    - Lambda 1
    - Lambda 2
    - Lambda 3
    - Lambda 4

NOMENCLATURE

Immunoglobulins are named based on the class, or subclass of the heavy chain and type or subtype of light chain. Unless it is stated precisely, you should assume that all subclass, types and subtypes are present. IgG means that all subclasses and types are present.

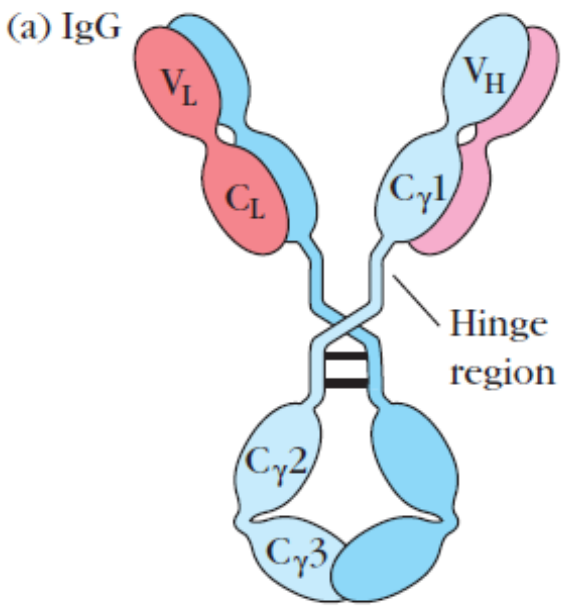
Career Avenues

HETEROGENEITY

Immunoglobulins considered as a population of molecules are normally very heterogeneous because they are composed of different classes and subclasses each of which has different types and subtypes of light chains. In addition, different immunoglobulin molecules can have different antigen binding properties because of different V_H and V_L regions.

IgG

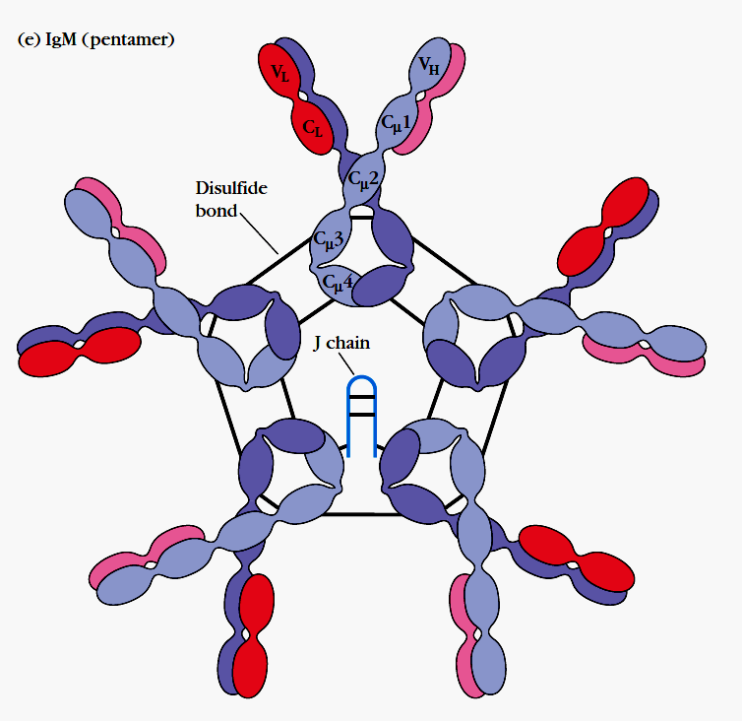
1. **Structure**: The structures of the IgG subclasses are presented in figure 7. All IgG's are monomers (7S immunoglobulin). The subclasses differ in the number of disulfide bonds and length of the hinge region.
2. **Properties**: IgG is the most versatile immunoglobulin because it is capable of carrying out all of the functions of immunoglobulin molecules.
   - IgG is the major Ig in serum - 75% of serum Ig is IgG
   - IgG is the major Ig in extra vascular spaces
   - Placental transfer - IgG is the only class of Ig that crosses the placenta. Transfer is mediated by a receptor on placental cells for the Fc region of IgG. Not all subclasses cross equally well; IgG2 does not cross well.
   - Fixes complement - Not all subclasses fix equally well; IgG4 does not fix complement
   - Binding to cells - Macrophages, monocytes, PMNs and some lymphocytes have Fc receptors for the Fc region of IgG. Not all subclasses bind equally well, IgG2 and IgG4 do not bind to Fc receptors. A consequence of binding to the Fc receptors on PMNs, monocytes and macrophages is that the cell can now internalize the antigen better. The antibody has prepared the antigen for eating by the phagocytic cells. The term **opsonin** is used to describe substances that enhance phagocytosis. IgG is a good opsonin. Binding of IgG to Fc receptors on other types of cells results in the activation of other functions.



Career Avenues

IgM

1. **Structure:** The structure of IgM is presented in figure. IgM normally exists as a pentamer (19S immunoglobulin) but it can also exist as a monomer. In the pentameric form all heavy chains are identical and all light chains are identical. Thus, the valence is theoretically 10. IgM has an extra domain on the mu chain (C_{H4}) and it has another protein covalently bound via a S-S bond called the J chain. This chain functions in polymerization of the molecule into a pentamer.



2. **Properties**

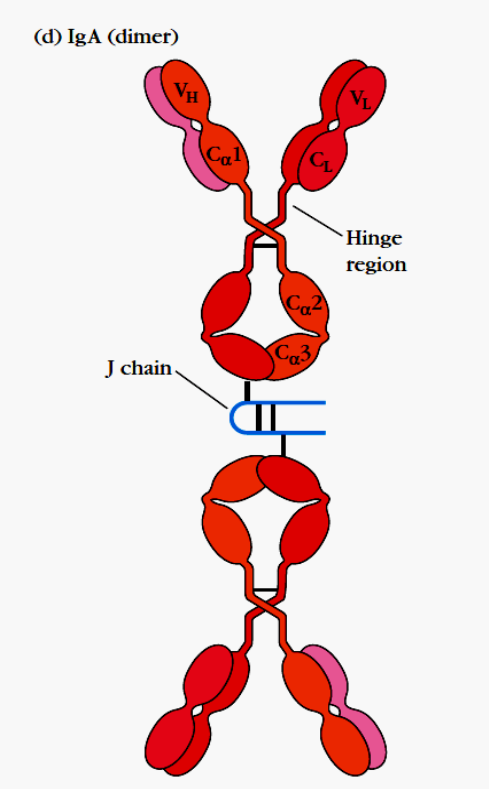
- IgM is the third most common serum Ig.
- IgM is the first Ig to be made by the fetus and the first Ig to be made by a virgin B cells when it is stimulated by antigen.
- As a consequence of its pentameric structure, IgM is a good complement fixing Ig. Thus, IgM antibodies are very efficient in leading to the lysis of microorganisms.
- As a consequence of its structure, IgM is also a good agglutinating Ig . Thus, IgM antibodies are very good in clumping microorganisms for eventual elimination from the body.
- IgM binds to some cells via Fc receptors.
- B cell surface Ig: Surface IgM exists as a monomer and lacks J chain but it has an extra 20 amino acids at the C-terminus to anchor it into the membrane. Cell surface IgM functions as a receptor for antigen on B cells.

IgA

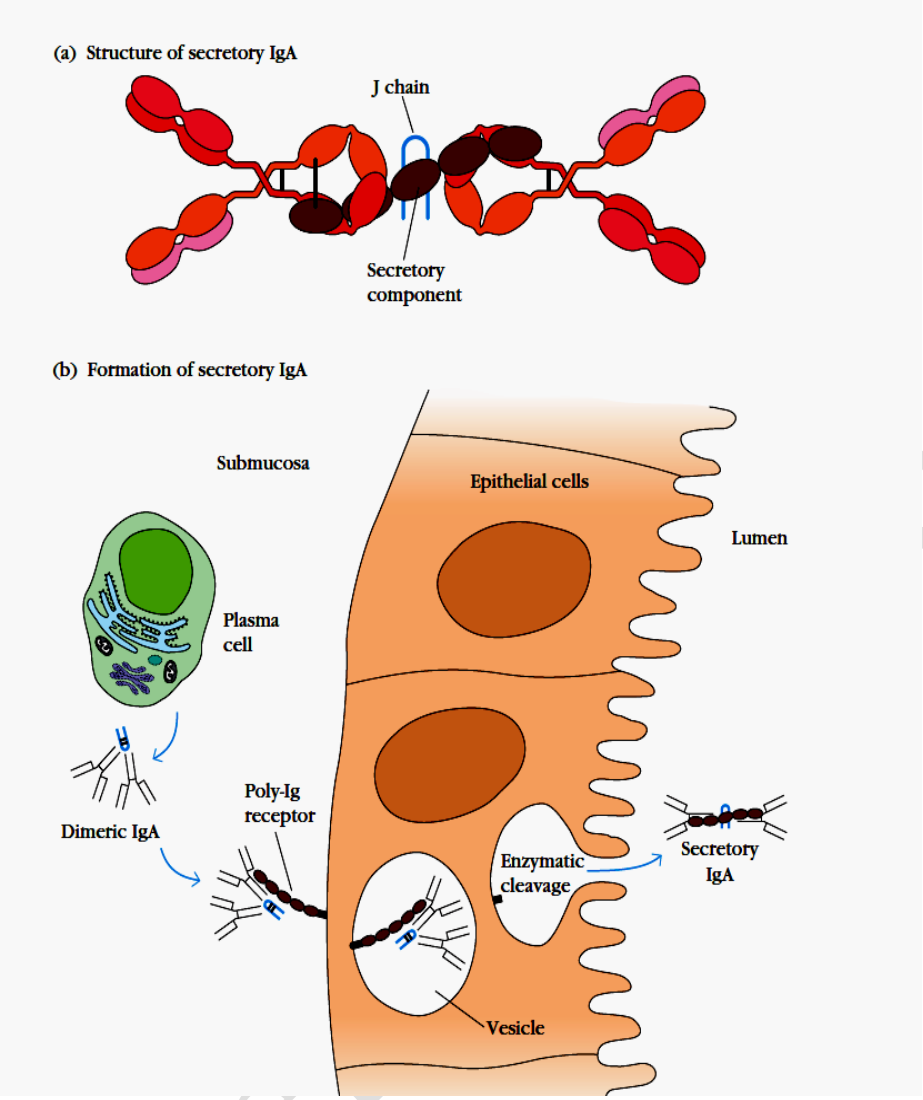
1. **Structure:** Serum IgA is a monomer but IgA found in secretions is a dimer as presented in Figure. When IgA exits as a dimer, a J chain is associated with it. When

Career Avenues

IgA is found in secretions is also has another protein associated with it called the secretory piece or T piece; sIgA is sometimes referred to as 11S immunoglobulin. Unlike the remainder of the IgA which is made in the plasma cell, the secretory piece is made in epithelial cells and is added to the IgA as it passes into the secretions (Figure 12). The secretory piece helps IgA to be transported across mucosa and also protects it from degradation in the secretions.



Career Avenues



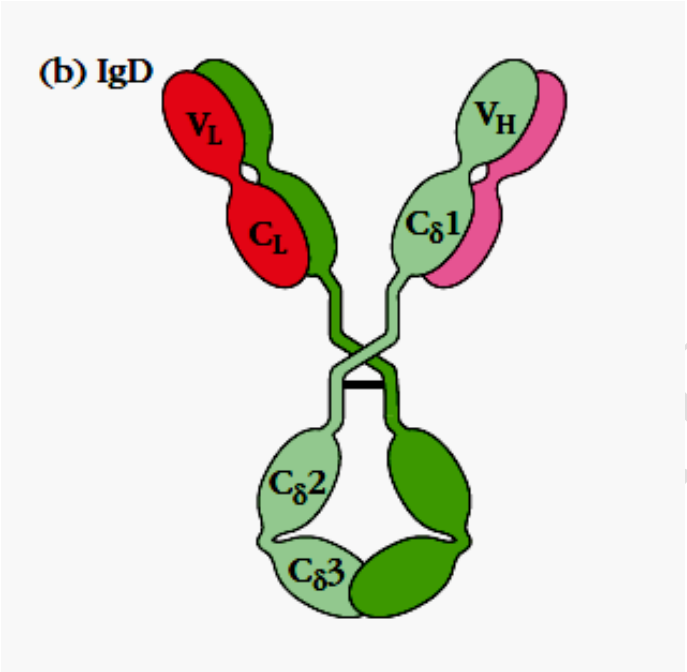
2. Properties

- IgA is the 2nd most common serum Ig.
- IgA is the major class of Ig in secretions - tears, saliva, colostrum, mucus. Since it is found in secretions secretory IgA is important in local (mucosal) immunity.
- Normally IgA does not fix complement, unless aggregated.
- IgA can binding to some cells - PMN's and some lymphocytes.

Career Avenues

IgD

1. **Structure:** The structure of IgD is presented in the Figure. IgD exists only as a monomer.



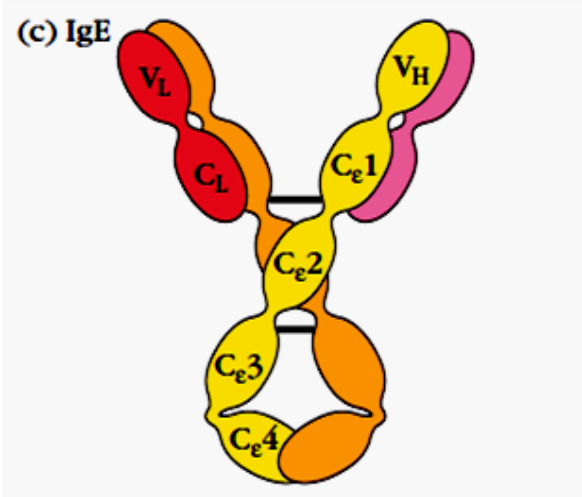
2. **Properties**

- IgD is found in low levels in serum; its role in serum uncertain.
- IgD is primarily found on B cell surfaces where it functions as a receptor for antigen. IgD on the surface of B cells has extra amino acids at C-terminal end for anchoring to the membrane. It also associates with the Ig-alpha and Ig-beta chains.
- IgD does not bind complement.

Career Avenues

IgE

1. **Structure**: The structure of IgE is presented in Figure. IgE exists as a monomer and has an extra domain in the constant region.



2. **Properties**

- IgE is the least common serum Ig since it binds very tightly to Fc receptors on basophils and mast cells even before interacting with antigen.
- Involved in allergic reactions - As a consequence of its binding to basophils an mast cells, IgE is involved in allergic reactions. Binding of the allergen to the IgE on the cells results in the release of various pharmacological mediators that result in allergic symptoms.
- IgE also plays a role in parasitic helminth diseases. Since serum IgE levels rise in parasitic diseases, measuring IgE levels is helpful in diagnosing parasitic infections. Eosinophils have Fc receptors for IgE and binding of eosinophils to IgE-coated helminths results in killing of the parasite.
- IgE does not fix complement.

ACTIONS OF ANTIBODY

In addition of binding antigen, antibody participate in a broad range of effector function- that will result in removal of the antigens and death of the pathogens. Because these effector functions result from interaction between heavy – chain constant region and other serum proteins or cell membrane receptors, not all class of antibodies have the similar functional property. Some major antibody mediated effector function given below:

OPSONIZATION

Opsonisation is the process by which microorganisms or other foreign particles are coated with antibody and/or complement, and thus prepared for recognition and ingestion by phagocytic cells. Opsonising antibodies, IgG, bind to Fc receptors on the surface of macrophages and neutrophils. This binding provides the phagocyte a method to capture antigens.

Career Avenues

TOXIN / VIRAL NEUTRALIZATION

Some gram positive and gram negative bacteria produce extracellular toxins that contribute to their pathogenic effects. Immunity against such toxins depends on the production of specific antibodies that inactivate the toxins. This process is called toxin neutralization. Once neutralized, the toxin antibody complex is either unable to attach to receptor sites on host target cells and is unable to enter the cell, or is ingested by macrophages. An antibody capable of neutralizing a toxin is called antitoxin. Similarly, antibody like IgG, IgM and IgA can bind to some viruses during their extracellular phage and inactivate them.

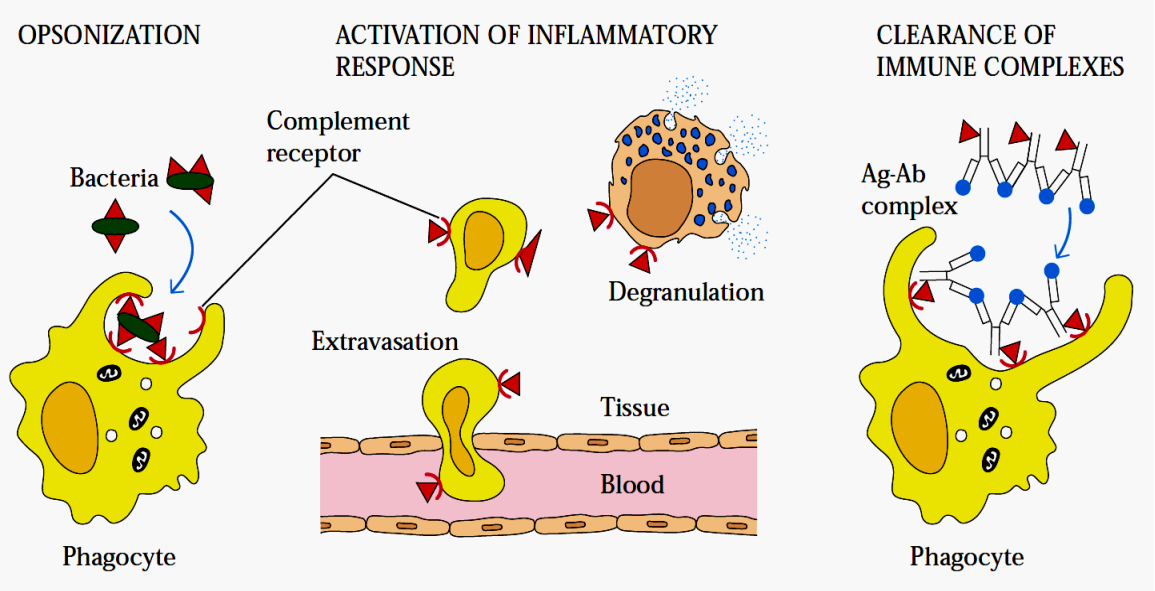
ACTIVATION OF COMPLEMENT

IgM and IgG can activate complement system.

IMMUNE COMPLEX FORMATION

The interaction of antigen antibody may result in a variety of consequences, including **precipitation** and **agglutination**. Precipitation results from the interaction of a soluble antibody with a soluble antigen to form an insoluble complex. Antibodies that thus aggregate soluble antigen are called precipitins.

The reaction of antibody with a multivalent antigen that is particulate (i.e. insoluble particle) result in the cross-linking of the various antigen particle by the antibodies. This cross-linking eventually results in the clumping or agglutination of the antigen particle by the antibodies.



ANTIBODY DEPENDENT CELL-MEDIATED CYTOTOXICITY (ADCC)

The NK cells function is to destroy malignant cells and cell infected with microorganisms. They recognize their targets with the help of binding with antibodies. They can bind to antibodied that coat infected or malignant cells; thus the antibody bridges the two cell types. This process is called ADCC, and can result in the death of the target cell.

TABLE 4-2 Properties and biological activities* of classes and subclasses of human serum immunoglobulins

Property/Activity	IgG1	IgG2	IgG3	IgG4	IgA1	IgA2	IgM‡	IgE	IgD
Molecular weight†	150,000	150,000	150,000	150,000	150,000–600,000	150,000–600,000	900,000	190,000	150,000
Heavy-chain component	γ1	γ2	γ3	γ4	α1	α2	μ	ϵ	δ
Normal serum level (mg/ml)	9	3	1	0.5	3.0	0.5	1.5	0.0003	0.03
In vivo serum half life (days)	23	23	8	23	6	6	5	2.5	3
Activates classical complement pathway	+	+/−	++	−	−	−	+++	−	−
Crosses placenta	+	+/−	+	+	−	−	−	−	−
Present on membrane of mature B cells	−	−	−	−	−	−	+	−	+
Binds to Fc receptors of phagocytes	++	+/−	++	+	−	−	?	−	−
Mucosal transport	−	−	−	−	++	++	+	−	−
Induces mast-cell degranulation	−	−	−	−	−	−	−	+	−

*Activity levels indicated as follows: ++ = high; + = moderate; +/− = minimal; − = none; ? = questionable.



GATE Questions

GATE 2008

Q.7 Serum IgM cannot activate the complement by itself because
(A) it does not have complement binding site
(B) it is planar in which complement binding sites in the Fc region are not accessible.
(C) it gets degraded and hence unable to activate the complement
(D) it needs metal ions to activate complement

GATE 2009

Q.8 IgG has four chains. Purified monoclonal IgG was subjected to electrophoresis. The number of bands visible by
a) Reducing SDS-PAGE b) Isoelectric focusing c) Native PAGE
are
(A) a-2; b-1; c-1 (B) a-1; b-1; c-2 (C) a-2; b-4; c-2 (D) a-4; b-2; c-4

GATE 2013

Q.9 Protein A, which has strong affinity to Fc region of immunoglobulin, is extracted from
(A)Saccharomyces cerevisiae (B)Staphylococcus aureus
(C)Streptococcus pyogenes (D)Streptococcus sanguis

Answers

7-B. 8-A. 9-B.

Career AV

ORIGIN OF ANTIBODY DIVERSITY

BACKGROUND

Immune system has the ability to respond to a vast number of foreign antigens

This is achieved majorly by the antibodies which bind to these antigens

These Antigen-Antibody reactions are highly specific and each antibody has a unique amino acid sequence in the variable region and can bind only a specific antigen

Theoretically it is believed that there can be close to 10^{10} antibody combining diversities in humans i.e. 10^{10} antibody specificities are possible.

Does this mean there are 10^{10} different immunoglobulin gene sequences coding for these variable regions which determine specificity? The answer is NO

In fact there are only ~30,000 genes altogether in the human genome

One of the major questions in immunology has been how can we make so many different antibody molecules.

Antibody diversity is generated by the following mechanisms.

- Multiple germ-line gene segments
- Combinatorial V-(D)-J joining
- Junctional flexibility
- P- region nucleotide addition (P-addition)
- N-region nucleotide addition (N-addition)
- Somatic hypermutation
- Combinatorial association of light and heavy chains

Multiple germ-line gene segments

The kappa and lambda light chains and the heavy chains are encoded by separate multigene families situated on different chromosomes. In germ-line DNA, each of these multigene families contains several coding sequences, called gene segments, separated by noncoding regions. During B-cell maturation, these gene segments are rearranged and brought together to form functional immunoglobulin genes.

The kappa and lambda light-chain families contain V, J, and C gene segments; the rearranged VJ segments encode the variable region of the light chains. The heavy-chain family contains V, D, J, and C gene segments; the rearranged VDJ gene segments encode the variable region of the heavy chain. In each gene family, C gene segments encode the constant regions. Each

Career Avenues

V gene segment is preceded at its 5' end by a small exon that encodes a short signal or leader (L) peptide that guides the heavy or light chain through the endoplasmic reticulum. The signal peptide is cleaved from the nascent light and heavy chains before assembly of the finished immunoglobulin molecule. Thus, amino acids encoded by this leader sequence do not appear in the immunoglobulin molecule.

Germ-line VDJ gene segments in Humans

Multiple-germ line segments	Heavy Chain	Light Chains	
		Kappa	Lambda
V	51	40	30
D	27	0	0
J	6	5	4

Multiple genes in the germ-line DNA contribute to the Antibody diversity to some extent

Combinatorial V-(D)-J joining

The contribution of multiple germ line gene segments to Ab diversity is magnified by the random rearrangement of these segments in somatic cells. The total number of possible combinations is conservatively estimated to be ~10^{10} different Ab specificities generated by the mammalian immune system. (Remember in humans the κ and λ loci contain roughly equal numbers of V genes**.**

V-J AND V-D-J JOINING

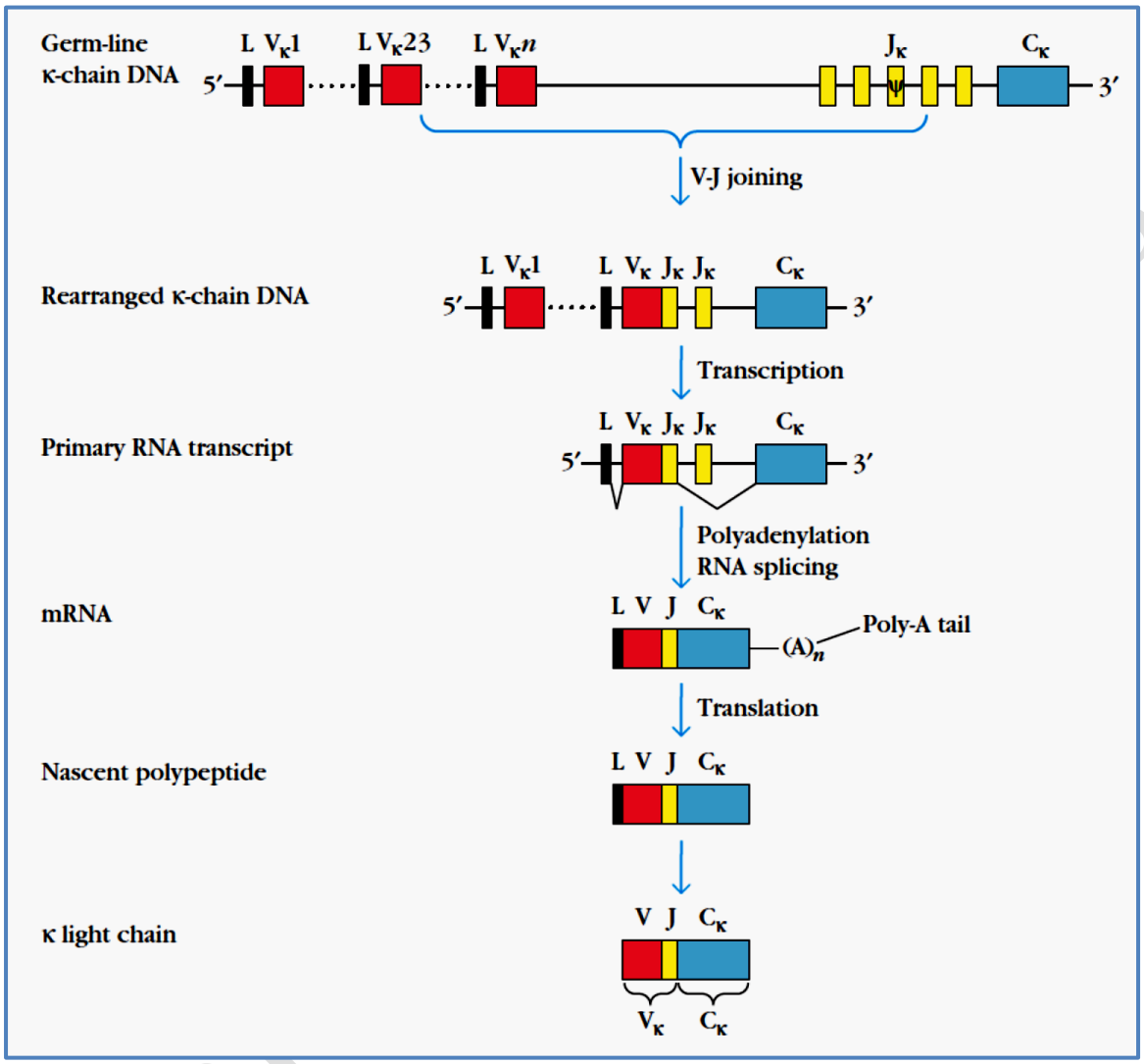
The region where the light chain V gene and J region or the heavy chain V gene and D and J regions come together is in the third hypervariable region. Since it is random which V and which J or D regions come together, there is a lot of diversity that can be generated by V-J and V-D-J joining.

Multiple-germ line segments	Heavy Chain	Light Chains	
		Kappa	Lambda
V	51	40	30
D	27	0	0
J	6	5	4
Combinatorial V-D-J and V-J joining(Possible no. of combinations)	51 x 27 x 6 = 8262	40 x 5 = 200	30 x 4 = 120

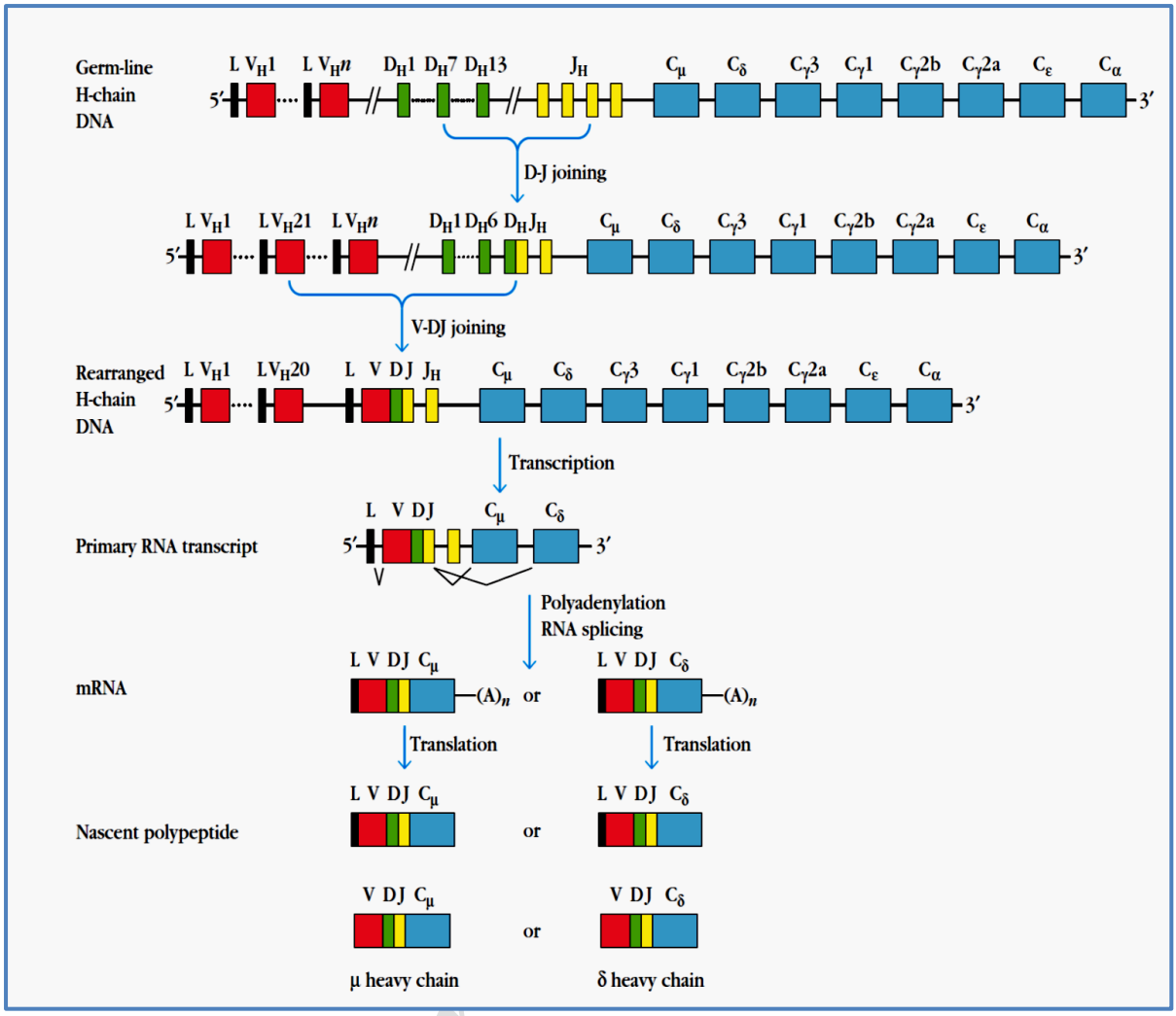
Career Avenues

Possible Combinatorial associations of Heavy and Light Chains is

$$8262 \times (200 \times 120) = 2.64 \times 10^{6}$$



Kappa Light chain gene rearrangement



Heavy chain gene rearrangement

Recombination only occurs between gene segments located on the same chromosome and it follows the special linking rule imposed by the RSSs. Only one joining event is needed for light chain genes whereas two are needed for H chains.

The most common mode of rearrangement involves the looping out and deletion of the DNA intervening between the two gene segments. The DNA is broken and re-ligated.

Career

Junctional flexibility

The enormous diversity generated by means of V,D, and J combinations is further augmented by a phenomenon known as junctional flexibility. This brings the potential repertoire up to 10^8 to 10^9 diversity.

The signal sequences join precisely to form a signal joint in a circular piece of DNA (where it has looped out) which is then lost from the genome when the cell divides. The joining of the V and J segments (ect) form what is called the coding joint and it is an imprecise joining and consequently generates more Ab variability.

Inaccurate or imprecise DNA rearrangement occurs because the nucleotide sequences at the 3' end of a V gene and the 5' end of a J segment in a light chain , or the ends of V,D, J gene segments in a H chain can each recombine at any of several nucleotides in the germline sequence. As long as the recombination does not generate nonfunctional DNA, different nucleotide and thus different aa sequences arise. The amino acid sequence variation generated by junctional flexibility has been shown to fall within the CDR3 (3rd HV region) in both the H and L chain. Since CDR3s are a major contributor to the Ag binding site, junctional flexibility can have a major impact in generating Ab diversity.

P- region nucleotide addition (P-addition)

P nucleotides are so called because they comprise palindromic sequences added to the ends of the gene segments. Nucleotides at the end of coding sequences are thought to be sealed to form a hairpin structure. This hairpin is later cleaved by an endonuclease. Cleavage sometimes occurs at a position that leaves a short single stranded region at the end of the coding sequences. Then subsequent addition of complementary nucleotides by repair enzymes occurs. Variation in the position at which the hairpin is cut thus leads to variation in the sequence of coding joint.

N- region nucleotide addition (N-addition)

Nucleotides called N sequences because they are non-template encoded in other words they are not present in the germline. They can be added to the junctions of rearranged VDJ genes after the hairpin has been cleaved by (only occurs in H chain DNA). This addition of new nucleotides is a random process with the possible addition of up to 20 nucleotides and is mediated by an enzyme called terminal deoxyribonucleotidyl transferase (TdT).

*N nucleotides are absent from light chains because the enz TdT is expressed only for a short period of time during the assembly of the H chain genes which occurs before the L chain gene are assembled.

Addition of nucleotides often disrupts the reading frame with roughly two in three rearrangements being nonproductive-with many B cell never succeeding to produce a functional Ab so diversity is achieved only with considerable waste.

(Junctional diversity is even more important in the T cell receptor).

Career Avenues

Somatic hypermutation

All Ab diversity discussed to this point has been due to mechanisms that operate during formation of specific variable regions by gene rearrangement. Once the functional variable region gene unit is formed a process called somatic hypermutation can take place.

In somatic hypermutation individual nucleotides in VJ or VDJ units are replaced with alternative bases thus most likely altering the specificity of the encoded Ig.

The rate of somatic mutation is a million fold higher than the spontaneous mutation rate in other genes and an average of one mutation will be introduced in every one to two cell divisions.

The mechanism of somatic hypermutation has not yet been determined. It has been suggested that mutations are introduced by an error prone DNA polymerase.

Combinatorial association of light and heavy chains

Humans there is potential to generate 8262 Heavy chain genes 320 Light chain genes (200 kappa and 120 Lambda)

If we assume that all combinations are possible then the potential number of Heavy and Light chain combinations would be ~2,644,240

Though the combinatorial diversity actually observed in an individual is much less, it is also a factor that contributes to the vast diversity of the antibodies

MULTISPECIFICITY

Due to cross reactions between antigenic determinants of similar structure an antibody can often react with more than one antigenic determinant. This is termed multispecificity. Multispecificity also contributes to antibody diversity.

Career Avenues



GATE QUESTIONS

GATE 2010

Q.10 Match the following in Group 1 with those in group 2

Group 1	**Group 2**
P. Isotype switching	1. V_H domain
Q. Clonal anergy	2. Non-responsive to self antigen
R. Class II MHC	3. Non-responsive T_H cells
S. Self tolerance	4. β_2- microglobulin

(A) P-1, Q-4, R-3, S-2 (B) P-2, Q-4, R-1, S-3

(C) P-1, Q-3, R-4, S-2 (D) P-2, Q-1, R-3, S-4

Answers
10-C.

Career Avenues

POLYCLONAL AND MONOCLONAL ANTIBODIES

Polyclonal antibodies (or antisera) are antibodies that are obtained from different B cell resources. They are a combination of immunoglobulin molecules secreted against a specific antigen, each identifying a different epitope. Polyclonal antibodies often recognize multiple epitopes, making them more tolerant of small changes in the nature of the antigen. Polyclonal antibodies are often the preferred choice for detection of denatured proteins. Polyclonal antibodies may be generated in a variety of species, including rabbit, goat, sheep, donkey, chicken and others, giving the users many options in experimental design. Polyclonal antibodies are sometimes used when the nature of the antigen in an untested species is not known. Polyclonal antibodies target multiple epitopes and so they generally provide more robust detection.

PRODUCTION OF PCA

These antibodies are typically produced by immunization of a suitable mammal, such as a mouse, rabbit or goat. Larger mammals are often preferred as the amount of serum that can be collected is greater. An antigen is injected into the mammal. This induces the B-lymphocytes to produce IgG immunoglobulins specific for the antigen. This polyclonal IgG is purified from the mammal's serum.

By contrast, monoclonal antibodies are derived from a single cell line.

MONOCLONAL ANTIBODIES

Monoclonal antibodies (**mAb** or **moAb**) are mono-specific antibodies that are the same because they are made by identical immune cells that are all clones of a unique parent cell. Because of their specificity, monoclonal antibodies are excellent as the primary antibody in an assay, or for detecting antigens in tissue, and will often give significantly less background staining than polyclonal antibodies. When compared to that of polyclonal antibodies, homogeneity of monoclonal antibodies is very high. If experimental conditions are kept constant, results from monoclonal antibodies will be highly reproducible, between experiments. Specificity of monoclonal antibodies makes them extremely efficient for binding of antigen within a mixture of related molecules, such as in the case of affinity purification.

PRODUCTION OF MCA

HYBRIDOMA TECHNOLOGY

Monoclonal antibodies are typically made by fusing myeloma cells with the spleen cells from a mouse that has been immunized with the desired antigen. However, recent advances have allowed the use of rabbit B-cells. Polyethylene glycol is used to fuse adjacent plasma membranes, but the success rate is low so a selective medium in which only fused cells can grow is used. This is because myeloma cells have lost the ability to synthesize hypoxanthine-guanine-phosphoribosyltransferase (HGPRT), an enzyme necessary for the salvage synthesis

Career Avenues

of nucleic acids. The absence of HGPRT is not a problem for these cells unless the de novo purine synthesis pathway is also disrupted. By exposing cells to aminopterin (a folic acid analogue, which inhibits dihydrofolatereductase, DHFR), they are unable to use the de novo pathway and become fully auxotrophic for nucleic acids requiring supplementation to survive.

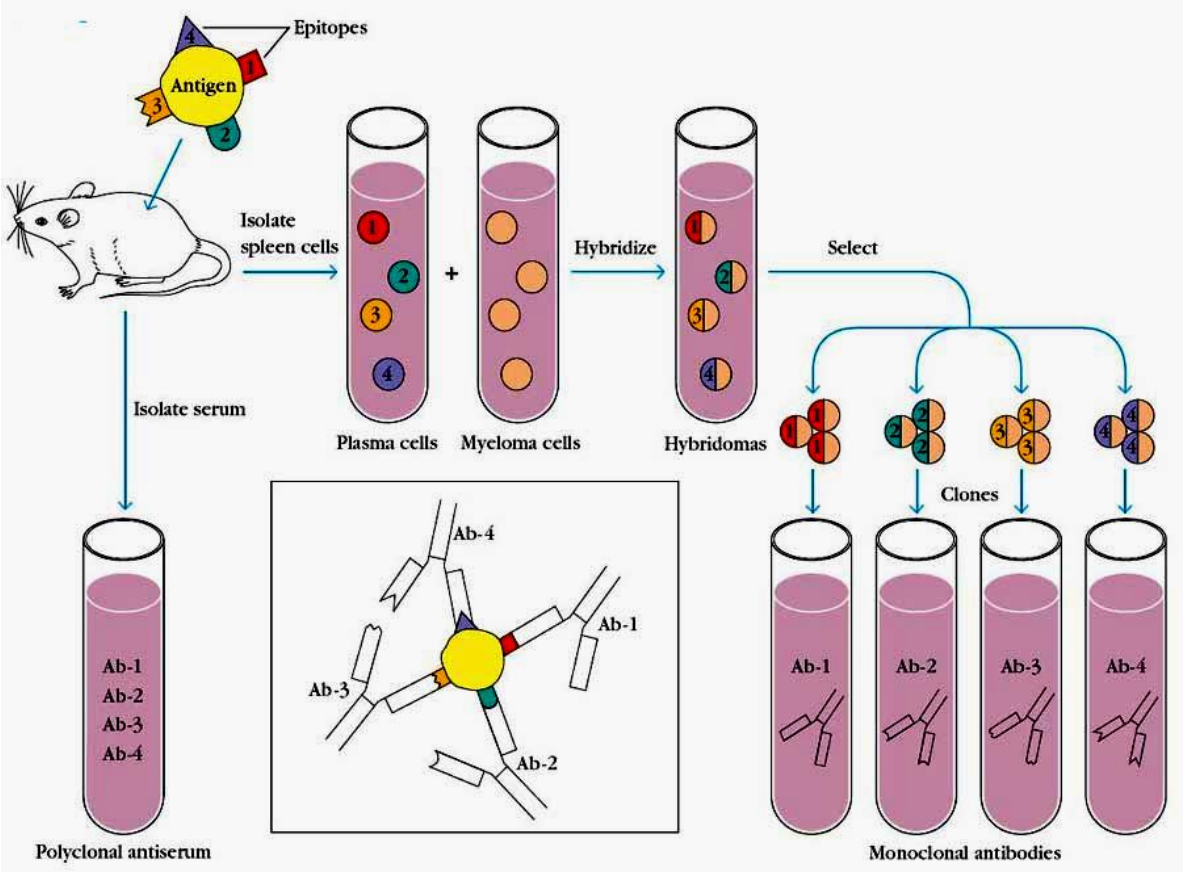
The selective culture medium is called HAT medium because it contains hypoxanthine, aminopterin, and thymidine. This medium is selective for fused (hybridoma) cells. Unfused myeloma cells cannot grow because they lack HGPRT, and thus cannot replicate their DNA. Unfused spleen cells cannot grow indefinitely because of their limited life span. Only fused hybrid cells, referred to as hybridomas, are able to grow indefinitely in the media because the spleen cell partner supplies HGPRT and the myeloma partner has traits that make it immortal (as it is a cancer cell).

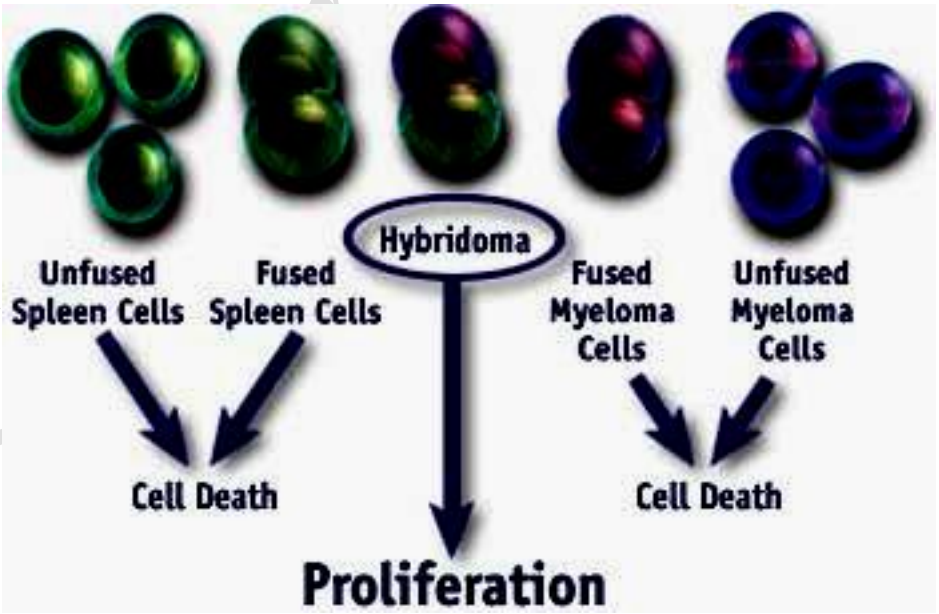
This mixture of cells is then diluted and clones are grown from single parent cells on microtitre wells. The antibodies secreted by the different clones are then assayed for their ability to bind to the antigen (with a test such as ELISA or Antigen Microarray Assay) or immuno-dot blot. The most productive and stable clone is then selected for future use.

The hybridomas can be grown indefinitely in a suitable cell culture medium. They can also be injected into mice (in the peritoneal cavity, surrounding the gut). There, they produce tumors secreting an antibody-rich fluid called ascites fluid.

The medium must be enriched during in-vitro selection to further favourhybridoma growth. This can be achieved by the use of a layer of feeder fibrocyte cells or supplement medium such as briclone. Culture-medium conditioned by macrophages can also be used. Production in cell culture is usually preferred as the ascites technique is painful to the animal. Where alternate techniques exist, this method (ascites) is considered unethical.

Career Avenues


Epitopes
Antigen
Isolate spleen cells
Plasma cells
Myeloma cells
Hybridize
Hybridomas
Select
Clones
Isolate serum
Ab-1
Ab-2
Ab-3
Ab-4
Polyclonal antiserum
Monoclonal antibodies


Hybridoma
Unfused Spleen Cells
Fused Spleen Cells
Fused Myeloma Cells
Unfused Myeloma Cells
Cell Death
Cell Death
Proliferation

APPLICATIONS OF MCA

DIAGNOSTIC TESTS

Once monoclonal antibodies for a given substance have been produced, they can be used to detect the presence of this substance. The Western blot test and immuno dot blot tests detect the protein on a membrane. They are also very useful in immunohistochemistry, which detect antigen in fixed tissue sections and immunofluorescence test, which detect the substance in a frozen tissue section or in live cells.

THERAPEUTIC TREATMENT

CANCER TREATMENT

One possible treatment for cancer involves monoclonal antibodies that bind only to cancer cell-specific antigens and induce an immunological response against the target cancer cell. Such mAb could also be modified for delivery of a toxin, radioisotope, cytokine or other active conjugate; it is also possible to design bispecific antibodies that can bind with their Fab regions both to target antigen and to a conjugate or effector cell. In fact, every intact antibody can bind to cell receptors or other proteins with its Fc region.



GATE QUESTIONS

GATE 2008

Q.11 In hybridoma technology, the myeloma cells used

(A) lack HGPRTase

(B) lack the ability to produce Ig

(C) lack both HGPRTase and ability to produce Ig

(D) lack thymidine kinase

GATE 2013

Q.12 The first humanized monoclonal antibody approved for the treatment of breast canceris

(A) Rituximab (B) Cetuximab

(C) Bevacizumab (D) Herceptin

Answers
11-A. 12-D.

Career Avenues

COMPLEMENT

COMPLEMENT FUNCTIONS

The term complement (C) is refer to a heat-labile serum component that was able to lyse bacteria and also contribute to host defenses in other ways as well. Complement can opsonize bacteria for enhanced phagocytosis; it can recruit and activate various cells including polymorphonuclear cells (PMNs) and macrophages; it can participate in regulation of antibody responses and it can aid in the clearance of immune complexes and apoptotic cells. Complement can also have detrimental effects for the host; it contributes to inflammation and tissue damage and it can trigger anaphylaxis.

Complement comprises over 20 different serum proteins (see Table 1) that are produced by a variety of cells including, hepatocytes, macrophages and gut epithelial cells. Some complement proteins bind to immunoglobulins or to membrane components of cells. Others are proenzymes that, when activated, cleave one or more other complement proteins. Upon cleavage some of the complement proteins yield fragments that activate cells, increase vascular permeability or opsonize bacteria.

Table 1. Proteins of the Complement system

Classical Pathway	**Lectin Pathway**	**Alternative Pathway**	**Lytic Pathway**
Activation Proteins: C1qrs, C2, C3, C4 Control Proteins: C1-INH, C4-BP	Mannan binding protein (MBP), mannan-asociated serine protease (MASP, MASP2)	C3, Factors B & D*, Properdin (P) Factors I* & H, decay accelerating factor (DAF), Complement receptor 1(CR1), *etc*.	C5, C6, C7, C8, C9 Protein S

Components underlined acquire enzymatic activity when activated.

Components marked with an asterisk have enzymatic activity in their native form.

PATHWAYS OF COMPLEMENT ACTIVATION

Complement activation can be divided into four pathways (figure 1): the classical pathway, the lectin pathway, the alternative pathway and the membrane attack (or lytic) pathway. Both classical and alternative pathways lead to the activation of C5 convertase and result in the production of C5b which is essential for the activation of the membrane attack pathway.

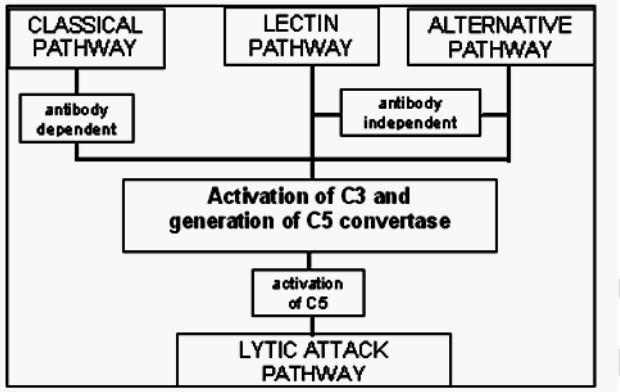
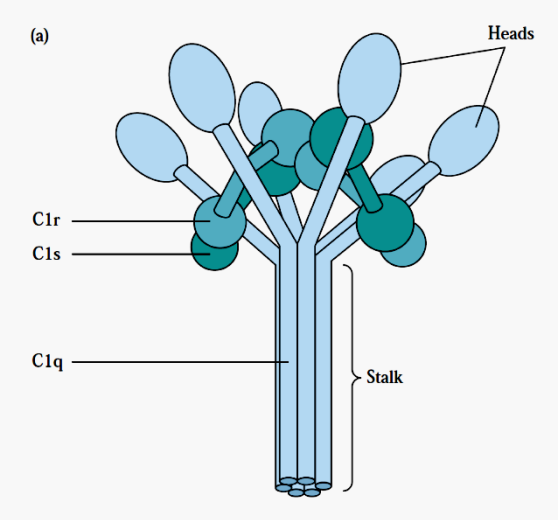


Figure 1. Pathways of complement activation

CLASSICAL PATHWAY

C1 ACTIVATION

C1, a multi-subunit protein containing three different proteins (C1q, C1r and C1s), binds to the Fc region of IgG and IgM antibody molecules that have interacted with antigen. C1 binding does not occur to antibodies that have not complexed with antigen and binding requires calcium and magnesium ions. The binding of C1 to antibody is via C1q and C1q must cross link at least two antibody molecules before it is firmly fixed. The binding of C1q results in the activation of C1r which in turn activates C1s. The result is the formation of an activated "C1qrs", which is an enzyme that cleaves C4 into two fragments C4a and C4b.



Career Avenues

C4 AND C2 ACTIVATION (GENERATION OF C3 CONVERTASE)

The C4b fragment binds to the membrane and the C4a fragment is released into the microenvironment. Activated "C1qrs" also cleaves C2 into C2a and C2b. C2a binds to the membrane in association with C4b, and C2b is released into the microenvironment. The resulting C4bC2a complex is a C3 convertase, which cleaves C3 into C3a and C3b.

C3 ACTIVATION (GENERATION OF C5 CONVERTASE)

C3b binds to the membrane in association with C4b and C2a, and C3a is released into the microenvironment. The resulting C4bC2aC3b is a C5 convertase. The generation of C5 convertase is the end of the classical pathway.

Several of the products of the classical pathway have potent biological activities that contribute to host defenses. Some of these products may also have detrimental effects if produced in an unregulated manner. Table 2 summarizes the biological activities of classical pathway components.

Table 2. Biological Activity of classical pathway products

Component	Biological Activity
C2b	**Prokinin**; cleaved by plasmin to yield kinin, which results in edema
C3a	**Anaphylotoxin**; can activate basophils and mast cells to degranulate resulting in increased vascular permeability and contraction of smooth muscle cells, which may lead to **anaphylaxis**
C3b	**Opsonin**; promotes phagocytosis by binding to complement receptors Activation of phagocytic cells
C4a	**Anaphylotoxin** (weaker than C3a)
C4b	**Opsonin**; promotes phagocytosis by binding to complement receptors

If the classical pathway were not regulated there would be continued production of C2b, C3a, and C4a. Thus, there must be some way to regulate the activity of the classical pathway. Table 3 summarizes the ways in which the classical pathway is regulated.

Table 3. Regulation of the Classical Pathway

Component	Regulation
All	**C1-INH**; dissociates C1r and C1s from C1q
C3a	**C3a inactivator (C3a-INA;Carboxypeptidase B)**;

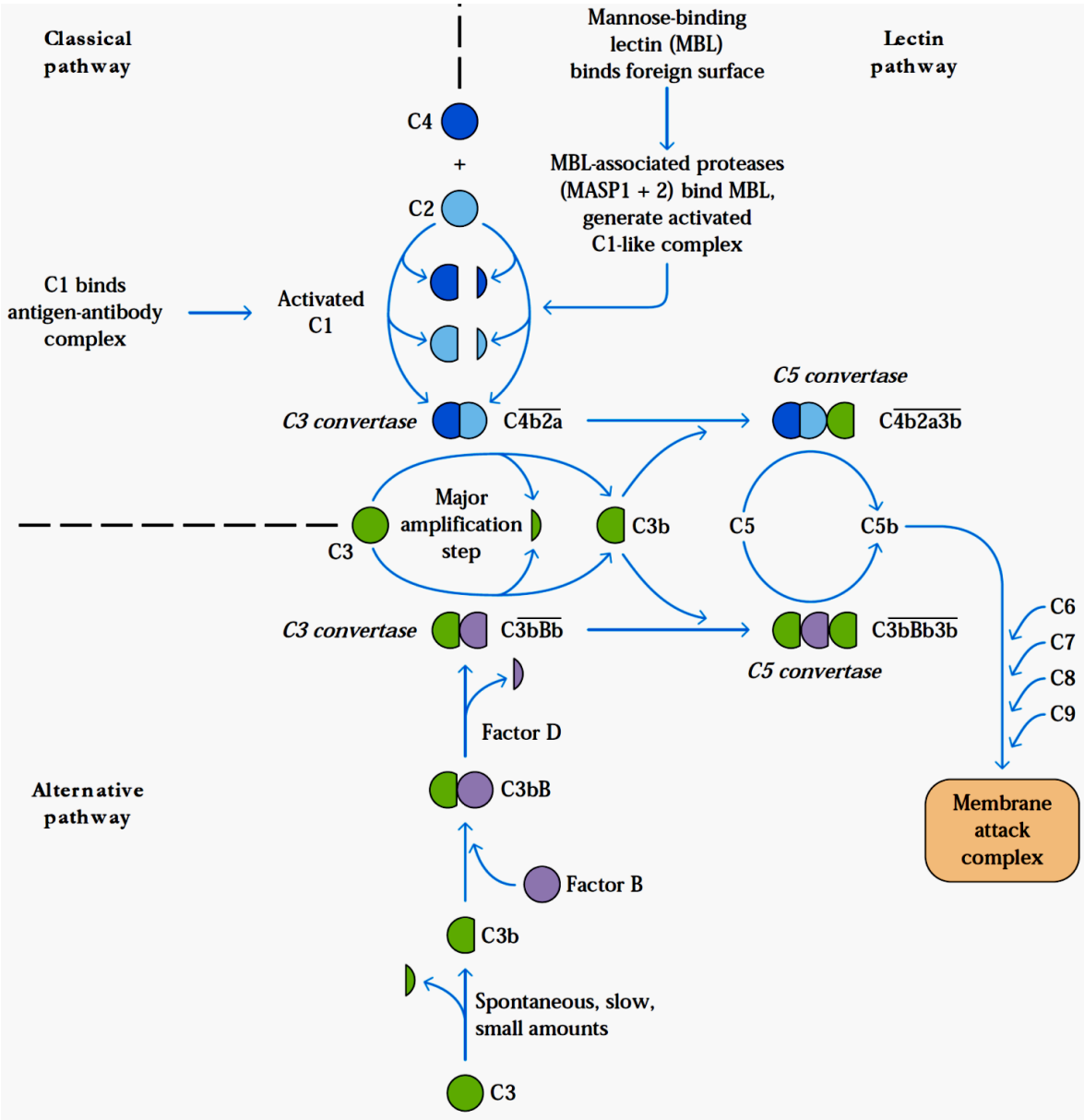
	inactivates C3a
C3b	**Factors H and I;** Factor H facilitates the degradation of C3b by Factor I
C4a	**C3-INA**
C4b	**C4 binding protein(C4-BP) and Factor I**; C4-BP facilitates degradation of C4b by Factor I; C4-BP also prevents association of C2a with C4b thus blocking the formation of C3 convertase

LECTIN PATHWAY

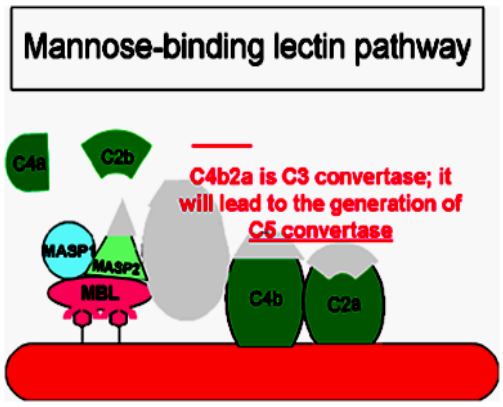
The lectin pathway (figure 3) is very similar to the classical pathway. It is initiated by the binding of mannose-binding lectin (MBL) to bacterial surfaces with mannose-containing polysaccharides (mannans). Binding of MBL to a pathogen results in the association of two serine proteases, MASP-1 and MASP-2 (MBL-associated serine proteases). MASP-1 and MASP-2 are similar to C1r and C1s, respectively and MBL is similar to C1q. Formation of the MBL/MASP-1/MASP-2 tri-molecular complex results in the activation of the MASPs and subsequent cleavage of C4 into C4a and C4b. The C4b fragment binds to the membrane and the C4a fragment is released into the microenvironment. Activated MASPs also cleave C2 into C2a and C2b. C2a binds to the membrane in association with C4b and C2b is released into the microenvironment. The resulting C4bC2a complex is a C3 convertase, which cleaves C3 into C3a and C3b. C3b binds to the membrane in association with C4b and C2a and C3a is released into the microenvironment. The resulting C4bC2aC3b is a C5 convertase. The generation of C5 convertase is the end of the lectin pathway.

The biological activities and the regulatory proteins of the lectin pathway are the same as those of the classical pathway.

Career Avenues



Overview of the complement activation pathways



ALTERNATE PATHWAY

The **alternative pathway** of the complement system is an innate component of the immune system's natural defense against infections, which can operate without antibody participation.

The alternative pathway is one of three complement pathways that opsonize and kill pathogens. The alternative pathway does not require a specific antibody to commence, and, so, can be effectively much faster than if antibody synthesis had to take place, as in the classical pathway. Rather, the C3 protein directly binds the microbe. The caveat of this faster activation is that only specific types of antigens can activate this pathway.

CASCADE

It is initiated by the spontaneous hydrolysis of C3, which is abundant in the plasma in the blood. "Tickover" occurs through the spontaneous cleavage of the thioester bond in C3 to form C3(H_2O).

This change in shape allows the binding of plasma protein Factor B, which allows Factor D to cleave Factor B into Ba and Bb.

Bb remains part of the C3(H_2O) to form C3(H_2O)Bb. This complex is also known as a fluid-phase C3 convertase. This convertase, although only produced in small amounts, can cleave multiple C3 proteins into C3a and C3b.

The alternative pathway C3-convertase consists of the activated B and D factors, forming an unstable compound that can become stable after binding properdin, a serum protein.

After the creation of C3 convertase, the complement system follows the same path regardless of the means of activation (alternative, classical, or MBL). Binding of another C3b-fragment to the C3-convertase of the alternative pathway creates a C5-convertase analoguous to the MBL or classical pathway.

The C5-convertase of the alternative pathway consists of C3bBbC3b also referred to as $C3b_2Bb$ (instead of C4b2a3b in the other pathways)

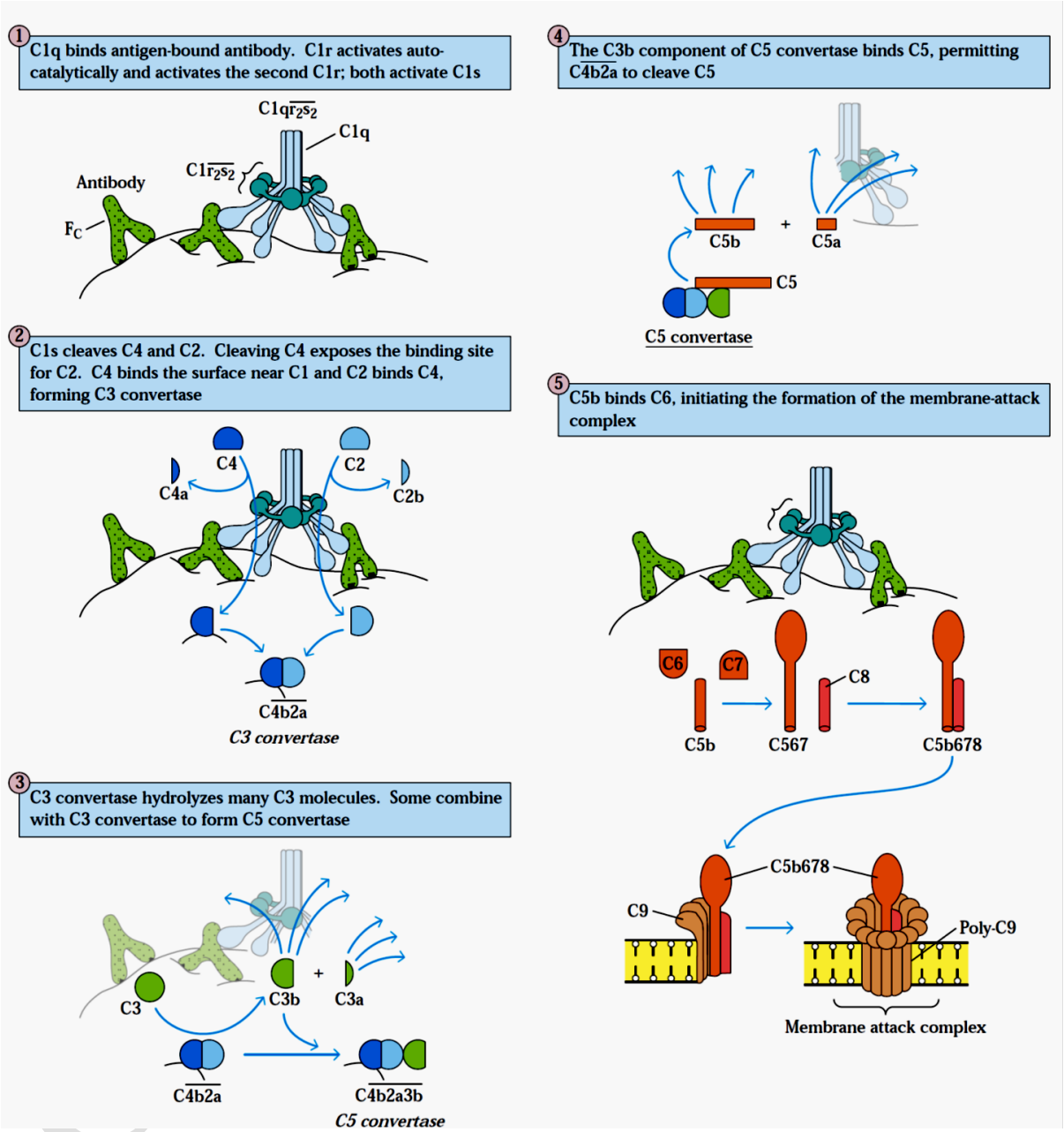
MEMBRANE ATTACK (LYTIC) PATHWAY

C5 convertase from the classical (C4b2a3b), lectin (C4b2a3b) or alternative (C3bBb3b) pathway cleaves C5 into C5a and C5b. C5a remains in the fluid phase and the C5b rapidly associates with C6 and C7 and inserts into the membrane. Subsequently C8 binds, followed by several molecules of C9. The C9 molecules form a pore in the membrane through which the cellular contents leak and lysis occurs. Lysis is not an enzymatic process; it is thought to be due to physical damage to the membrane. The complex consisting of C5bC6C7C8C9 is referred to as the membrane attack complex (MAC).

C5a generated in the lytic pathway has several potent biological activities. It is the most potent anaphylotoxin. In addition, it is a chemotactic factor for neutrophils and stimulates the respiratory burst in them and it stimulates inflammatory cytokine production by macrophages. Its activities are controlled by inactivation by carboxypeptidase B (C3-INA).

Career Avenues

Some of the C5b67 complex formed can dissociate from the membrane and enter the fluid phase. If this were to occur it could then bind to other nearby cells and lead to their lysis. The damage to bystander cells is prevented by Protein S (vitronectin). Protein S binds to soluble C5b67 and prevents its binding to other cells.



BIOLOGICALLY ACTIVE PRODUCTS OF COMPLEMENT ACTIVATION

Activation of complement results in the production of several biologically active molecules which contribute to resistance, anaphylaxis and inflammation.

KININ PRODUCTION

C2b generated during the classical pathway of C activation is a prokinin which becomes biologically active following enzymatic alteration by plasmin. Excess C2b production is prevented by limiting C2 activation by C1 inhibitor (C1-INH) also known as serpin which displaces C1rs from the C1qrs complex (Figure 10). A genetic deficiency of C1-INH results in an overproduction of C2b and is the cause of hereditary angioneurotic edema. This condition can be treated with Danazol which promotes C1-INH production or with ε-amino caproic acid which decreases plasmin activity.

ANAPHYLOTOXINS

C4a, C3a and C5a (in increasing order of activity) are all anaphylotoxins which cause basophil/mast cell degranulation and smooth muscle contraction. Undesirable effects of these peptides are controlled by carboxypeptidase B (C3a-INA).

CHEMOTACTIC FACTORS

C5a and MAC (C5b67) are both chemotactic. C5a is also a potent activator of neutrophils, basophils and macrophages and causes induction of adhesion molecules on vascular endothelial cells.

OPSONINS

C3b and C4b in the surface of microorganisms attach to C-receptor (CR1) on phagocytic cells and promote phagocytosis.

OTHER BIOLOGICALLY ACTIVE PRODUCTS OF C ACTIVATION

Degradation products of C3 (iC3b, C3d and C3e) also bind to different cells by distinct receptors and modulate their functions.

There are known genetic deficiencies of most individual C complement components, but C3 deficiency is most serious and fatal. Complement deficiencies also occur in immune complex diseases (e.g., SLE) and acute and chronic bacterial, viral and parasitic infections.

Career Avenues



GATE QUESTIONS

GATE 2011

Q.13 The activation of the complement components occurs via three distinct pathways. Which of the following component(s) is specific to the 'Alternate Pathway'?

(A) Factor B and D
(B) Mannose binding protein
(C) Clqr2s2
(D) C2

Answers:

13 – A.

Career Avenues

ANTIGEN-ANTIBODY REACTION

The specific association of antigens and antibodies is dependent on hydrogen bonds, hydrophobic interactions, electrostatic forces, and van der Waals forces. These are all bonds of a weak, non-covalent nature, yet some of the associations between antigen and antibody can be quite strong. Like antibodies, antigens can be multivalent, either through multiple copies of the same epitope, or through the presence of multiple epitopes that are recognized by multiple antibodies. Interactions involving multivalency can produce more stabilized complexes, however multivalency can also result in steric difficulties, thus reducing the possibility for binding. All antigen-antibody binding is reversible, however, and follows the basic thermodynamic principles of any reversible bimolecular interaction:

$$K_A = \frac{[Ab-Ag]}{[Ab][Ag]}$$

where K_A is the affinity constant, Ab and Ag are the molar concentrations of unoccupied binding sites on the antibody or antigen respectively, and Ab-Ag is the molar concentration of the antibody-antigen complex.

The time taken to reach equilibrium is dependent on the rate of diffusion and the affinity of the antibody for the antigen, and can vary widely. The affinity constant for antibody-antigen binding can span a wide range, extending from below 10^5 mol^{-1} to above 10^{12} mol^{-1}. Affinity constants can be affected by temperature, pH and solvent.

Affinity constants can be determined for monoclonal antibodies, but not for polyclonal antibodies, as multiple bondings take place between polyclonal antibodies and their antigens. Quantitative measurements of antibody affinity for antigen can be made by equilibrium dialysis. . Repeated equilibrium dialyses with a constant antibody concentration but varying ligand concentration are used to generate Scatchard plots, which give information about affinity valence and possible cross-reactivity.

Affinity describes the strength of interaction between antibody and antigen at single antigenic sites. Within each antigenic site, the variable region of the antibody "arm" interacts through weak non-covalent forces with antigen at numerous sites; the more interactions, the stronger the affinity. Avidity is perhaps a more informative measure of the overall stability or strength of the antibody-antigen complex. It is controlled by three major factors: antibody-epitope affinity; the valence of both the antigen and antibody; and the structural rearrangement of the interacting parts. Ultimately these factors define the specificity of the antibody, that is, the likelihood that the particular antibody is binding to a precise antigen epitope. Cross-reactivity refers to an antibody or population of antibodies binding to epitopes on other antigens. This can be caused either by low avidity or specificity of the antibody or by multiple distinct antigens having identical or very similar epitopes. Cross-reactivity is sometimes desirable

Career Avenues

when one wants general binding to a related group of antigens or when attempting cross-species labeling when the antigen epitope sequence is not highly conserved in evolution.

Immunochemical techniques capitalize upon the extreme specificity, at the molecular level, of each immunoglobulin for its antigen, even in the presence of high levels of contaminating molecules. The multivalancy of most antigens and antibodies enables them to interact to form a precipitate. Examples of experimental applications that use antibodies are Western Blot, Immunohistochemistry and Immunocytochemistry, Enzyme-Linked Immunosorbent Assay (ELISA), Immunoprecipitation, and Flow Cytometry.

NATURE OF ANTIGEN-ANTIBODY REACTIONS

A. Lock and Key Concept

The combining site of an antibody is located in the Fab portion of the molecule and is constructed from the hypervariable regions of the heavy and light chains. X-Ray crystallography studies of antigen-antibody interactions show that the antigenic determinant nestles in a cleft formed by the combining site of the antibody as illustrated in Figure 1. Thus, our concept of antigen-antibody reactions is one of a key (i.e. the antigen) which fits into a lock (i.e. the antibody).

B. Non-covalent Bonds

The bonds that hold the antigen to the antibody combining site are all non-covalent in nature. These include hydrogen bonds, electrostatic bonds, Van der Waals forces and hydrophobic bonds. Multiple bonding between the antigen and the antibody ensures that the antigen will be bound tightly to the antibody.

C. Reversibility

Since antigen-antibody reactions occur via non-covalent bonds, they are by their nature reversible.

AFFINITY AND AVIDITY

A. Affinity

Antibody affinity is the strength of the reaction between a single antigenic determinant and a single combining site on the antibody. It is the sum of the attractive and repulsive forces operating between the antigenic determinant and the combining site of the antibody. Affinity is the equilibrium constant that describes the antigen-antibody reaction. Most antibodies have a high affinity for their antigens.

B. Avidity

Avidity is a measure of the overall strength of binding of an antigen with many antigenic determinants and multivalent antibodies. Avidity is influenced by both the valence of the antibody and the valence of the antigen. Avidity is more than the sum of the individual affinities. To repeat, affinity refers to the strength of binding between a single antigenic determinant and an individual antibody combining site whereas avidity refers to the overall strength of binding between multivalent antigens and antibodies.

Career Avenues

SPECIFICITY AND CROSS REACTIVITY

A. Specificity

Specificity refers to the ability of an individual antibody combining site to react with only one antigenic determinant or the ability of a population of antibody molecules to react with only one antigen. In general, there is a high degree of specificity in antigen-antibody reactions. Antibodies can distinguish differences in:

- the primary structure of an antigen,
- isomeric forms of an antigen, and
- secondary and tertiary structure of an antigen.

B. Cross reactivity

Cross reactivity refers to the ability of an individual antibody combining site to react with more than one antigenic determinant or the ability of a population of antibody molecules to react with more than one antigen. Cross reactions arise because the cross reacting antigen shares an epitope in common with the immunizing antigen or because it has an epitope which is structurally similar to one on the immunizing antigen (multispecificity).

Career Avenues

REGULATION OF IMMUNE RESPONSES

The effective immune response is an outcome of the interplay between antigen and a network of immunologically competent cells. The immune response is controlled by many mechanisms which restore the immune system to a resting state when responsiveness to a given antigen is no longer required.

Effective immune responses are usually resulted from the interactions between pathogens and a network of immunologic elements. All these responses, which are created as a result of a challenge with infective microbe, are matter to several different control mechanisms. These mechanisms return the immune system to the resting state after removal of the microbe is occurred and the immunologic actions of these elements to the removed microbe are no longer needed.

The immune response, like all biological systems, is subject to a variety of control mechanisms. These mechanisms restore the immune system to a resting state when responsiveness to a given antigen is no longer required. An effective immune response is an outcome of the interplay between antigen and a network of immunological competent cells. The nature immune of immune response, both qualitatively and quantitatively, is determined by many factors, including the form and route of administration of the antigen, the antigen-presenting cell (APC), the genetic background of the individual and any history of previous exposure to the antigen in question or to a cross-reacting antigen. Specific antibodies may also modulate the immune response to an antigen.

- The immune response is subject to a variety of control mechanisms which serve to restore the immune system to a resting state when the response to a given antigen is no longer required.
- Many factors govern the outcome of any immune response. These include the antigen itself, its dose and route of administration, and the genetic background of the individual responding to antigenic challenge.
- Immunoglobulins can influence the immune response positively as anti-idiotype or through immune complex formation. They may also negatively influence immune responses by reducing antigenic challenge or by feedback inhibition of B cells.
- The antigen-presenting cell may affect the immune response through its ability to provide co-stimulation to T cells. Different types of antigen-presenting cell promote different modes of immune response.
- T cells regulate the immune response. Cytokine production by T cells influences the type of immune response elicited by antigen. CD4+T cells can deviate immune responses to TH1 (T-1 helper) or TH2 type responses. Regulatory T cells may belong to the CD4 or CD8 subpopulations. They can inhibit responses by production of suppressive cytokines such as interleukin 10 (IL-10) and transforming growth factor-β (TGFβ).

Career Avenues

- Selective migration of lymphocyte subsets to different sites can modulate the local type of immune response, since TH1 cells and TH2 cells respond to different sets of chemokines.
- Genetic factors which influence the immune system include both MHC-linked and non-MHC linked genes. They affect the level of immune response, susceptibility to infection and autoimmune disease. Defects in many of these genes lead to immunodeficiency or abnormal immune responses.
- The neuroendocrine system influences immune responses. Corticosteroids in particular down- regulate TH1 responses and macrophage activation.

The nature of the immune response both quantitatively and qualitatively is regulated by many factors, these include:

A. REGULATION BY ANTIGENS

Antigens include those which induce immune responses, and those which cause the state of tolerance.

T cells and B cells are triggered by antigen after effective engagement of their antigen-specific receptors together with appropriate co-stimulation. In the case of the T cell, this engagement is not with antigen itself, but with processed antigenic peptide bound to MHC class I or class II molecules on APCs. The nature of an antigen its dose and the route of administration have all been shown to have a profound influence on the outcome of an immune response. An effective immune response removes antigen from the system. Repeated antigen exposure is required to maintain T- and B-cell proliferation of during an effective immune response there is often a dramatic expansion of specifically reactive effector cells. At the end of an immune response reduced antigen exposure results in a reduced expression of IL-2 and its receptor leading to apoptosis of the antigen-specific T cells. The majority of antigen-specific cells therefore die at the end of an immune response and a minor population of long- lived antigen-specific T and B cells survives to give rise to the memory population of antigen-specific cells.

The nature of the antigen influences the type of immune response that occurs

Different antigens elicit different kinds of immune response. Polysaccharide capsule antigens of bacteria generally induce IgM responses, whereas proteins can induce both cell-mediated and humoral immune responses. Intracellular organisms such as some bacteria, parasites or viruses induce a cell-mediated immune response, whereas soluble protein antigens induce a humoral response. A cell-mediated immune response is also induced by agents such as silica. However, some antigens (e.g. those of intracellular microorganisms) may not be cleared so effectively, leading to a sustained immune response that has pathological consequences.

Career Avenues

Large doses of antigen can induce tolerance

Very large doses of antigen often result in specific T- and sometimes B-cell tolerance. It has been shown that administration of antigen to neonatal mice often results in tolerance to this antigen. It has been speculated that this may be the result of the immaturity of the immune system. However, recent studies have shown that neonatal mice can develop efficient immune responses and that non-responsiveness may in some cases be attributable, not to be the immaturity of T cells, but to immune deviation whereby a non-protective type II cytokine response dominated a protective type I response. T-independent polysaccharide antigens have been shown to generate tolerance in B cells after administration in high doses.

The route of administration of an antigen can determine whether or not an immune response occurs.

THE ANTIGEN-PRESENTING CELL

The nature of the APC initially presenting the antigen may determine whether responsiveness or tolerance ensues. Effective activation of T cells requires the expression of co-stimulatory molecules on the surface of the APC. Thus presentations by dendritic cells or activated macrophages, which express high levels of MHC, class II in addition to co-stimulatory molecules, results in highly effective T cell activation.

B. REGULATION BY ANTIBODIES

1. Blocking the antigenic determinants.
2. Antibody-feedback mechanism (increased ab affinity).

C. REGULATION BY IMMUNE COMPLEXES

Immune complexes may enhance or suppress immune responses
One of the ways in which antibody (either IgM or IgG) might act to modulate the immune response involves an Fc-development mechanism ad immune-complex formation with antigen. Immune complexes can inhibit or augment the immune response.

1. **Inhibition**: B-cells Fc receptors are crossed-linked to its ag receptor, it will inhibits the signals of B-cell activation.
2. **Augmentation**: When the immune complex is presented to B-cells via APC.

Career Avenues

D. REGULATION BY LYMPHOCYTES

T-cells clearly modulate the immune response in a positive sense by providing T-cell help. Furthermore, the kind of help which is generated (TH1 or TH2) affects the nature of the immune response, favoring either humoral or cell-mediated immunity. In addition, there is clear evidence that

- T cells are capable of down-regulating immune responses.
- CD4+T cells can prevent the induction of autoimmunity.
- TH cell subsets are involved in the regulation of immunoglobulin production.
- CD8+T cells can transfer resistance and tolerance.

E. REGULATION OF THE IMMUNE RESPONSE BY CD4+T IS A NORMAL PHYSIOLOGICAL PROCESS

The role of such CD4+ or CD8+T cells mediated regulatory effects in normal physiology has been questioned. However, the observation that CD4+T cells which are able to prevent autoimmunity are present in unmanipulated normal animals supports their importance in homeostasis. Peripheral T cell lymphopenia can lead to the development of autoimmune disease.

F. REGULATION BY NK AND NK T-CELLS

Natural killer (NK) cells make cytokines and chemokines and thus play an important role in the innate immune response to infections and tumors. Their production of immunogulatory cytokines and chemokines at early stages in the immune response influences the characteristics of the subsequent adaptive immune response and thus can influence the outcome of the immune response. These cells play a key role in the early immune response to intracellular pathogens.

G. REGULATION BY LOCALIZATION OF CELLS

The spatial and temporal production of chemokines by different cell types is an important mechanism of immune regulation. There is god evidence to suggest that recruitment of TH1 and TH2 cells is differentially controlled thus ensuring the maintenance of locally polarized immune responses. The expression of different chemokine receptors on TH1 cells (CXCR3 and CCR5) and TH2 cells (CCR3, CCR4, CCR8) allows chemotactic signals to produce the differential localization of T cell subsets to sites of inflammation.

Career Avenues

CYTOKINES

T lymphocytes secrete various protein molecules called cytokines. These molecules act as immune-regulatory hormones and play an important roles in the regulation of immune responses.

Regulation of hematopoiesis is accomplished in various ways, which include:

- Control of the levels and types of cytokines produced by bone-marrow stromal cells
- The production of cytokines with hematopoietic activity by other cell types, such as activated T cells and macrophages
- The regulation of the expression of receptors for hematopoietically active cytokines in stem cells and progenitor cells
- The removal of some cells by the controlled induction of cell death.

REGULATION OF SERUM LEVELS

In humans, the **neonatal Fc receptor (FcRN)** transfers IgGs from mother to fetus during gestation and also plays a role in the regulation of IgG serum levels

CTLA-4

CTLA-4 in the regulation of lymphocyte activation and proliferation.

REGULATION OF B-CELL DEVELOPMENT

A number of transcription factors that regulate expression of various gene products at different stages of B-cell development have been identified. Among these are NF-6B, BSAP,Ets-1, c-Jun, Ikaros, Oct-2, Pu.1, EBF, BCF, and E2A.

Career Avenues

IMMUNE TOLERANCE

Immune tolerance or **immunological tolerance** is the process by which the immune system does not attack an antigen. It can be either 'natural' or 'self-tolerance', where the body does not mount an immune response to self-antigens, or 'induced tolerance', where tolerance to external antigens can be created by manipulating the immune system.

- Tolerance is antigenic specific and results from the recognition of antigens by specific lymphocytes.
- Normal individuals are tolerant of their own antigens (self antigen) - Self-tolerance.
- Foreign antigens may be administered in ways that preferentially inhibit immune response by inducing tolerance in specific lymphocytes-antigen induction.
- It is an antigen-induced, active process.
- Like immunologic memory, it is antigen specific, it can exist in B cells, T cells or both ; its easier to induce and lasts longer in T cells than in B cell.

FACTORS AFFECTING TOLERANCE ROLE OF ANTIGEN

FACTORS WHICH AFFECT RESPONSE	FAVOR IMMUNE RESPONSE	FAVOR TOLERANCE
Physical form of antigen	Large aggregated, complex molecules, properly processed	Soluble, aggregate-free, simple small molecule, not processed
Route of injection	Subcutaneous or intramuscular	Oral, or sometimes, intravenous
Dose of antigen	Optimal dose	Very large or very small dose

FACTORS AFFECTING TOLERANCE : THE ROLE OF HOST

FACTORS THAT AFFECT RESPONSE	FAVOR IMMUNE RESPONSE	FAVOR TOLERANCE
Age of responding animal	Fully differentiated; memory T & B cells	Newborn(mice), immunologically immature
Differentiation state of cells	Older, immunologically mature	Relative undifferentiated B cell with only IgM, T cells in the thymic cortex

Career Avenues

It occurs in three forms: central tolerance, peripheral tolerance and acquired tolerance.

CENTRAL TOLERANCE

Central tolerance occurs during lymphocyte development and operates in the thymus and bone marrow. Here, T and B lymphocytes that recognize self-antigens are deleted before they develop into fully immunocompetent cells, preventing autoimmunity. This process is most active in fetal life, but continues throughout life as immature lymphocytes are generated.

In mammals the process occurs in the thymus (T cells) and bone marrow (B cells), when maturing lymphocytes are exposed to self-antigens. Self-antigens are present in both organs due to endogenous expression within the organ and importation of antigen due to circulation from peripheral sites. In the case of T cell central tolerance, additional sources of antigen are made available in the thymus by the action of the transcription factor AIRE.

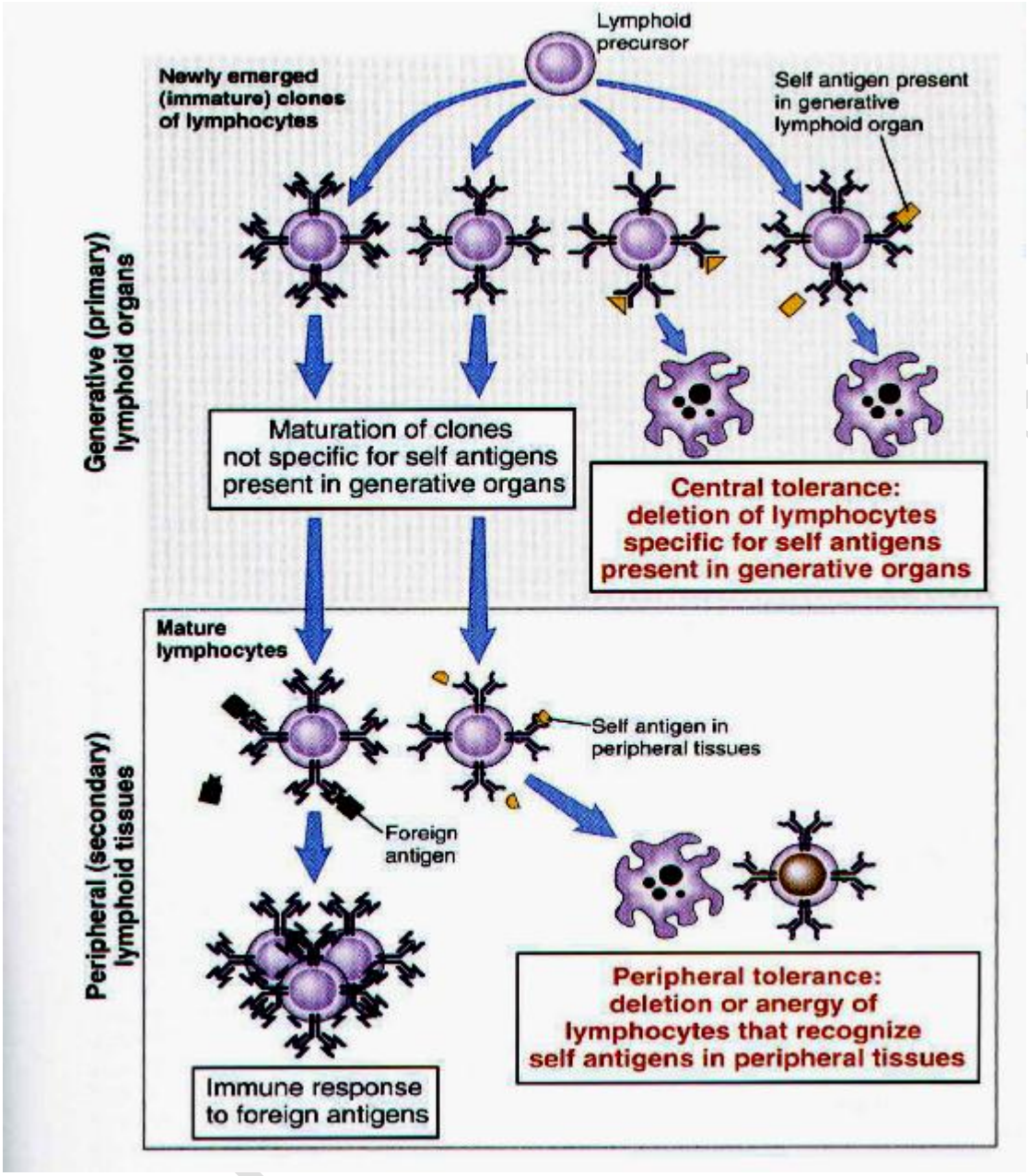
Positive selection occurs first when naive T-cells are exposed to antigens in the thymus. T-cells which have receptors with sufficient affinity for self-MHC molecules are selected. Other cells that do not show sufficient affinity to self-antigens will undergo a deletion process known as **death by neglect** which involves apoptosis of the cells. The positive selection is a classic example of the importance of some degree of auto-reactiveness. This does not occur in B-cells.

Negative selection of T-cells with a very high affinity of self-MHC molecules are induced to anergy, or lineage divergence to form T-regulatory cells.

PERIPHERAL TOLERANCE

Peripheral tolerance is immunological tolerance developed after T and B cells mature and enter the periphery.

Career Avenues



Central and Peripheral tolerance to self antigens: Immature lymphocytes specific for self antigens may encounter these antigens in the generative lymphoid organs and are then deleted. Mature self-reactive lymphocytes may be inactivated or deleted by encounter with self antigens in peripheral tissues. The illustration depicts B lymphocytes, but the same processes occur with T lymphocytes.

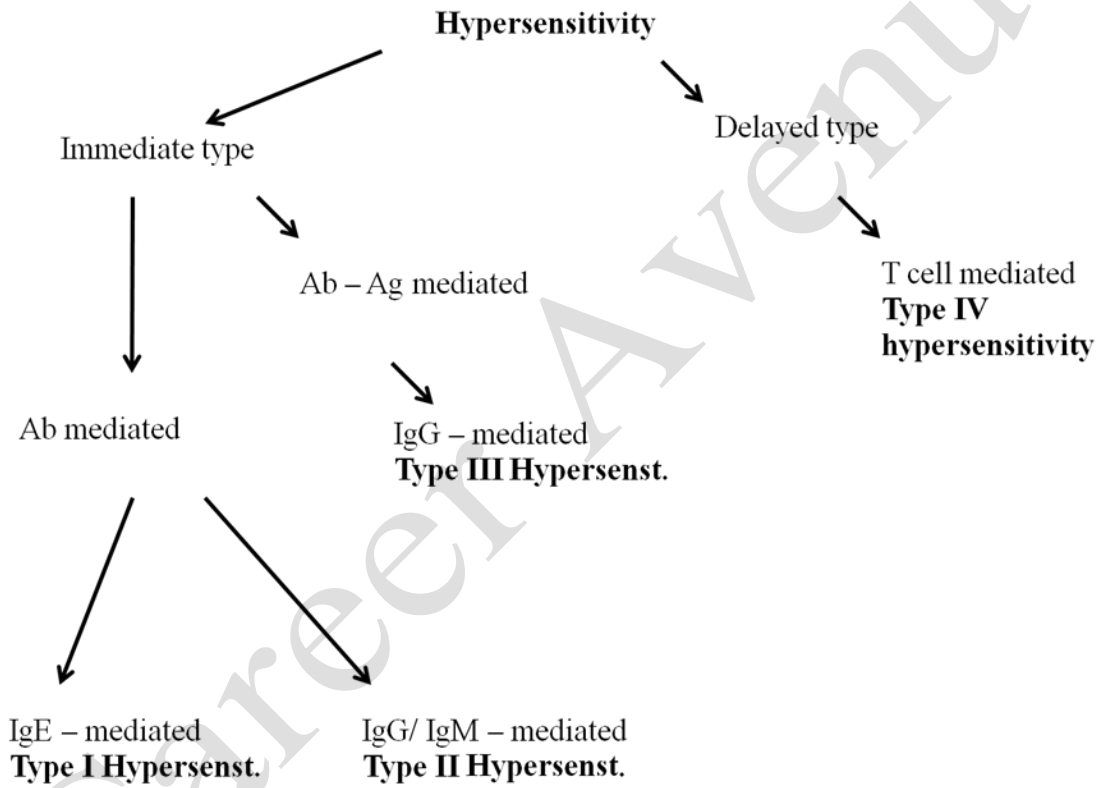
ACQUIRED OR INDUCED TOLERANCE

Acquired or induced tolerance refers to the immune system's adaptation to external antigens characterized by a specific non-reactivity of the lymphoid tissues to a given antigen that in other circumstances would likely induce cell-mediated or humoral immunity. One of the most important natural kinds of acquired tolerance is immune tolerance in pregnancy, where the fetus and the placenta must be tolerated by the maternal immune system.

HYPERSENSITIVITY

Occasionally, the immune system responds inappropriately to the presence of antigen. These responses are refered to as *hypersensitivities*. There are four different types of hypersensitivities that result from different alterations of the immune system. These types are classified as:

- Type I: Immediate Hypersensitivity
- Type II: Cytotoxic Hypersensitivity
- Type III: Immune Complex Hypersensitivity
- Type IV: Delayed Hypersensitivity



TYPE I HYPERSENSITIVITY

Type I or Immediate Hypersensitivity can be illustrated by considering the following experiment:

1. First, a guinea pig is injected intravenously with an antigen. For this example, bovine serum albumin (BSA, a protein) will be used. After two weeks, the same antigen will be re-injected into the same animal. Within a few minutes, the animal begins to suffocate and dies by a process called *anaphylactic shock.*
2. Instead of re-injecting the immunized guinea pig, serum is transferred from this pig to a "naive" (unimmunized) pig. When this second guinea pig is now injected with BSA,

Career Avenues

it also dies of anaphylactic shock. However, if the second pig is injected with a different antigen (e.g. egg white albumin), the pig shows no reaction.

3. If immune cells (T-cells and macrophages instead of serum) are transferred from the immunized pig to a second pig, the result is very different; injection of the second pig with BSA has no effect.

Examples of Type I hypersensitivities include allergies to penicillin, insect bites, molds, etc. A person's sensitivity to these allergens can be tested by a cutaneous reaction. If the specific antigen in question is injected intradermally and the patient is sensitive, a specific reaction known as *wheal and flare* can be observed within 15 minutes. Individuals who are hypersensitive to such allergens must avoid contact with large inocula to prevent anaphylactic shock.

TYPE II HYPERSENSITIVITY

Type II or Cytotoxic Hypersensitivity also involves antibody-mediated reactions. However, the immunoglobulin class (isotype) is generally IgG. In addition, this process involves K-cells rather than mast cells. K-cells are, of course, involved in antibody-dependent cell-mediated cytotoxicity (ADCC). Type II hypersensitivity may also involve complement that binds to cell-bound antibody. The difference here is that the antibodies are specific for (or able to cross-react with) "self" antigens. When these circulating antibodies react with a host cell surface, tissue damage may result.

There are many examples of Type II hypersensitivity. These include:

- **Pemphigus:** IgG antibodies that react with the intracellular substance found between epidermal cells.
- **Autoimmune hemolytic anemia (AHA):** This disease is generally inspired by a drug such as penicillin that becomes attached to the surface of red blood cells (RBC) and acts as hapten for the production of antibody which then binds the RBC surface leading to lysis of RBCs.
- **Goodpasture's syndrome:** Generally manifested as a glomerulonephritis, IgG antibodies that react against glomerular basement membrane surfaces can lead to kidney destruction.

TYPE III HYPERSENSITIVITY

Type III or Immune Complex hypersensitivity involves circulating antibody that reacts with free antigen. These circulating complexes can then become deposited on tissues. Tissue deposition may lead to reaction with complement, causing tissue damage. this type of hypersensitivity develops as a result of systematic exposure to an antigen and is dependent on i) the type of antigen and antibody and ii) the size of the resulting complex. More specifically, complexes that are too small remain in circulation; complexes too large are removed by the glomerulus; intermediate complexes may become lodged in the glomerulus leading to kidney damage.

One example of a Type III hypersensitivity is **serum sickness**, a condition that may develop when a patient is injected with a large amount of e.g. antitoxin that was produced in an animal. After about 10 days, anti-antitoxin antibodies react with the antitoxin forming

Career Avenues

immune complexes that deposit in tissues. Type III hypersensitivities can be ascertained by intradermal injection of the antigen, followed by the observance of an "Arthus" reaction (swelling and redness at site of injection) after a few hours.

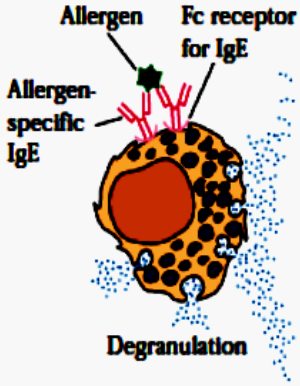
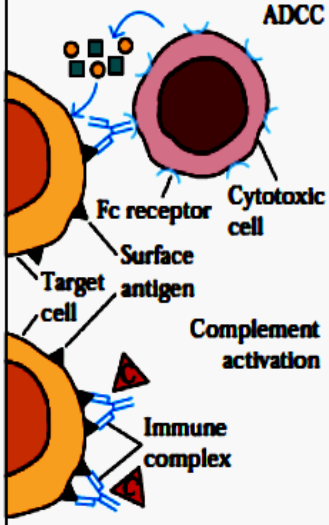
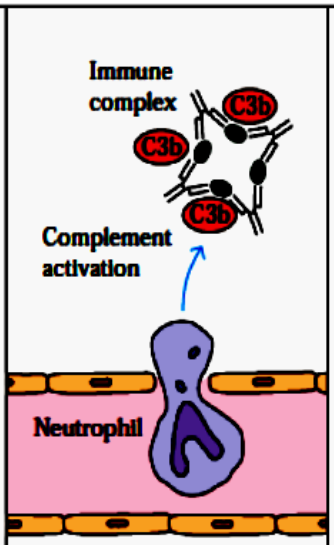
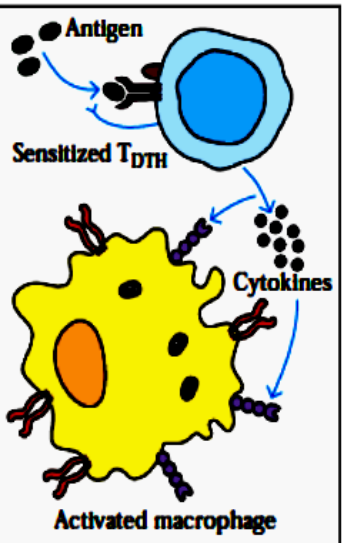
TYPE IV HYPERSENSITIVITY

Type IV or Delayed Hypersensitivity can be illustrated by considering the following experiment:

1. First, a guinea pig is injected with a sub-lethal dose of *Mycobacterium tuberculosis* (MT). Following recovery of the animal, injection of a lethal dose of MT under the skin produces only erythema (redness) and induration (hard spot) at the site of injection 1-2 days later.
2. Instead of reinjecting the immunized guinea pig, serum is transferred from this pig to a "naive" (unimmunized) pig. When this second guinea pig is now injected with MT, it dies of the infection.
3. If immune cells (T-cells and macrophages instead of serum) are transferred from the immunized pig to a second pig, the result is very different; injection of the second pig with MT causes only erythema and induration at the site of injection 1-2 days later.
4. In a separate experiment, if the immunized guinea pig is injected with a lethal dose of *Listeria monocytogenes* (LM) instead of MT, it dies of the infection. However, if the pig is simultaneously injected with both LM and MT, it survives.

Career Avenues

COMPARATIVE REPRESENTATION OF ALL TYPE OF HYPERSENSITIVITY

 **Type I**	 **Type II**	 **Type III**	 **Type IV**
IgE-Mediated Hypersensitivity	**IgG-Mediated Cytotoxic Hypersensitivity**	**Immune Complex-Mediated Hypersensitivity**	**Cell-Mediated Hypersensitivity**
Ag induces crosslinking of IgE bound to mast cells and basophils with release of vasoactive mediators	**Ab directed against cell surface antigens meditates cell destruction via complement activation or ADCC**	**Ag-Ab complexes deposited in various tissues induce complement activation and an ensuing inflammatory response mediated by massive infiltration of neutrophils**	**Sensitized T_H1 cells release cytokines that activate macrophages or T_C cells which mediate direct cellular damage**
Typical manifestations include systemic anaphylaxis and localized anaphylaxis such as hay fever, asthma, hives, food allergies, and eczema	**Typical manifestations include blood transfusion reactions, erythroblastosis fetalis, and autoimmune hemolytic anemia**	**Typical manifestations include localized Arthus reaction and generalized reactions such as serum sickness, necrotizing vasculitis, glomerulnephritis, rheumatoid arthritis, and systemic lupus erythematosus**	**Typical manifestations include contact dermatitis, tubercular lesions and graft rejection**

Graft Versus Host Reaction

IMMUNOLOGIC BASIS OF GRAFT REJECTION

The degree of immune response to a graft varies with the type of graft. The following terms are used to denote different types of transplants:

- **Autograft** is self-tissue transferred from one body site to another in the same individual. Transferring healthy into a burned area in burn patients and use of healthy blood vessels to replace blocked coronary arteries are examples of frequently used autografts.
- **Xenograft** is grafts between members of different species (also known as heterologous, xenogeneic or heterografts).

- **Allograft** is grafts between two members of the same species (also known as allogeneic or homograft).
- **Isograft** is grafts between members of the same species with identical genetic makeup (identical twins or inbred animals).

Autografts and isografts are usually accepted, owing to the genetic identity between graft and host.

PRINCIPLES OF TRANSPLANTATION

An immunocompetent host recognizes the foreign antigens on grafted tissues (or cells) and mounts an immune response which results in rejection. On the other hand, if an immunocompromised host is grafted with foreign immunocompetent lymphoid cells, the immunoreactive T-cells in the graft recognize the foreign antigens on the host tissue, leading to damage of the host tissue.

HOST-VERSUS-GRAFT-REACTION

The duration of graft survival follows the order, xeno- < allo- < iso- = auto- graft. The time of rejection also depends on the antigenic disparity between the donors and recipient. MHC antigens are the major contributors in rejection, but the minor histocompatibility antigens also play a role. Rejection due to disparity in several minor histocompatibility antigens may be as quick or quicker than rejection mediated by an MHC antigen. As in other immune responses, there is immunological memory and secondary response in graft rejection. Thus, once a graft is rejected by a recipient, a second graft from the same donor, or a donor with the same histocompatibility antigens, will be rejected in a much shorter time.

GRAFT-VERSUS-HOST (GVH) REACTION

Histocompatible lymphoid cells, when injected into an immunocompromised host, are readily accepted. However, the immunocompetent T lymphocytes among the grafted cells recognize the alloantigens and, in response, they proliferate and progressively cause damage to the host tissues and cells. This condition is known as graft-versus-host (GVH) disease and is often fatal.

Career Avenues

AUTO-IMMUNITY

Autoimmunity can be defined as breakdown of mechanisms responsible for self-tolerance and induction of an immune response against components of the self. In numerous autoimmune diseases it is well recognized that products of the immune system cause damage to the self.

AID can be passively transferred by the transfer of auto antibody and sensitised lymphocyte.

CAUSE OF AID

SEQUESTERED ANTIGEN

Lymphoid cells may not be exposed to some self-antigens during their differentiation, because they may be late-developing antigens or may be confined to specialized organs (*e.g*., testes, sperm, brain, eye, *etc*.). A release of antigens from these organs resulting from accidental traumatic injury or surgery can result in the stimulation of an immune response and initiation of an autoimmune disease.

NEO ANTIGEN

Neo antigens are altered or modified antigens. These antigen are modified by physical agent (i.e. irridiation), chemical agents (i.e. drugs) or microbial agents as intracellular viruses.

The disease produced by neo antigens: dermatitis, anemia, leukopenia and thrombocytopenia.

CROSS-REACTIVE ANTIGEN

Many species share organ specific antigen. For example antigen of human brain and that of sheep brain is cross reactive antigen.

LOSS OF IMMUNE REGULATION

Over reactivity or lower reactivity of T and B cells has been reported to cause AID.

Classification of auto immune diseases:

- Haemolytic auto immune disease
- Localised auto immune disease
- Systemic auto immune disease

A) HAEMOLYTIC AUTO IMMUNE DISEASE

It causes destruction of component of blood. For example: haemolytic anemia and thrombocytopenia.

Career Avenues

Haemolytic anemia is antibody mediated AID. It involves two kind of antibodies; cold antibody and warm antibody. Cold antibodies are of IgM type and active at 4oC while warm antibodies are of IgG type and activate at 40oC.

Thrombocytopenia is also an antibody mediated AID. This disease is due to generation of anti-platelets antibody.

B) LOCALIZED AUTO IMMUNE DISEASE

Example: Thyrotoxicosis or Grave's disease (due to over production of thyroxin)

Addison's disease and myasthenia gravis (due to production of antibody against muscles)

C) SYSTEMIC AUTO IMMUNE DISEASE

Example: Rheumatoid arthritis: it is Ig M mediated AID and has increased level of T cels and macrophages.

Hashimoto's disease: due to hyposecretion of thyroxin. It is T cell mediated AID and involves IgG and Ig M type antibodies.

DIAGNOSIS

Diagnosis of autoimmune diseases is based on symptoms and detection of antibodies (and/or very early T cells) reactive against antigens of tissues and cells involved. Antibodies against cell/tissue associated antigens are detected by immunofluorescence. Antibodies against soluble antigens are normally detected ELISA or radioimmunoassay (see table above). In some cases, a biological /biochemical assay may be used (*e.g.*, Graves diseases, pernicious anemia).

TREATMENT

The goals of treatment of autoimmune disorders are to reduce symptoms and control the autoimmune response while maintaining the body's ability to fight infections. Treatments vary widely and depend on the specific disease and symptoms: Anti-inflammatory (corticosteroid) and immunosuppressive drug therapy (such as cyclophosphamide, azathioprine, cyclosporine) is the present method of treating autoimmune diseases. Extensive research is being carried out to develop innovative treatments which include: anti-TNF alpha therapy against arthritis, feeding antigen orally to trigger tolerance, anti-idiotype antibodies, antigen peptides, anti-IL2 receptor antibodies, anti-CD4 antibodies, anti-TCR antibodies, etc.

Career Avenues



GATE QUESTIONS

GATE 2007

Q. 14 Systemic lupus erythematosus (SLE), an autoimmune disease, is characterized by the presence of

(A) Anti-DNA antibodies
(B) Anti-thyroglobulin antibodies
(C) Anti-insulin antibodies
(D) Anti-collagen antibodies

Answers:

14- A.

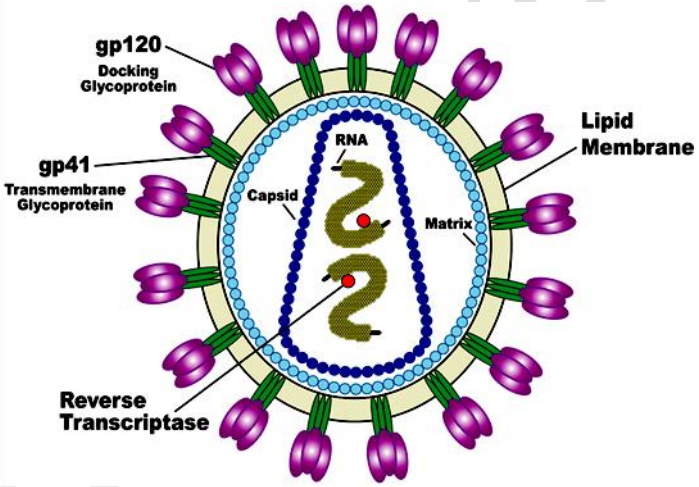
Career Avenues

HIV

STRUCTURE

HIV is different in structure from other retroviruses. It is around 120 nm in diameter (around 60 times smaller than a red blood cell) and roughly spherical.

HIV-1 is composed of two copies of single-stranded RNA enclosed by a conical capsid comprising the viral protein p24, typical of lentiviruses (Figure 1). The RNA component is 9749 nucleotides long. This is in turn surrounded by a plasma membrane of host-cell origin. The single-strand RNA is tightly bound to the nucleocapsid proteins, p6, p7 and enzymes that are indispensable for the development of the virion, such as reverse transcriptase and integrase. The nucleocapsid (p7 and p6) associates with the genomic RNA (one molecule per hexamer) and protects the RNA from digestion by nucleases. A matrix composed of an association of the viral protein p17 surrounds the capsid, ensuring the integrity of the virion particle. Also enclosed within the virion particle are Vif, Vpr, Nef, p7 and viral Protease .The envelope is formed when the capsid buds from the host cell, taking some of the host-cell membrane with it. The envelope includes the glycoproteins gp120 and gp41.



As a result of its role in virus-cell attachment, the structure of the virus envelope spike, consisting of gp120 and gp41, is of particular importance. It is hoped that determining the envelope spike's structure would contribute to scientific understanding of the virus and its replication cycle, and help in the creation of a cure. The first model of its structure was compiled in 2006 using cryo-electron microscopy and suggested that three copies of gp120-gp41 heterodimers are thought to form a trimer as the envelope spike. However, published shortly after was evidence for a single-stalk "mushroom" model, with a head consisting of a trimer gp120s and a gp41 stem, which appears as a compact structure with no obvious separation between the three monomers, anchoring it to the envelope. There are various possibilities as to the source of this difference, as it is unlikely that the viruses imaged by the two groups were structurally different. More recently, further evidence backing up the heterodimer trimer-based model has been found

GENOME ORGANIZATION

HIV has several major genes coding for structural proteins that are found in all retroviruses, and several nonstructural ("accessory") genes that are unique to HIV. The *gag* gene provides the basic physical infrastructure of the virus, and *pol* provides the basic mechanism by which retroviruses reproduce, while the others help HIV to enter the host cell and enhance its reproduction. Though they may be altered by mutation, all of these genes except *tev* exist in all known variants of HIV.

- gag (group-specific antigen): codes for the Gag polyprotein, which is processed during maturation to MA (matrix protein, p17); CA (capsid protein, p24); SP1 (spacer peptide 1, p2); NC (nucleocapsid protein, p7); SP2 (spacer peptide 2, p1) and p6.
- pol: codes for viral enzymesreverse transcriptase, integrase, and HIV protease.
- env (for "envelope"): codes for gp160, the precursor to gp120 and gp41, proteins embedded in the viral envelope which enable the virus to attach to and fuse with target cells.
- Transactivators: tat, rev, vpr
- Other regulators: vif, nef, vpu
- *tev*: This gene is only present in a few HIV-1 isolates. It is a fusion of parts of the *tat*, *env*, and rev genes, and codes for a protein with some of the properties of tat, but little or none of the properties of rev.

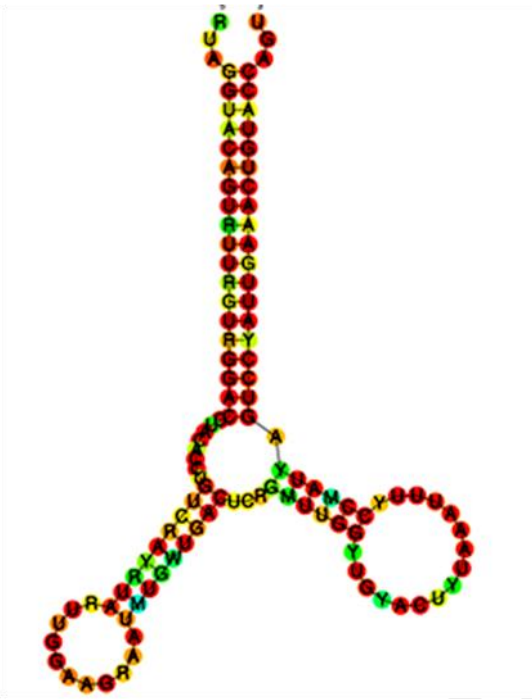
RNA SECONDARY STRUCTURE

Several conserved secondary structure elements have been identified within the HIV RNA genome. These include the trans-activating responsive (TAR) element located within the 5' end of the genome and the HIV Rev response element (RRE) within the env gene. Another RNA structure that has been identified is gag stem loop 3 (GSL3), thought to be involved in viral packaging.[13][14] RNA secondary structures have been proposed to affect the HIV life cycle by altering the function of HIV protease and reverse transcriptase, although not all elements identified have been assigned a function.

An RNA secondary structure determined by 2' hydroxyl acetylation and primer extension (SHAPE) analysis has shown to contain three stem loops and is located between the HIV protease and reverse transcriptase genes. This *cis* regulatory RNA has been shown to be conserved throughout the HIV family and is thought to influence the viral life cycle.[15]

The complete structure of an HIV-1 genome, extracted from infectious virions has been solved to single-nucleotide resolution

Career Avenues



HIV LIFE CYCLE

A) ENTRY

HIV can only replicate (make new copies of itself) inside human cells. The process typically begins when a virus particle bumps into a cell that carries on its surface a special protein called CD4. The spikes on the surface of the virus particle stick to the CD4 and allow the viral envelope to fuse with the cell membrane. The contents of the HIV particle are then released into the cell, leaving the envelope behind.

B) REVERSE TRANSCRIPTION AND INTEGRATION

Once inside the cell, the HIV enzyme reverse transcriptase converts the viral RNA into DNA, which is compatible with human genetic material. This DNA is transported to the cell's nucleus, where it is spliced into the human DNA by the HIV enzyme integrase. Once integrated, the HIV DNA is known as provirus.

C) TRANSCRIPTION AND TRANSLATION

HIV provirus may lie dormant within a cell for a long time. But when the cell becomes activated, it treats HIV genes in much the same way as human genes. First it converts them into messenger RNA (using human enzymes). Then the messenger RNA is transported outside the nucleus, and is used as a blueprint for producing new HIV proteins and enzymes.

Career Avenues

D) ASSEMBLY, BUDDING AND MATURATION

Among the strands of messenger RNA produced by the cell are complete copies of HIV genetic material. These gather together with newly made HIV proteins and enzymes to form new viral particles, which are then released from the cell. The enzyme protease plays a vital role at this stage of the HIV life cycle by chopping up long strands of protein into smaller pieces, which are used to construct mature viral cores.

The newly matured HIV particles are ready to infect another cell and begin the replication process all over again. In this way the virus quickly spreads through the human body. And once a person is infected, they can pass HIV on to others in their bodily fluids.

Career Avenues

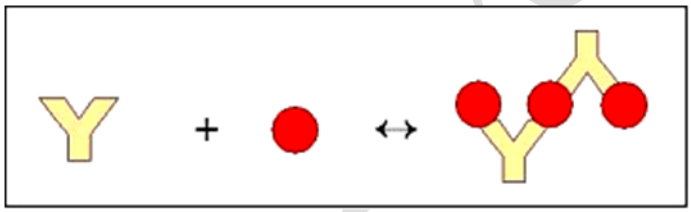
PRINCIPLE AND TECHNIQUES OF IMMUNOLOGY

A) AGGLUTINATION TESTS

AGGLUTINATION/HEMAGGLUTINATION

When the antigen is particulate, the reaction of an antibody with the antigen can be detected by agglutination (clumping) of the antigen. The general term agglutinin is used to describe antibodies that agglutinate particulate antigens. When the antigen is an erythrocyte the term hemagglutination is used. All antibodies can theoretically agglutinate particulate antigens but IgM, due to its high valence, is particularly good agglutinin and one sometimes infers that an antibody may be of the IgM class if it is a good agglutinating antibody.

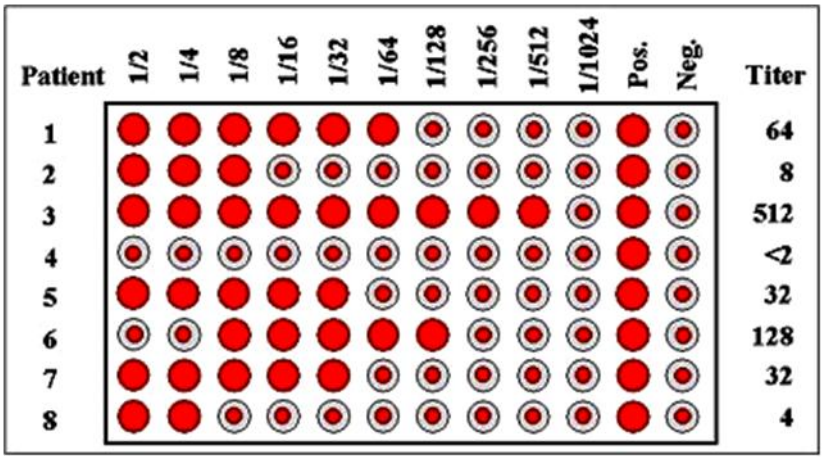
Qualitative agglutination test: Agglutination tests can be used in a qualitative manner to assay for the presence of an antigen or an antibody. The antibody is mixed with the particulate antigen and a positive test is indicated by the agglutination of the particulate antigen.



For example, a patient's red blood cells can be mixed with antibody to a blood group antigen to determine a person's blood type. In a second example, a patient's serum is mixed with red blood cells of a known blood type to assay for the presence of antibodies to that blood type in the patient's serum.

Quantitative agglutination test: Agglutination tests can also be used to measure the level of antibodies to particulate antigens. In this test, serial dilutions are made of a sample to be tested for antibody and then a fixed number of red blood cells or bacteria or other such particulate antigen is added. Then the maximum dilution that gives agglutination is determined. The maximum dilution that gives visible agglutination is called the titer. The results are reported as the reciprocal of the maximal dilution that gives visible agglutination. Following figure illustrates a quantitative hemagglutination test.

Career Avenues



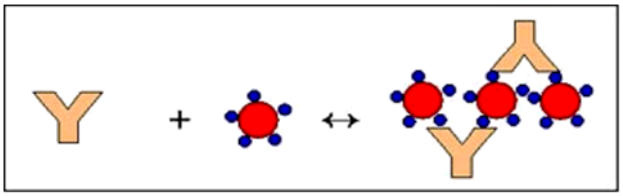
Prozone effect - Occasionally, it is observed that when the concentration of antibody is high (i.e. lower dilutions), there is no agglutination and then, as the sample is diluted, agglutination occurs (See Patient 6 in above given figure). The lack of agglutination at high concentrations of antibodies is called the prozone effect. Lack of agglutination in the prozone is due to antibody excess resulting in very small complexes that do not clump to form visible agglutination.

APPLICATIONS OF AGGLUTINATION TESTS

- Determination of blood types or antibodies to blood group antigens.
- To assess bacterial infections

For example, a rise in titer of an antibody to a particular bacterium indicates an infection with that bacterial type. N.B. a fourfold rise in titer is generally taken as a significant rise in antibody titer.
Practical considerations: Although the test is easy to perform, it is only semi-quantitative.

Passive hemagglutination: The agglutination test only works with particulate antigens. However, it is possible to coat erythrocytes with a soluble antigen (e.g. viral antigen, a polysaccharide or a hapten) and use the coated red blood cells in an agglutination test for antibody to the soluble antigen (Following figure). This is called passive hemagglutination. The test is performed just like the agglutination test. Applications include detection of antibodies to soluble antigens and detection of antibodies to viral antigens.

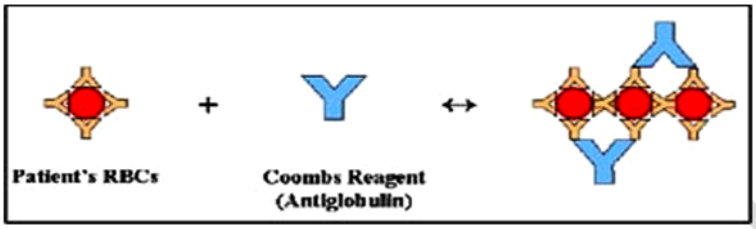


COOMB'S TEST (ANTIGLOBULIN TEST)

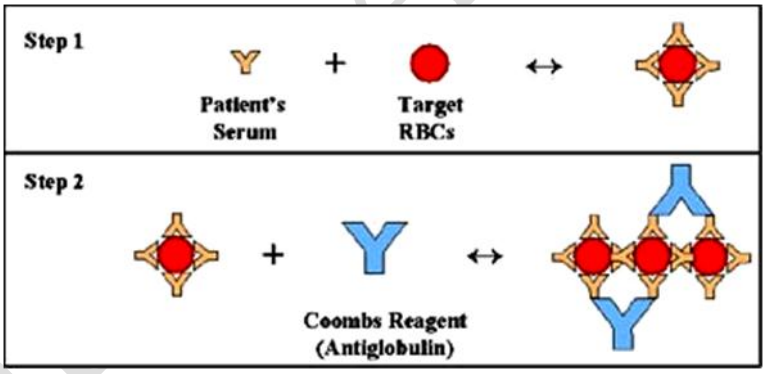
Direct Coomb's Test: When antibodies bind to erythrocytes, they do not always result in agglutination. This can result from the antigen/antibody ratio being in antigen excess or antibody excess or in some cases electrical charges on the red blood cells preventing the

Career Avenues

effective cross linking of the cells. These antibodies that bind to but do not cause agglutination of red blood cells are sometimes referred to as incomplete antibodies. In no way is this meant to indicate that the antibodies are different in their structure, although this was once thought to be the case. Rather, it is a functional definition only. In order to detect the presence of non-agglutinating antibodies on red blood cells, one simply adds a second antibody directed against the immunoglobulin (antibody) coating the red cells. This anti-immunoglobulin can now cross link the red blood cells and result in agglutination. This test is illustrated in following figure and is known as the Direct Coomb's test.



Indirect Coomb's Test: If it is necessary to know whether a serum sample has antibodies directed against a particular red blood cell and you want to be sure that you also detect potential non- agglutinating antibodies in the sample, an Indirect Coomb's test is performed (Following figure). This test is done by incubating the red blood cells with the serum sample, washing out any unbound antibodies and then adding a second anti-immunoglobulin reagent to cross link the cells.



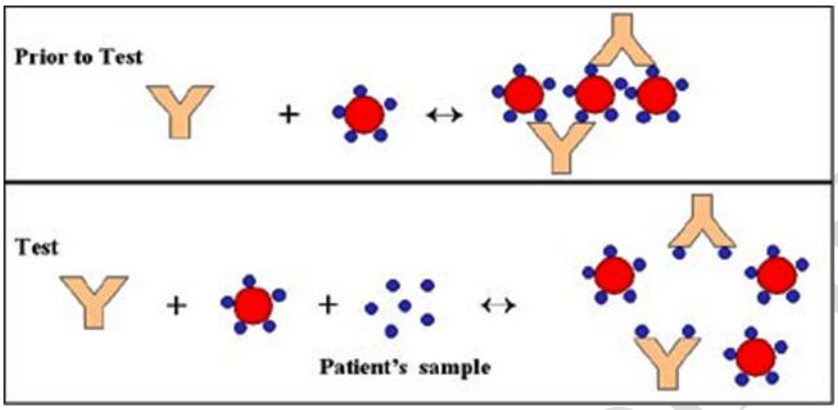
APPLICATIONS

These include detection of anti-rhesus factor (Rh) antibodies. Antibodies to the Rh factor generally do not agglutinate red blood cells. Thus, red cells from Rh^+ children born to Rh^- mothers, who have anti-Rh antibodies, may be coated with these antibodies. To check for this, a direct Coombs test is performed. To see if the mother has anti-Rh antibodies in her serum an Indirect Coombs test is performed.

Hemagglutination Inhibition: The agglutination test can be modified to be used for the measurement of soluble antigens. This test is called hemagglutination inhibition. It is called hemagglutination inhibition because one measures the ability of soluble antigen to inhibit the agglutination of antigen-coated red blood cells by antibodies. In this test, a fixed amount of

Career Avenues

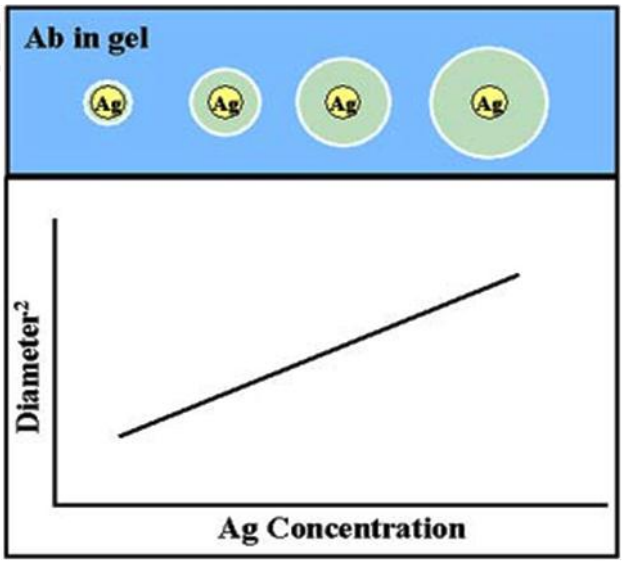
antibodies to the antigen in question is mixed with a fixed amount of red blood cells coated with the antigen (see passive hemagglutination above). Also included in the mixture are different amounts of the sample to be analyzed for the presence of the antigen. If the sample contains the antigen, the soluble antigen will compete with the antigen coated on the red blood cells for binding to the antibodies, thereby inhibiting the agglutination of the red blood cells as illustrated in following figure.



By serially diluting the sample, you can quantitate the amount of antigen in your unknown sample by its titer. This test is generally used to quantitate soluble antigens and is subject to the same practical considerations as the agglutination test.

B) PRECIPITATION TESTS

Radial Immunodiffusion (Mancini): In radial immunodiffusion antibody is incorporated into the agar gel as it is poured and different dilutions of the antigen are placed in holes punched into the agar. As the antigen diffuses into the gel, it reacts with the antibody and when the equivalence point is reached a ring of precipitation is formed as illustrated in following figure.



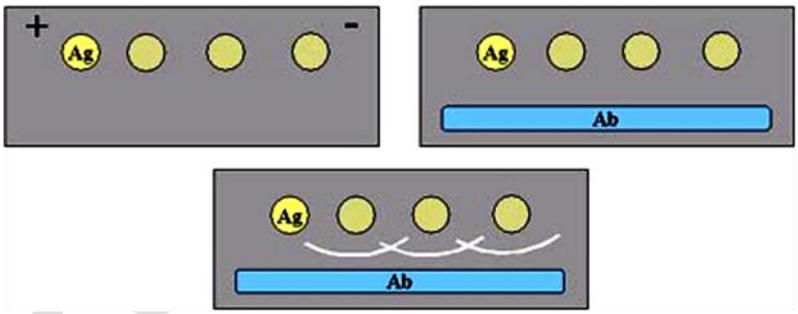
Career Avenues

The diameter of the ring is proportional to the log of the concentration of antigen since the amount of antibody is constant. Thus, by running different concentrations of a standard antigen one can generate a standard cure from which one can quantitate the amount of an antigen in an unknown sample. Thus, this is a quantitative test. If more than one ring appears in the test, more than one antigen/antibody reaction has occurred. This could be due to a mixture of antigens or antibodies. This test is commonly used in the clinical laboratory for the determination of immunoglobulin levels in patient samples.

IMMUNOELECTROPHORESIS

In immunoelectrophoresis, a complex mixture of antigens is placed in a well punched out of an agar gel and the antigens are electrophoresed so that the antigen are separated according to their charge. After electrophoresis, a trough is cut in the gel and antibodies are added. As the antibodies diffuse into the agar, precipitin lines are produced in the equivalence zone when an antigen/antibody reaction occurs as illustrated in following figure.

This tests is used for the qualitative analysis of complex mixtures of antigens, although a crude measure of quantity (thickness of the line) can be obtained. This test is commonly used for the analysis of components in a patient' serum. Serum is placed in the well and antibody to whole serum in the trough. By comparisons to normal serum, one can determine whether there are deficiencies on one or more serum components or whether there is an overabundance of some serum component (thickness of the line). This test can also be used to evaluate purity of isolated serum proteins.



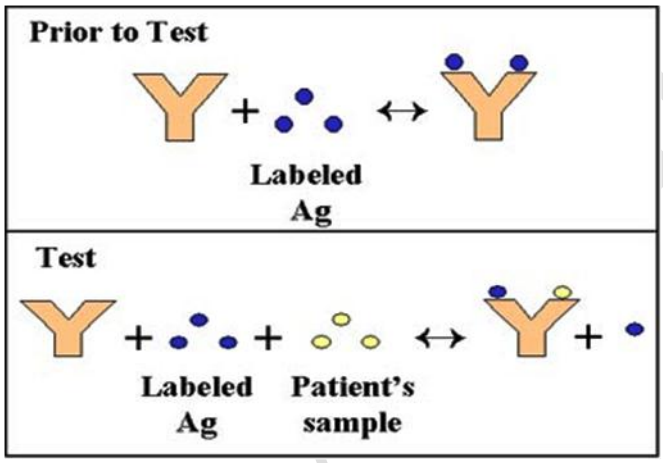
Countercurrent electrophoresis: In this test the antigen and antibody are placed in wells punched out of an agar gel and the antigen and antibody are electrophoresed into each other where they form a precipitation line as illustrated in following figure. This test only works if conditions can be found where the antigen and antibody have opposite charges. This test is primarily qualitative, although from the thickness of the band you can get some measure of quantity. Its major advantage is its speed.



Radioimmunoassay (RIA)/Enzyme Linked Immunosorbent Assay (ELISA)

Radioimmunoassays (RIA) are assays that are based on the measurement of radioactivity associated with immune complexes. In any particular test, the label may be on either the antigen or the antibody. Enzyme Linked Immunosorbent Assays (ELISA) are those that are based on the measurement of an enzymatic reaction associated with immune complexes. In any particular assay, the enzyme may be linked to either the antigen or the antibody.

Competitive RIA/ELISA for Ag Detection: The method and principle of RIA and ELISA for the measurement of antigen is shown in following figure. By using known amounts of a standard unlabeled antigen, one can generate a standard curve relating radioactivity (cpm) (Enzyme) bound versus amount of antigen. From this standard curve, one can determine the amount of an antigen in an unknown sample.



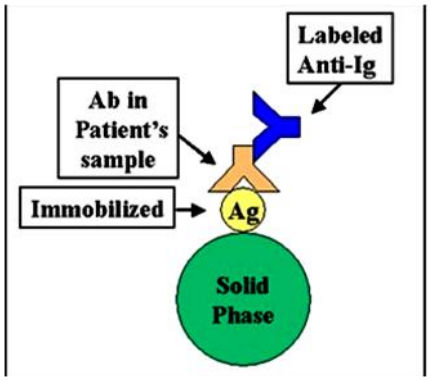
The key to the assay is the separation of the immune complexes from the remainder of the components. This has been accomplished in many different ways and serves as the basis for the names given to the assay:

Precipitation with ammonium sulphate: Ammonium sulphate (33-50% final concentration) will precipitate immunoglobulins but not many antigens. Thus, this can be used to separate the immune complexes from free antigen. This has been called the Farr Technique.

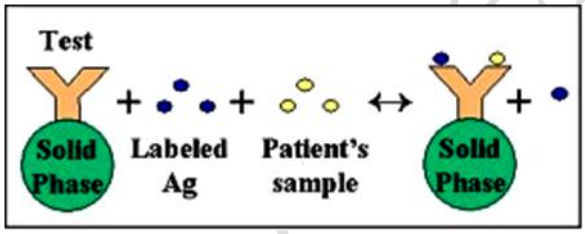
Anti-immunoglobulin antibody: The addition of a second antibody directed against the first antibody can result in the precipitation of the immune complexes and thus the separation of the complexes from free antigen.

Immobilization of the Antibody: The antibody can be immobilized onto the surface of a plastic bead or coated onto the surface of a plastic plate and thus the immune complexes can easily be separated from the other components by simply washing the beads or plate (following figure 17). This is the most common method used today and is referred to as Solid phase RIA or ELISA. In the clinical laboratory, competitive RIA and ELISA are commonly used to quantitate serum proteins, hormones, drugs metabolites.

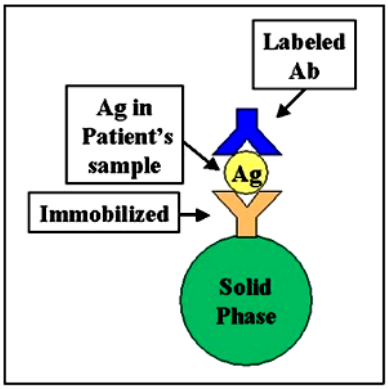
Career Avenues



Non-competitive RIA/ELISA for Ag or Ab: Non-competitive RIA and ELISAs are also used for the measurement of antigens and antibodies. In given figure, the bead is coated with the antigen and is used for the detection of antibody in the unknown sample.



The amount of labeled second antibody bound is related to the amount of antibody in the unknown sample. This assay is commonly employed for the measurement of antibodies of the IgE class directed against particular allergens by using a known allergen as antigen and anti-IgE antibodies as the labeled reagent. It is called the RAST test (radioallergosorbent test). In following figure, the bead is coated with antibody and is used to measure an unknown antigen. The amount of labeled second antibody that binds is proportional to the amount of antigen that bound to the first antibody.



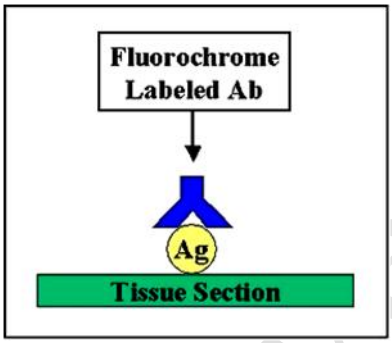
Career Avenues

C) TESTS FOR CELL ASSOCIATED ANTIGENS

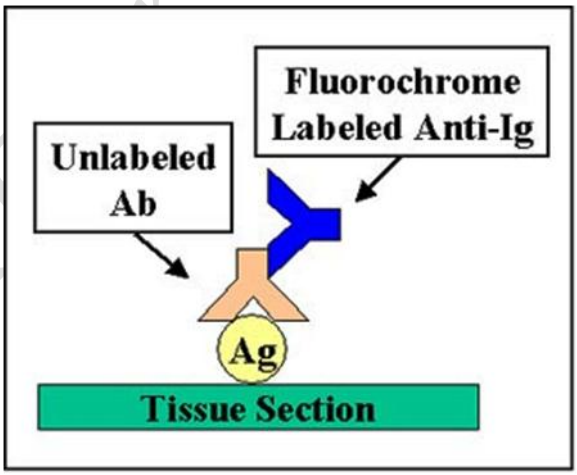
Immunofluorescence

Immunofluorescence is a technique whereby an antibody labeled with a fluorescent molecule (fluorescein or rhodamine or one of many other fluorescent dyes) is used to detect the presence of an antigen in or on a cell or tissue by the fluorescence emitted by the bound antibody.

Direct Immunofluorescence: In direct immunofluorescence, the antibody specific to the antigen is directly tagged with the fluorochrome (following figure).



Indirect Immunofluorescence: In indirect immunofluorescence, the antibody specific for the antigen is unlabeled and a second anti-immunoglobulin antibody directed toward the first antibody is tagged with the fluorochrome (following figure). Indirect fluorescence is more sensitive than direct immunofluorescence since there is amplification of the signal.

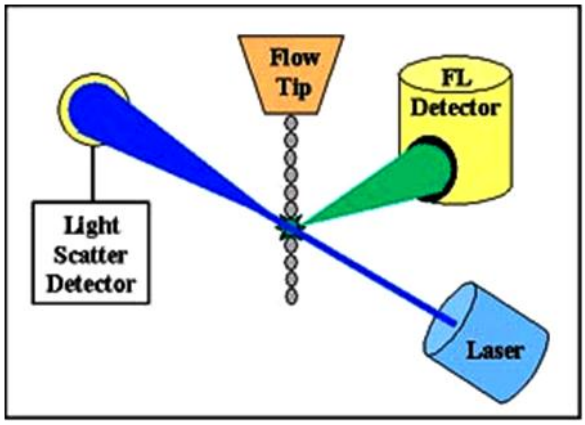


Flow Cytometry: Flow cytometry is commonly used in the clinical laboratory to identify and enumerate cells bearing a particular antigen. Cells in suspension are labeled with a fluorescent tag by either direct or indirect immunofluorescence. The cells are then analyzed on the flow cytometer.

Following figure illustrates the principle of flow cytometry. In a flow cytometer, the cells exit a flow cell and are illuminated with a laser beam. The amount of laser light that is scattered off the cells as they passes through the laser can be measured, which gives information

Career Avenues

concerning the size of the cells. In addition, the laser can excite the fluorochrome on the cells and the fluorescent light emitted by the cells can be measured by one or more detectors.



Enzyme-Linked Immunosorbent Assay

Enzyme-linked immunosorbent assay, commonly known as **ELISA** (or EIA), is similar in principle to RIA but depends on an enzyme rather than a radioactive label. An enzyme conjugated with an antibody reacts with a colorless substrate to generate a colored reaction product. Such a substrate is called **a chromogenic substrate.** A number of enzymes have been employed for ELISA, including alkaline phosphatase, Horse radish peroxidase, and β-galactosidase. These assays approach the sensitivity of RIAs and have the advantage of being safer and less costly.

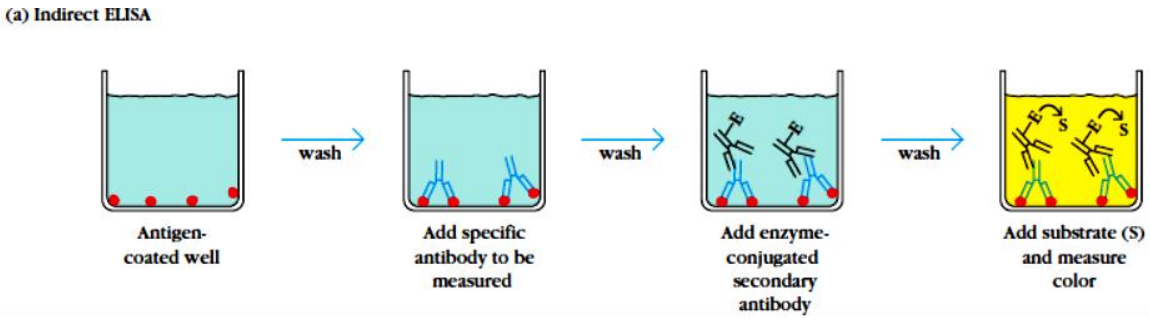
There are numerous variants of ELISA. A number of variations of ELISA have been developed, allowing qualitative detection or quantitative measurement of either antigen or antibody. Each type of ELISA can be used qualitatively to detect the presence of antibody or antigen. Alternatively, a standard curve based on known concentrations of antibody or antigen is prepared, from which the unknown concentration of a sample can be determined.

INDIRECT ELISA

Antibody can be detected or quantitatively determined with an indirect ELISA. Serum or some other sample containing primary antibody (Ab1) is added to an antigen-coated microtiter well and allowed to react with the antigen attached to the well. After any free Ab1 is washed away, the presence of antibody bound to the antigen is detected by adding an enzyme-conjugated secondary anti-isotype antibody (Ab2), which binds to the primary antibody. Any free Ab2 then is washed away, and a substrate for the enzyme is added. The amount of colored reaction product that forms is measured by specialized spectrophotometric plate readers, which can measure the absorbance of all of the wells of a 96-well plate in seconds.

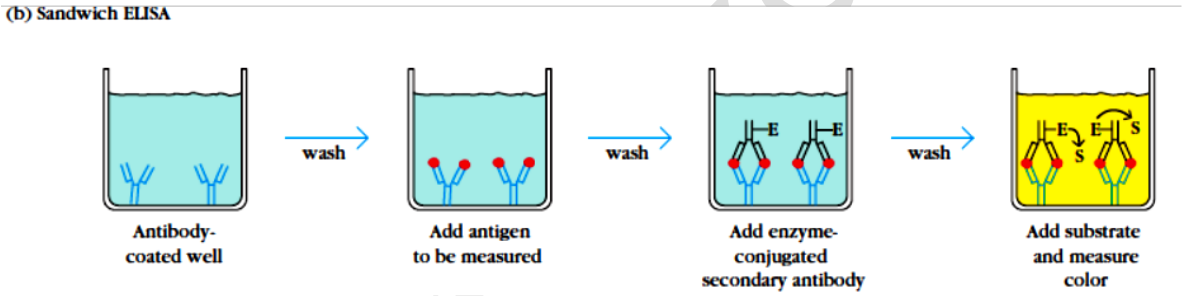
Indirect ELISA is the method of choice to detect the presence of serum antibodies against human immunodeficiency virus (HIV), the causative agent of AIDS. In this assay, recombinant envelope and core proteins of HIV are adsorbed as solid-phase antigens to microtiter wells. Individuals infected with HIV will produce serum antibodies to epitopes on these viral proteins. Generally, serum antibodies to HIV can be detected by indirect ELISA within 6 weeks of infection.

Career Avenues



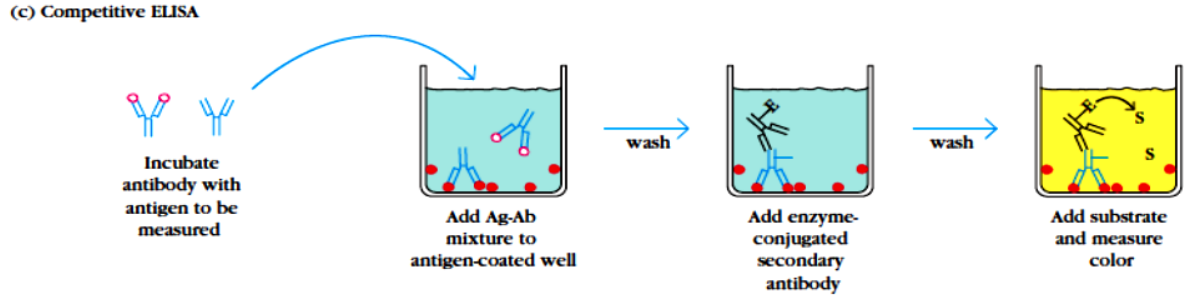
SANDWICH ELISA

Antigen can be detected or measured by a sandwich ELISA. In this technique, the antibody (rather than the antigen) is immobilized on a microtiter well. A sample containing antigen is added and allowed to react with the immobilized antibody. After the well is washed, a second enzyme- linked antibody specific for a different epitope on the antigen is added and allowed to react with the bound antigen. After any free second antibody is removed by washing, substrate is added, and the colored reaction product is measured.



COMPETITIVE ELISA

Another variation for measuring amounts of antigen is competitive ELISA. In this technique, antibody is first incubated in solution with a sample containing antigen. The antigen-antibody mixture is then added to an antigencoated microtiter well. The more antigen present in the sample, the less free antibody will be available to bind to the antigen-coated well. Addition of an enzyme-conjugated secondary antibody (Ab2) specific for the isotype of the primary antibody can be used to determine the amount of primary antibody bound to the well as in an indirect ELISA. In the competitive assay, however, the higher the concentration of antigen in the original sample, the lower the absorbance.



KEY NOTES FOR IMMUNOLOGY

- Immunity is the state of protection against foreign organisms or substances (antigens). Vertebrates have two types of immunity, innate and adaptive.
- Innate immunity is not specific to any one pathogen but rather constitutes a first line of defense, which includes anatomic, physiologic, endocytic and phagocytic, and inflammatory barriers.
- Innate and adaptive immunity operate in cooperative and interdependent ways. The activation of innate immune responses produces signals that stimulate and direct subsequent adaptive immune responses.
- Adaptive immune responses exhibit four immunologic attributes: specificity, diversity, memory, and self/nonself recognition.
- The high degree of specificity in adaptive immunity arises from the activities of molecules (antibodies and T-cell receptors) that recognize and bind specific antigens.
- Antibodies recognize and interact directly with antigen. Tcell receptors recognize only antigen that is combined with either class I or class II major histocompatibility complex (MHC) molecules.
- The two major subpopulations of T lymphocytes are the CD4-T helper (TH) cells and CD8-T cytotoxic (TC) cells. TH cells secrete cytokines that regulate immune response upon recognizing antigen combined with class II MHC. TC cells recognize antigen combined with class I MHC and give rise to cytotoxic T cells (CTLs), which display cytotoxic ability.
- Exogenous (extracellular) antigens are internalized and degraded by antigen-presenting cells (macrophages, B cells, and dendritic cells); the resulting antigenic peptides complexed with class II MHC molecules are then displayed on the cell surface.
- Endogenous (intracellular) antigens (e.g., viral and tumor proteins produced in altered self-cells) are degraded in the cytoplasm and then displayed with class I MHC molecules on the cell surface.
- The immune system produces both humoral and cell-mediated responses. The humoral response is best suited for elimination of exogenous antigens; the cell-mediated response, for elimination of endogenous antigens.
- While an adaptive immune system is found only in vertebrates, innate immunity has been demonstrated in organisms as different as insects, earthworms, and higher plants.
- Dysfunctions of the immune system include common maladies such as allergy or asthma. Loss of immune function leaves the host susceptible to infection; in autoimmunity, the immune system attacks host cells or tissues.

Career Avenues

CELLS AND ORGAN OF IMMUNE SYSTEM

- The cells that participate in the immune response are white blood cells, or leukocytes. The lymphocyte is the only cell to possess the immunologic attributes of specificity, diversity, memory, and self/nonself recognition.
- Many of the body's cells, tissues, and organs arise from the progeny of different stem-cell populations. The division of a stem cell can result in the production of another stem cell and a differentiated cell of a specific type or group.
- All leukocytes develop from a common multipotent hematopoietic stem cell during hematopoiesis. Various hematopoietic growth factors (cytokines) induce proliferation and differentiation of the different blood cells. The differentiation of stem cells into different cell types requires the expression of different lineage-determining genes. A number of transcription factors play important roles in this regard.
- Hematopoiesis is closely regulated to assure steady-state levels of each of the different types of blood cell. Cell division and differentiation of each of the lineages is balanced by programmed cell death.
- There are three types of lymphocytes: B cells, T cells, and natural killer cells (NK cells). NK cells are much less abundant than B and T cells, and most lack a receptor that is specific for a particular antigen. However, a subtype of NK cells, NK1-T cells, have both T-cell receptors and many of the markers characteristic of NK cells. The three types of lymphoid cells are best distinguished on the basis of function and the presence of various membrane molecules.
- Naive B and T lymphocytes (those that have not encountered antigen) are small resting cells in the G0 phase of the cell cycle.After interacting with antigen, these cells enlarge into lymphoblasts that proliferate and eventually differentiate into effector cells and memory cells.
- Macrophages and neutrophils are specialized for the phagocytosis and degradation of antigens. Phagocytosis is facilitated by opsonins such as antibody, which increase the attachment of antigen to the membrane of the phagocyte.
- Activated macrophages secrete various factors that regulate the development of the adaptive immune response and mediate inflammation. Macrophages also process and present antigen bound to class II MHC molecules, which can then be recognized by TH cells.
- Basophils and mast cells are nonphagocytic cells that release a variety of pharmacologically active substances and play important roles in allergic reactions.
- Dendritic cells capture antigen.With the exception of follicular dendritic cells, these cells express high levels of class II MHC molecules. Along with macrophages and B cells, dendritic cells play an important role in TH-cell activation by processing and presenting antigen bound to class II MHC molecules and by providing the required co-stimulatory signal. Follicular dendritic cells, unlike the others, facilitate B-cell activation but play no role in T-cell activation.
- The primary lymphoid organs provide sites where lymphocytes mature and become antigenically committed. T lymphocytes mature within the thymus, and B lymphocytes

Career Avenues

arise and mature within the bone marrow of humans, mice, and several other animals, but not all vertebrates.

- Primary lymphoid organs are also places of selection where many lymphocytes that react with self-antigens are eliminated. Furthermore, the thymus eliminates thymocytes that would mature into useless T cells because their T-cell receptors are unable to recognize self-MHC.
- The lymphatic system collects fluid that accumulates in tissue spaces and returns this fluid to the circulation via the left subclavian vein. It also delivers antigens to the lymph nodes, which interrupt the course of lymphatic vessels.
- Secondary lymphoid organs capture antigens and provide sites where lymphocytes become activated by interaction with antigens. Activated lymphocytes undergo clonal proliferation and differentiation into effector cells.
- There are several types of secondary lymphoid tissue: lymph nodes, spleen, the loose clusters of follicles, and Peyer's patches of the intestine, and cutaneous-associated lymphoid tissue. Lymph nodes trap antigen from lymph, spleen traps blood-borne antigens, intestinal-associated lymphoid tissues (as well as other secondary lymphoid tissues) interact with antigens that enter the body from the gastrointestinal tract, and cutaneous-associated lymphoid tissue protects epithelial tissues.
- An infection that begins in one area of the body eventually involves cells, organs, and tissues that may be distant from the site of pathogen invasion. Antigen from distant sites can arrive at lymph nodes via lymph and dendritic cells, thereby assuring activation of T cells and B cells and release of these cells and their products to the circulation. Inflammatory processes bring lymphocytes and other leukocytes to the site of infection. Thus, although dispersed through- out the body, the components of the immune system communicate and collaborate to produce an effective response to infection.
- Vertebrate orders differ greatly in the kinds of lymphoid organs, tissues, and cells they possess. The most primitive vertebrates, the jawless fishes, have only gut-associated lymphoid tissues, lack B and T cells, and cannot mount adaptive immune responses. Jawed vertebrates possess a greater variety of lymphoid tissues, have B and T cells, and display adaptive immunity.

ANTIGEN AND ANTIBODY

- All immunogens are antigens but not all antigens are immunogens.
- Immunogenicity is determined by many factors including foreignness, molecular size, chemical composition, complexity, dose, susceptibility to antigen processing and presentation, the genotype of the recipient animal (in particular, its MHC genes), route of administration, and adjuvants.
- The sizes of B-cell epitopes range widely. Some are quite small (e.g., small peptides or small organic molecules), and are often bound in narrow grooves or deep pockets of the antibody. Protein B-cell epitopes are much larger and interact with a larger, flatter complementary surface on the antibody molecule.
- T-cell epitopes are generated by antigen processing, which fragments protein into small peptides that combine with class I or class II MHC molecules to form peptide-MHC

Career Avenues

complexes that are displayed on the surface of cells. T-cell activation requires the formation of a ternary complex between a T cell's TCR and peptide-MHC on antigen presenting or altered self cells.

- Haptens are small molecules that can bind to antibodies but cannot by themselves induce an immune response. However, the conjugate formed by coupling a hapten to a large carrier protein is immunogenic and elicits production of anti-hapten antibodies when injected into an animal. Such injections also produce anti-carrier and antihapten/ carrier antibodies as well.
- In the body, the formation of hapten-carrier conjugates is the basis of allergic responses to drugs such as penicillin.
- The innate immune system uses pattern-recognition receptors to recognize and respond to broad structural motifs that are highly conserved within microbial species but are generally absent from the host.
- An antibody molecule consists of two identical light chains and two identical heavy chains, which are linked by disulfide bonds. Each heavy chain has an amino-terminal variable region followed by a constant region.
- In any given antibody molecule, the constant region contains one of five basic heavy-chain sequences (α, δ, ε, γ, and μ) called isotypes and one of two basic light-chain sequences (lambda (λ) and kappa (κ)) called types.
- The heavy-chain isotype determines the class of an antibody (IgM; IgG; IgD; IgA; and IgE).
- The five antibody classes have different effector functions, average serum concentrations, and half-lives.
- Each of the domains in the immunoglobulin molecule has a characteristic tertiary structure called the immunoglobulin fold. The presence of an immunoglobulin fold domain also identifies many other nonantibody proteins as members of the immunoglobulin superfamily.
- Within the amino-terminal variable domain of each heavy and light chain are three complementarity-determining regions (CDRs). These polypeptide regions contribute the antigen-binding site of an antibody, determining its specificity.
- Immunoglobulins are expressed in two forms: secreted antibody that is produced by plasma cells, and membrane- bound antibody that associates with Ig-_/Ig-_ heterodimers to form the B-cell antigen receptor present on the surface of B cells.
- The three major effector functions that enable antibodies to remove antigens and kill pathogens are: opsonization, which promotes antigen phagocytosis by macrophages and neutrophils; complement activation, which activates a pathway that leads to the generation of a collection of proteins that can perforate cell membranes; and antibody dependent cell-mediated cytotoxicity (ADCC), which can kill antibody-bound target cells.
- Unlike polyclonal antibodies that arise from many B cell clones and have a heterogeneous collection of binding sites, a monoclonal antibody is derived from a single B cell clone and is a homogeneous collection of binding sites.

Career Avenues

T-CELL RECEPTOR, T-CELL MATURATION, ACTIVATION AND DIFFERENTIATION

- Most T-cell receptors, unlike antibodies, do not react with soluble antigen but rather with processed antigen bound to a self-MHC molecule; certain γδ receptors recognize antigens not processed and presented with MHC.
- T-cell receptors, first isolated by means of clonotypic monoclonal antibodies, are heterodimers consisting of an α and β chain or a γ and δ chain.
- The membrane-bound T-cell receptor chains are organized into variable and constant domains. TCR domains are similar to those of immunoglobulins and the V region has hypervariable regions.
- TCR germ-line DNA is organized into multigene families corresponding to the α, β, γ, δ, chains. Each family contains multiple gene segments.
- The mechanisms that generate TCR diversity are generally similar to those that generate antibody diversity, although somatic mutation does not occur in TCR genes, as it does in immunoglobulin genes.
- The T-cell receptor is closely associated with the CD3, a complex of polypeptide chains involved in signal transduction.
- T cells express membrane molecules, including CD4, CD8, CD2, LFA-1, CD28, and CD45R, that play accessory roles in T-cell function or signal transduction.
- Formation of the ternary complex TCR-antigen-MHC requires binding of a peptide to the MHC molecule and binding of the complex by the T-cell receptor.
- Interactions between TCR and MHC class I/peptide differ from those with MHC class II/peptide in the contact points between the TCR and MHC molecules.
- The γδ T-cell receptor is distinguished by ability to bind native antigens and by differences in the orientation of the variable and constant regions.
- In addition to reaction with self MHC plus foreign antigens, T cells also respond to foreign MHC molecules, a reaction that leads to rejection of allogeneic grafts.
- Progenitor T cells from the bone marrow enter the thymus and rearrange their TCR genes. In most cases these thymocytes rearrange αβ TCR genes and become αβ T cells. A small minority rearrange γδ TCR genes and become γδ T cells.
- The earliest thymocytes lack detectable CD4 and CD8 and are referred to as double-negative cells. During development, the majority of double-negative thymocytes develop into $CD4^+$ $CD8^-$ αβ T cells or $CD4^-CD8^+$ αβ T cells.
- Positive selection in the thymus eliminates T cells unable to recognize self-MHC and is the basis of MHC restriction. Negative selection eliminates thymocytes bearing high affinity receptors for self-MHC molecules alone or self-antigen plus self-MHC and produces self-tolerance.
- TH-cell activation is initiated by interaction of the TCR CD3 complex with a peptide-MHC complex on an antigen- presenting cell. Activation also requires the activity of

Career Avenues

accessory molecules, including the co-receptors CD4 and CD8. Many different intracellular signal-transduction pathways are activated by the engagement of the TCR.

- T-cells that express CD4 recognize antigen combined with a class II MHC molecule and generally function as TH cells; T cells that express CD8 recognize antigen combined with a class I MHC molecule and generally function as TC cells.
- In addition to the signals mediated by the T-cell receptor and its associated accessory molecules (signal 1), activation of the TH cell requires a co-stimulatory signal (signal 2) provided by the antigen-presenting cell. The co-stimulatory signal is commonly induced by interaction between molecules of the B7 family on the membrane of the APC with CD28 on the TH cell. Engagement of CTLA-4, a close relative of CD28, by B7 inhibits T-cell activation.
- TCR engagement with antigenic peptide-MHC may induce activation or clonal anergy. The presence or absence of the co-stimulatory signal (signal 2) determines whether activation results in clonal expansion or clonal anergy.
- Naive T cells are resting cells (G0) that have not encountered antigen. Activation of naive cells leads to the generation of effector and memory T cells. Memory T cells, which are more easily activated than naive cells, are responsible for secondary responses. Effector cells are short lived and perform helper, cytotoxic, or delayed-type hypersensitivity functions.

B-CELL GENERATION, ACTIVATION AND DIFFERENTIATION

- B cells develop in bone marrow and undergo antigen induced activation and differentiation in the periphery. Activated B cells can give rise to antibody-secreting plasma cells or memory B cells.
- During B-cell development, sequential Ig-gene rearrangements transform a pro-B cell into an immature B cell expressing mIgM with a single antigenic specificity. Further development yields mature naive B cells expressing both mIgM and mIgD.
- When a self-reactive BCR is expressed in the bone marrow, negative selection of the self-reactive immature B cells occurs. The selected cells are deleted by apoptosis or undergo receptor editing to produce non-self-reactive mIg. B cells reactive with self-antigens encountered in the periphery are rendered anergic.
- In the periphery, the antigen-induced activation and differentiation of mature B cells generates an antibody response. The antibody response to proteins and most other antigens requires TH cells. These are thymus-dependent or simply T-dependent (TD) responses. Responses to some antigens, such as certain bacterial cell-wall products (e.g., LPS) and polymeric molecules with repeating epitopes, do not require TH cells and are independent (TI) antigens. The vast majority of antigens are dependent.
- B-cell activation is the consequence of signal-transduction process triggered by engagement of the B-cell receptor that ultimately leads to many changes in the cell, including changes in the expression of specific genes.
- Activation induced by TD antigens requires contact dependent help delivered by the interaction between CD40 on B cells and CD40L on activated TH cells. The CD40/

Career Avenues

CD40L interaction is essential for B-cell survival, the formation of germinal centers, the generation of memory-cell populations, and somatic hypermutation.

- The properties of the primary and secondary antibody responses differ. The primary response has a long lag period, a logarithmic rise in antibody formation, a short plateau, and then a decline. IgM is the first antibody class produced, followed by a gradual switch to other classes, such as IgG. The secondary response has a shorter lag time, a more rapid logarithmic phase, a longer plateau phase, and a slower decline than the primary response. Mostly IgG and other isotypes are produced in the secondary response rather than IgM, and the average affinity of antibody produced is higher.
- Within a week or so of exposure to a TD antigen, germinal centers form. Germinal centers are sites of somatic hypermutation of rearranged immunoglobulin genes. Germinal centers are the sites of affinity maturation, formation of memory B cells, class switching, and plasma-cell formation.

CYTOKINES

- Cytokines are low-molecular-weight proteins that are produced and secreted by a variety of cell types. They play major roles in the induction and regulation of the cellular interactions involving cells of the immune, inflammatory and hematopoietic systems.
- The biological activities of cytokines exhibit pleiotropy, redundancy, synergy, antagonism, and, in some instances, cascade induction.
- There are over 200 different cytokines, most of which fall into one of the following families: hematopoietins, interferons, chemokines, and tumor necrosis factors.
- Cytokines act by binding to cytokine receptors, most of which can be classified as immunoglobulin superfamily receptors, class I cytokine receptors, class II cytokine receptors, members of the TNF receptor family, and chemokines receptors.
- A cytokine can only act on a cell that expresses a receptor for it. The activity of particular cytokines is directed to specific cells by regulation of the cell's profile of cytokine receptors.
- Cytokine-induced multimerization of class I and class I cytokine receptors activates a JAK/STAT signal-transduction pathway.
- Antigen stimulation of TH cells in the presence of certain cytokines can lead to the generation of subpopulations of helper T cells known as TH1 and TH2. Each subset displays characteristic and different profiles of cytokine secretion.
- The cytokine profile of TH1 cells supports immune responses that involve the marshalling of phagocytes, CTLs, and NK cells to eliminate intracellular pathogens.TH2 cells produce cytokines that support production of particular immunoglobulin isotypes and IgE-mediated responses.

Career Avenues

MAJOR HISTOCOMPATIBILITY COMPLEX

- The major histocompatibility complex (MHC) comprises a stretch of tightly linked genes that encode proteins associated with intercellular recognition and antigen presentation to T-lymphocytes.
- A group of linked MHC genes is generally inherited as a unit from parents; these linked groups are called haplotypes.
- MHC genes are polymorphic in that there are large numbers of alleles for each gene, and they are polygenic in that there are a number of different MHC genes.
- Class I MHC molecules consist of a large glycoprotein chain with 3 extracellular domains and a transmembrane segment, and β2-microglobulin, a protein with a single domain.
- Class II MHC molecules are composed of two noncovalently associated glycoproteins, the α and β chain, encoded by separate MHC genes.
- X-ray crystallographic analyses reveal peptide-binding clefts in the membrane-distal regions of both class I and class II MHC molecules.
- Both class I and class II MHC molecules present antigen to T cells. Class I molecules present processed endogenous antigen to CD8 T cells. Class II molecules present processed exogenous antigen to CD4 T cells.
- Certain conserved motifs in peptides influence their ability to interact with the membrane-distal regions of class I and class II MHC molecules.
- Class I molecules are expressed on most nucleated cells; class II antigens are restricted to B cells, macrophages, and dendritic cells.
- The class III region of the MHC encodes molecules that include a diverse group of proteins that play no role in antigen presentation.
- Detailed maps of the human and mouse MHC reveal the presence of genes involved in antigen processing, including proteasomes and transporters.
- Studies with mouse strains have shown that MHC haplotype influences immune responsiveness and the ability to present antigen.
- Increased susceptibility to a number of diseases, predominantly, but not exclusively, of an autoimmune nature, has been linked to certain MHC alleles.

ANTIGEN PROCESSING AND PRESENTATION

- T-cells recognize antigen displayed within the cleft of a self-MHC molecule on the membrane of a cell.
- In general, CD4+ TH cells recognize antigen with class II MHC molecules on antigen-processing cells.
- CD8+ TC cells recognize antigen with class I MHC molecules on target cells.
- Complexes between antigenic peptides and MHC molecules are formed by degradation of a protein antigen in one of two different antigen-processing pathways.

Career Avenues

- Endogenous antigens are degraded into peptides within the cytosol by proteasomes and assemble with class I molecules in the RER.
- Exogenous antigens are internalized and degraded within the acidic endocytic compartments and subsequently pair with class II molecules.
- Peptide binding to class II molecules involves replacing a fragment of invariant chain in the binding cleft by a process catalyzed by nonclassic MHC molecule HLA-DM.
- Presentation of nonpeptide (lipid and glycolipid) antigens derived from bacteria involves the class I–like CD1 molecules.

THE COMPLEMENT SYSTEM

- The complement system comprises a group of serum proteins, many of which exist in inactive forms.
- Complement activation occurs by the classical, alternative, or lectin pathways, each of which is initiated differently.
- The three pathways converge in a common sequence of events that leads to generation of a molecular complex that causes cell lysis.
- The classical pathway is initiated by antibody binding to a cell target; reactions of IgM and certain IgG subclasses activate this pathway.
- Activation of the alternative and lectin pathways is antibody- independent. These pathways are initiated by reaction of complement proteins with surface molecules of microorganisms.
- In addition to its key role in cell lysis, the complement system mediates opsonization of bacteria, activation of inflammation, and clearance of immune complexes.
- Interactions of complement proteins and protein fragments with receptors on cells of the immune system control both innate and acquired immune responses.
- Because of its ability to damage the host organism, the complement system requires complex passive and active regulatory mechanisms.
- Clinical consequences of inherited complement deficiencies range from increases in susceptibility to infection to tissue damage caused by immune complexes.

CELL-MEDIATED EFFECTOR RESPONSES

- The cell-mediated branch of the immune system involves two types of antigen-specific effector cells: cytotoxic T lymphocytes (CTLs) and CD4+ T cells that mediate DTH reactions. Compared with naive TH and TC cells, the effector cells are more easily activated, express higher levels of cell-adhesion molecules, exhibit different trafficking patterns, and produce both soluble and membrane effector molecules.
- The first phase of the CTL-mediated immune response involves the activation and differentiation of TC cells, called CTL precursors (CTL-Ps).

Career Avenues

- Antigen-specific CD8+ populations can be identified and tracked by labeling with MHC tetramers.
- The second phase of the CTL-mediated response involves several steps: TCR-MHC mediated recognition of target cells, formation of CTL/target-cell conjugates, reorientation of CTL cytoplasmic granules toward the target cell, granule release, formation of pores in the target-cell membrane, dissociation of CTL from the target, and the death of the target cell.
- CTLs induce cell death via two mechanisms: the perforin-granzyme pathway and the Fas/FasL pathway.
- Various nonspecific (non-MHC dependent) cytotoxic cells (NK cells, neutrophils, eosinophils, macrophages) can also kill target cells. Many of these cells bind to the Fc region of antibody on target cells and subsequently release lytic enzymes, perforin, or TNF, which damage the target-cell membrane, a process, called antibody-dependent cell mediated cytotoxicity (ADCC).
- NK cells mediate lysis of tumor cells and virus-infected cells by perforin-induced pore formation, a mechanism similar to one of those employed by CTLs.
- The expression of relatively high levels of class I MHC molecules on normal cells protects them against NK cell–mediated killing. NK cell killing is regulated by the balance between positive signals generated by the engagement of activating receptors (NKR-P1 and others) and negative signals from inhibitory receptors (CD94/NKG2 and the KIR family).

ANTIGEN-ANTIBODY INTERACTIONS: PRINCIPLES AND APPLICATIONS

- Antigen-antibody interactions depend on four types of noncovalent interactions: hydrogen bonds, ionic bonds, hydrophobic interactions, and van der Waals interactions.
- The affinity constant, which can be determined by Scatchard analysis, provides a quantitative measure of the strength of the interaction between an epitope of the antigen and a single binding site of an antibody. The avidity reflects the overall strength of the interactions between a multivalent antibody molecule and a multivalent antigen molecule at multiple sites.
- The interaction of a soluble antigen and precipitating antibody in a liquid or gel medium forms an Ag-Ab precipitate. Electrophoresis can be combined with precipitation in gels in a technique called immunoelectrophoresis.
- The interaction between a particulate antigen and agglutinating antibody (agglutinin) produces visible clumping, or agglutination that forms the basis of simple, rapid, and
- sensitive immunoassays.
- Radioimmunoassay (RIA) is a highly sensitive and quantitative procedure that utilizes radioactively labeled antigen or antibody.
- The enzyme-linked immunosorbent assay (ELISA) depends on an enzyme-substrate reaction that generates acolored reaction product. ELISA assays that employ

Career Avenues

chemiluminescence instead of a chromogenic reaction are the most sensitive immunoassays available.

- In Western blotting, a protein mixture is separated by electrophoresis; then the protein bands are electrophoretically transferred onto nitrocellulose and identified with labelled antibody or labeled antigen.
- Fluorescence microscopy using antibodies labeled with fluorescent molecules can be used to visualize antigen on or within cells.
- Flow cytometry provides an unusually powerful technology for the quantitative analysis and sorting of cell populations labeled with one or more fluorescent antibodies.

HYPERSENSITIVITY

- Hypersensitive reactions are inflammatory reactions within the humoral or cell-mediated branches of the immune system that lead to extensive tissue damage or even death. The four types of hypersensitive reaction generate characteristic effector molecules and clinical manifestations.
- A type I hypersensitive reaction is mediated by IgE antibodies, whose Fc region binds to receptors on mast cells or blood basophils. Crosslinkage of the fixed IgE by allergen leads to mast cell or basophil degranulation with release of pharmacologically active mediators. The principal effects of these mediators are smooth-muscle contraction and vasodilation. Clinical manifestations of type I reactions include potentially life-threatening systemic anaphylaxis and localized responses such as hay fever and asthma.
- A type II hypersensitive reaction occurs when antibody reacts with antigenic determinants present on the surface of cells, leading to cell damage or death through complement mediated lysis or antibody-dependent cell-mediated cytotoxicity (ADCC). Transfusion reactions and hemolytic disease of the newborn are type II reactions.
- A type III hypersensitive reaction is mediated by the formation of immune complexes and the ensuing activation of complement. Complement split products serve as immune effector molecules that elicit localized vasodilation and chemotactically attract neutrophils. Deposition of immune complexes near the site of antigen entry can induce an Arthus reaction, in which lytic enzymes released by the accumulated neutrophils and the complement membrane attack complex cause localized tissue damage.
- A type IV hypersensitive reaction involves the cell-mediated branch of the immune system. Antigen activation of sensitized TH1 cells induces release of various cytokines that cause macrophages to accumulate and become activated. The net effect of the activation of macrophages is to release lytic enzymes that cause localized tissue damage.

IMMUNE RESPONSE TO INFECTIOUS DISEASE

- Innate immune responses form the initial defense against pathogens. These include physical barriers, such as skin, as well as the nonspecific production of complement components and certain cytokines in response to infection by various pathogens.
- The immune response to viral infections involves both humoral and cell-mediated components. Antibody to a viral receptor can block viral infections of host cells. However, a number of viruses, including influenza, are able to mutate their receptor

Career Avenues

molecules and thus evade the humoral antibody response. Once a viral infection has been established, cell-mediated immunity appears to be more important than humoral.

- The immune response to extracellular bacterial infections is generally mediated by antibody. Antibody can induce localized production of immune effector molecules of the complement system, thus facilitating development of an inflammatory response.
- Antibody can also activate complement- mediated lysis of the bacterium, neutralize toxins, and serve as an opsonin to increase phagocytosis. Some bacteria secrete protease enzymes that cleave IgA dimers, thus reducing the effectiveness of IgA in the mucous secretions. Other bacteria escape phagocytosis by producing surface capsules or proteins that inhibit adherence to phagocytes, by secreting toxins that kill phagocytes, or by their ability to survive within phagocytes. Host defense against intracellular bacteria depends largely on CD4+ T-cell–mediated responses.
- Both humoral and cell-mediated immune responses have been implicated in immunity to protozoan infections. In general, humoral antibody is effective against blood-borne stages of the protozoan life-cycle, but once protozoans have infected host cells, cell-mediated immunity is necessary. Protozoans escape the immune response through several mechanisms.
- Some—notably, *Trypanosoma brucei*—are covered by a glycoprotein coat that is constantly changed by a genetic switch mechanism. Others (including *Plasmodium*, the causative agent of malaria) slough off their glycoprotein coat after antibody has bound to it.
- Helminths are large parasites that normally do not multiply within cells. Because few of these organisms are carried by an affected individual, immune-system exposure to helminths is limited; consequently, only a low level of immunity is induced. Although helminths generally are attacked by antibody-mediated defenses, these may be ineffective. A cell-mediated response by CD4+ T cells plays a critical role in the response to *Schistosoma.*
- Emerging and re-emerging pathogens include some that are newly described and others that had been thought to be controlled by public-health practices. Factors leading to the emergence of such pathogens include increased travel and intense crowding of some populations.

VACCINE

- A state of immunity can be induced by passive or active immunization:
    - Short-term passive immunization is induced by transfer of preformed antibodies.
    - Infection or inoculation achieves long-term active immunization.
- Three types of vaccines are currently used in humans: attenuated (avirulent) microorganisms, inactivated (killed) microorganisms, or purified macromolecules.
- Protein components of pathogens expressed in cell culture may be effective vaccines.
- Recombinant vectors, including viruses or bacteria, engineered to carry genes from infectious microorganisms, maximize cell-mediated immunity to the encoded antigens.

Career Avenues

- Plasmid DNA encoding a protein antigen from a pathogen can serve as an effective vaccine inducing both humoral and cell-mediated immunity.
- Realizing the optimum benefit of vaccines will require cheaper manufacture and proved delivery methods for existing vaccines.

Career Avenues